Problem 2-1

Determine the magnitude of the resultant force $\mathbf{F_R = F_1 + F_2}$ and its direction, measured counterclockwise from the positive *x* axis.

Given:

$F_1 = 600$ N

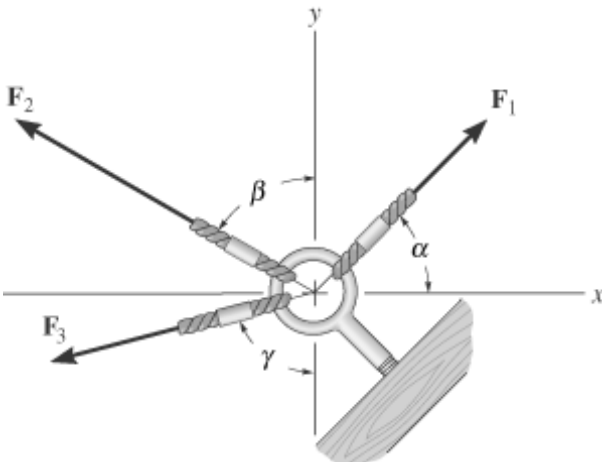
$F_2 = 800$ N

$F_3 = 450$ N

$\alpha = 45$ deg

$\beta = 60$ deg

$\gamma = 75$ deg



Solution:

$\psi = 90 \text{ deg} - \beta + \alpha$

$F_R = \sqrt{F_1^2 + F_2^2 - 2\, F_1\, F_2 \cos(\psi)}$

$F_R = 867$ N

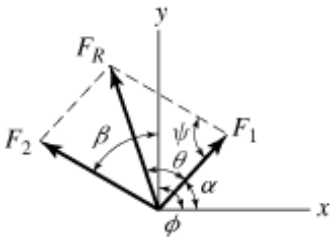
$\dfrac{F_R}{\sin(\psi)} = \dfrac{F_2}{\sin(\theta)}$

$\theta = \operatorname{asin}\left(F_2 \dfrac{\sin(\psi)}{F_R}\right)$

$\theta = 63.05 \text{ deg}$

$\phi = \theta + \alpha$

$\phi = 108 \text{ deg}$



Problem 2-2

Determine the magnitude of the resultant force and its direction measured counterclockwise from the positive *x* axis.

Given:

$F_1 = 80$ lb

$F_2 = 60$ lb

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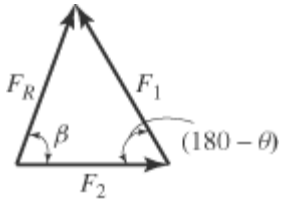
$\theta = 120 \text{ deg}$

Solution:

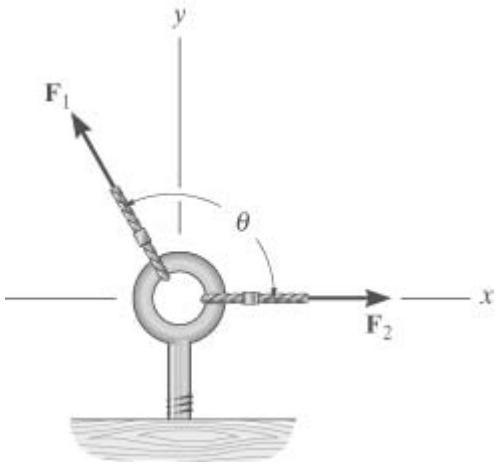
$$F_R = \sqrt{F_1^2 + F_2^2 - 2\,F_1\,F_2\cos(180 \text{ deg} - \theta)}$$

$F_R = 72.1 \text{ lb}$

$$\beta = \operatorname{asin}\left(F_1\,\frac{\sin(180 \text{ deg} - \theta)}{F_R}\right)$$



$\beta = 73.9 \text{ deg}$



Problem 2-3

Determine the magnitude of the resultant force $\mathbf{F_R} = \mathbf{F_1} + \mathbf{F_2}$ and its direction, measured counterclockwise from the positive *x* axis.

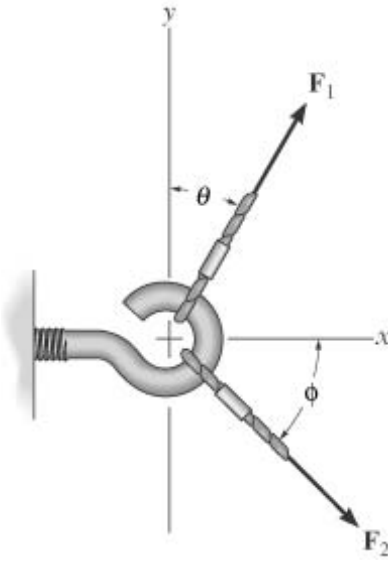
Given:

$F_1 = 250 \text{ lb}$

$F_2 = 375 \text{ lb}$

$\theta = 30 \text{ deg}$

$\phi = 45 \text{ deg}$



Solution:

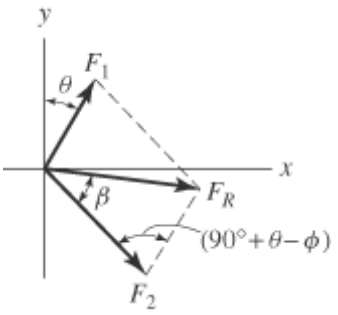
$$F_R = \sqrt{F_1^2 + F_2^2 - 2\,F_1\,F_2\cos(90 \text{ deg} + \theta - \phi)}$$

$F_R = 178 \text{ kg}$

$$\frac{F_R}{\sin(90 \text{ deg} + \theta - \phi)} = \frac{F_1}{\sin(\beta)}$$

$$\beta = \operatorname{asin}\left(\frac{F_1}{F_R}\,\sin(90 \text{ deg} + \theta - \phi)\right)$$

$\beta = 37.89 \text{ deg}$



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Angle measured ccw from *x* axis

$360\ \text{deg} - \phi + \beta = 353\,\text{deg}$

Problem 2-4

Determine the magnitude of the resultant force $\mathbf{F_R} = \mathbf{F_1} + \mathbf{F_2}$ and its direction, measured counterclockwise from the positive *u* axis.

Given:

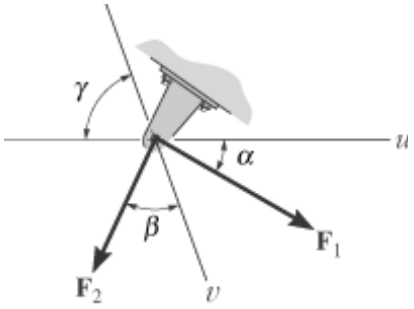
$F_1 = 300\ \text{N}$

$F_2 = 500\ \text{N}$

$\alpha = 30\ \text{deg}$

$\beta = 45\ \text{deg}$

$\gamma = 70\ \text{deg}$



Solution:

$$F_R = \sqrt{F_1^2 + F_2^2 - 2\,F_1\,F_2\cos(180\ \text{deg} - \beta - \gamma + \alpha)}$$

$F_R = 605\ \text{N}$

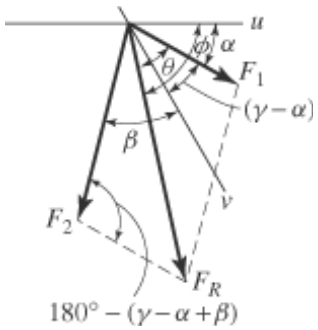
$$\frac{F_R}{\sin(180\ \text{deg} - \beta - \gamma + \alpha)} = \frac{F_2}{\sin(\theta)}$$

$$\theta = \text{asin}\left(F_2\,\frac{\sin(180\ \text{deg} - \beta - \gamma + \alpha)}{F_R}\right)$$

$\theta = 55.40\,\text{deg}$

$\phi = \theta + \alpha$

$\phi = 85.4\,\text{deg}$



Problem 2-5

Resolve the force $\mathbf{F_1}$ into components acting along the *u* and *v* axes and determine the magnitudes of the components.

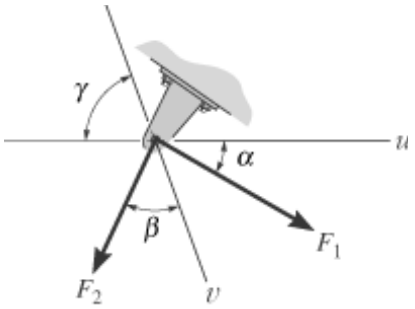
Given:

$F_1 = 300\ \text{N} \qquad \alpha = 30\ \text{deg}$

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$F_2 = 500\ \text{N} \quad \beta = 45\ \text{deg}$

$\gamma = 70\ \text{deg}$



Solution:

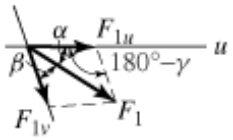
$$\frac{F_{1u}}{\sin(\gamma - \alpha)} = \frac{F_1}{\sin(180\ \text{deg} - \gamma)}$$

$$F_{1u} = F_1 \frac{\sin(\gamma - \alpha)}{\sin(180\ \text{deg} - \gamma)}$$

$F_{1u} = 205\ \text{N}$

$$\frac{F_{1v}}{\sin(\alpha)} = \frac{F_1}{\sin(180\ \text{deg} - \gamma)}$$

$$F_{1v} = F_1 \frac{\sin(\alpha)}{\sin(180\ \text{deg} - \gamma)}$$



$F_{1v} = 160\ \text{N}$

Problem 2-6

Resolve the force $\mathbf{F_2}$ into components acting along the *u* and *v* axes and determine the magnitudes of the components.

Given:

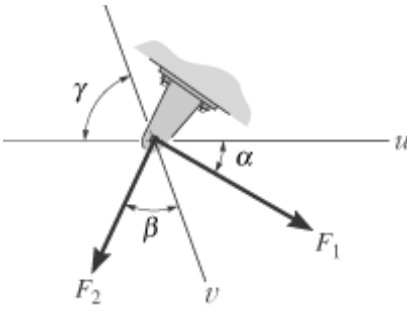
$F_1 = 300\ \text{N}$

$F_2 = 500\ \text{N}$

$\alpha = 30\ \text{deg}$

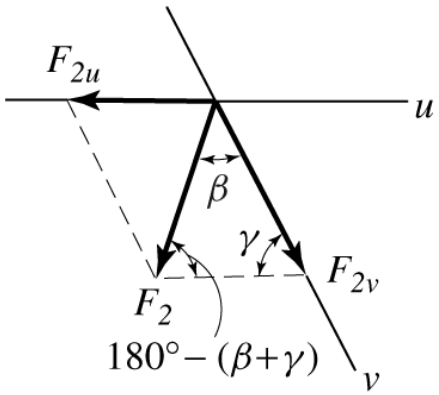
$\beta = 45\ \text{deg}$

$\gamma = 70\ \text{deg}$



Solution:

$$F_{2u} = F_2\left(\frac{\sin(\beta)}{\sin(\gamma)}\right)$$



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$F_{2u} = 376.2\ \text{N}$

$$F_{2v} = F_2\left[\frac{\sin\left[180\ \text{deg} - (\beta + \gamma)\right]}{\sin(\gamma)}\right]$$

$F_{2v} = 482.2\ \text{N}$

Problem 2-7

Determine the magnitude of the resultant force $\mathbf{F_R} = \mathbf{F_1} + \mathbf{F_2}$ and its direction measured counterclockwise from the positive u axis.

Given:

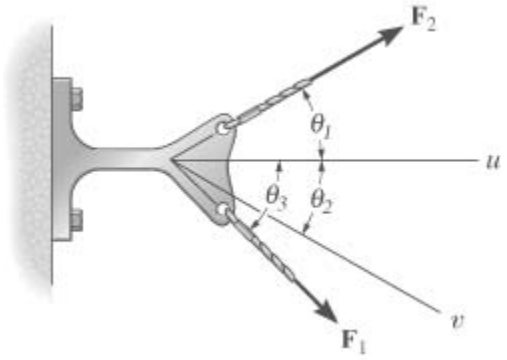
$F_1 = 25\ \text{lb}$

$F_2 = 50\ \text{lb}$

$\theta_1 = 30\ \text{deg}$

$\theta_2 = 30\ \text{deg}$

$\theta_3 = 45\ \text{deg}$



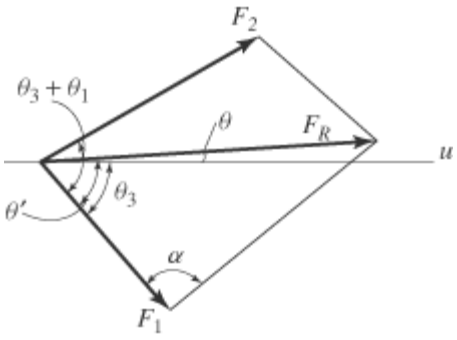
Solution:

$\alpha = 180\ \text{deg} - (\theta_3 + \theta_1)$

$$F_R = \sqrt{F_2^2 + F_1^2 - 2\,F_1\,F_2\cos(\alpha)}$$

$F_R = 61.4\,\text{lb}$

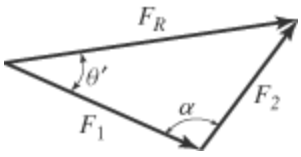
$$\frac{\sin(\theta')}{F_2} = \frac{\sin(\alpha)}{F_R} \qquad \theta' = \text{asin}\left(\sin(\alpha)\,\frac{F_2}{F_R}\right)$$



$\theta' = 51.8\,\text{deg}$

$\theta = \theta' - \theta_3$

$\theta = 6.8\,\text{deg}$



Problem 2-8

Resolve the force $\mathbf{F_1}$ into components acting along the u and v axes and determine the components.

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Given:

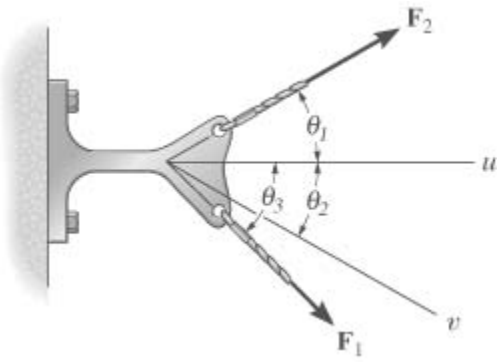
$F_1 = 25$ lb

$F_2 = 50$ lb

$\theta_1 = 30$ deg

$\theta_2 = 30$ deg

$\theta_3 = 45$ deg

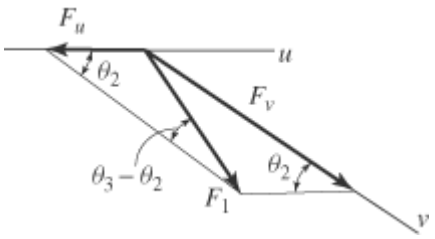


Solution:

$$\frac{-F_u}{\sin(\theta_3 - \theta_2)} = \frac{F_1}{\sin(\theta_2)}$$

$$F_u = \frac{-F_1 \sin(\theta_3 - \theta_2)}{\sin(\theta_2)}$$

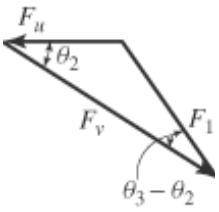
$F_u = -12.9$ lb



$$\frac{F_v}{\sin(180 \text{ deg} - \theta_3)} = \frac{F_1}{\sin(\theta_2)}$$

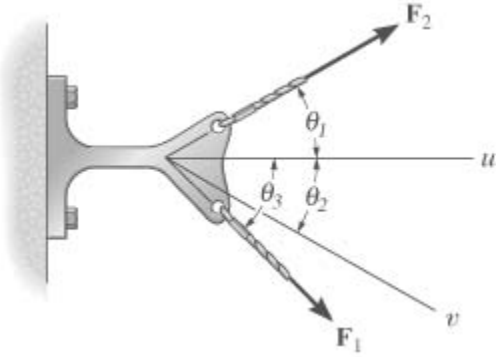
$$F_v = \frac{F_1 \sin(180 \text{ deg} - \theta_3)}{\sin(\theta_2)}$$

$F_v = 35.4$ lb



Problem 2-9

Resolve the force $\mathbf{F_2}$ into components acting along the *u* and *v* axes and determine the components.



Given:

$F_1 = 25$ lb

$F_2 = 50$ lb

$\theta_1 = 30$ deg

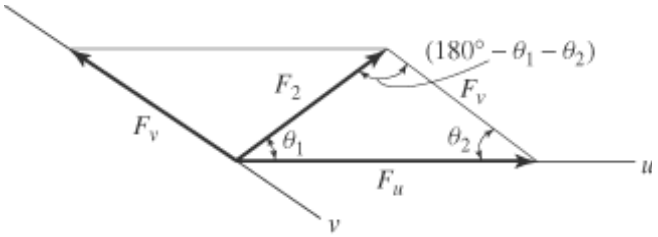
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$\theta_2 = 30 \text{ deg}$

$\theta_3 = 45 \text{ deg}$

Solution:

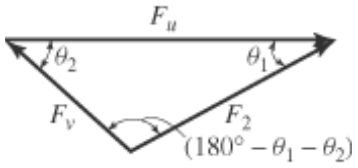
$$\frac{F_u}{\sin\left[180 \text{ deg} - (\theta_1 + \theta_2)\right]} = \frac{F_2}{\sin(\theta_2)}$$



$$F_u = \frac{F_2 \sin\left[180 \text{ deg} - (\theta_1 + \theta_2)\right]}{\sin(\theta_2)}$$

$F_u = 86.6 \text{ lb}$

$$\frac{-F_v}{\sin(\theta_1)} = \frac{F_2}{\sin(\theta_2)}$$



$$F_v = \frac{-F_2 \sin(\theta_1)}{\sin(\theta_2)}$$

$F_v = -50 \text{ lb}$

Problem 2-10

Determine the components of the **F** force acting along the *u* and *v* axes.

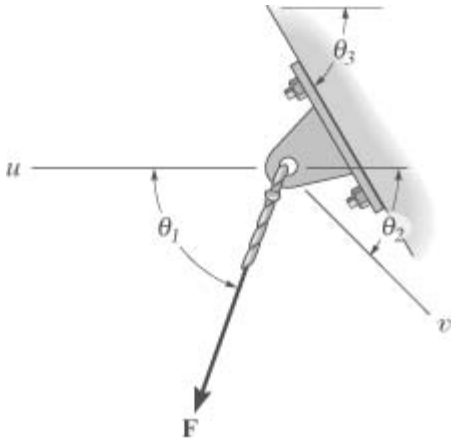
Given:

$\theta_1 = 70 \text{ deg}$

$\theta_2 = 45 \text{ deg}$

$\theta_3 = 60 \text{ deg}$

$F = 250 \text{ N}$



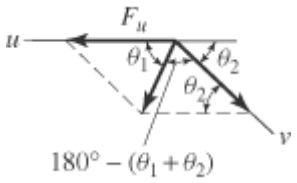
Solution:

$$\frac{F_u}{\sin\left[180 \text{ deg} - (\theta_1 + \theta_2)\right]} = \frac{F}{\sin(\theta_2)}$$

$$F_u = \frac{F \sin\left[180 \text{ deg} - (\theta_1 + \theta_2)\right]}{\sin(\theta_2)} \qquad F_u = 320 \text{ N}$$

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$$\frac{F_v}{\sin(\theta_1)} = \frac{F}{\sin(\theta_2)}$$



$$F_v = \frac{F\sin(\theta_1)}{\sin(\theta_2)}$$

$F_v = 332$ N

Problem 2-11

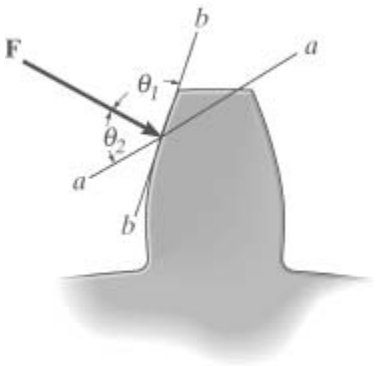
The force **F** acts on the gear tooth**.** Resolve this force into two components acting along the lines *aa* and *bb.*

Given:

$F = 20$ lb

$\theta_1 = 80$ deg

$\theta_2 = 60$ deg

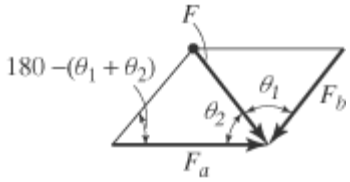


Solution:

$$\frac{F}{\sin\left[180 \text{ deg} - (\theta_1 + \theta_2)\right]} = \frac{F_a}{\sin(\theta_1)}$$

$$F_a = \frac{F\sin(\theta_1)}{\sin\left[180 \text{ deg} - (\theta_1 + \theta_2)\right]}$$

$F_a = 30.6$ lb



$$\frac{F}{\sin\left[180 \text{ deg} - (\theta_1 + \theta_2)\right]} = \frac{F_b}{\sin(\theta_2)}$$

$$F_b = \frac{F\sin(\theta_2)}{\sin\left[180 \text{ deg} - (\theta_1 + \theta_2)\right]}$$

$F_b = 26.9$ lb

Problem 2-12

The component of force **F** acting along line *aa* is required to be F_a. Determine the magnitude of **F** and its component along line *bb*.

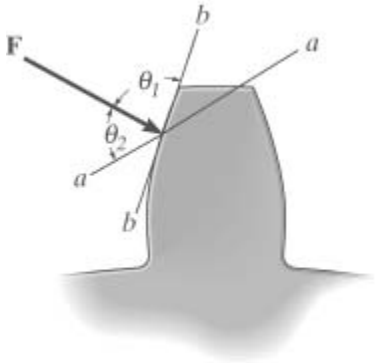
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Given:

$F_a = 30$ lb

$\theta_1 = 80$ deg

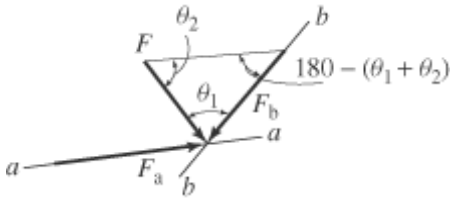
$\theta_2 = 60$ deg



Solution:

$$\frac{F_a}{\sin(\theta_1)} = \frac{F}{\sin[180\ \text{deg} - (\theta_1 + \theta_2)]}$$

$$F = F_a \left(\frac{\sin(180\ \text{deg} - \theta_1 - \theta_2)}{\sin(\theta_1)} \right) \qquad F = 19.6\,\text{lb}$$



$$\frac{F_a}{\sin(\theta_1)} = \frac{F_b}{\sin(\theta_2)}$$

$$F_b = \frac{F_a \sin(\theta_2)}{\sin(\theta_1)} \qquad F_b = 26.4\,\text{lb}$$

Problem 2-13

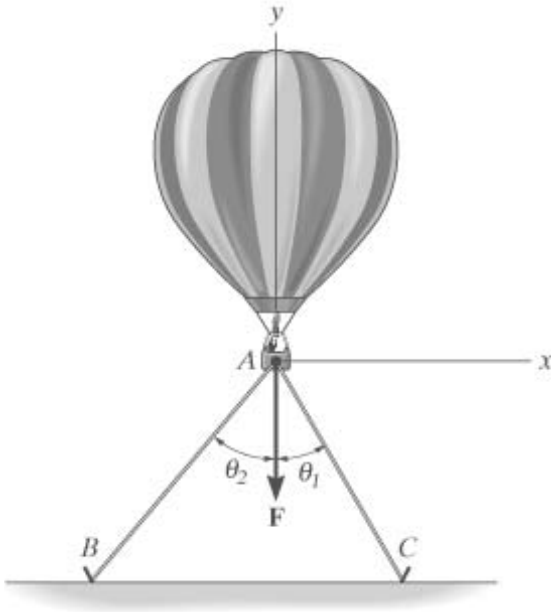
A resultant force **F** is necessary to hold the ballon in place. Resolve this force into components along the tether lines *AB* and *AC*, and compute the magnitude of each component.

Given:

$F = 350$ lb

$\theta_1 = 30$ deg

$\theta_2 = 40$ deg

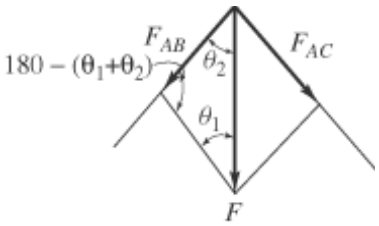


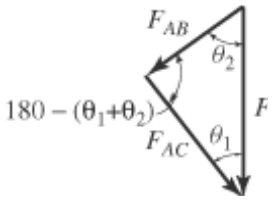
Solution:

$$\frac{F_{AB}}{\sin(\theta_1)} = \frac{F}{\sin[180\ \text{deg} - (\theta_1 + \theta_2)]}$$

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$$F_{AB} = F\left[\frac{\sin(\theta_1)}{\sin\left[180\ \text{deg} - (\theta_1 + \theta_2)\right]}\right]$$





$F_{AB} = 186\,\text{lb}$

$$\frac{F_{AC}}{\sin(\theta_2)} = \frac{F}{\sin\left[180\ \text{deg} - (\theta_1 + \theta_2)\right]}$$

$$F_{AC} = F\left[\frac{\sin(\theta_2)}{\sin\left[180\ \text{deg} - (\theta_1 + \theta_2)\right]}\right]$$

$F_{AC} = 239\,\text{lb}$

Problem 2-14

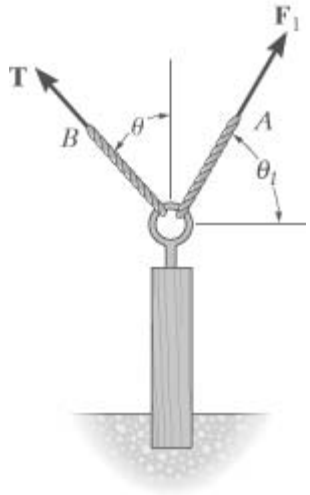
The post is to be pulled out of the ground using two ropes A and B. Rope A is subjected to force F_1 and is directed at angle θ_1 from the horizontal. If the resultant force acting on the post is to be F_R, vertically upward, determine the force $\mathbf{T}$ in rope B and the corresponding angle θ.

Given:

$F_R = 1200\ \text{lb}$

$F_1 = 600\ \text{lb}$

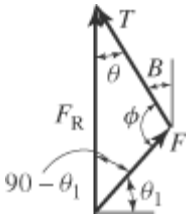
$\theta_1 = 60\ \text{deg}$



Solution:

$$T = \sqrt{F_1^{\,2} + F_R^{\,2} - 2\,F_1\,F_R\cos(90\ \text{deg} - \theta_1)}$$

$T = 744\,\text{lb}$



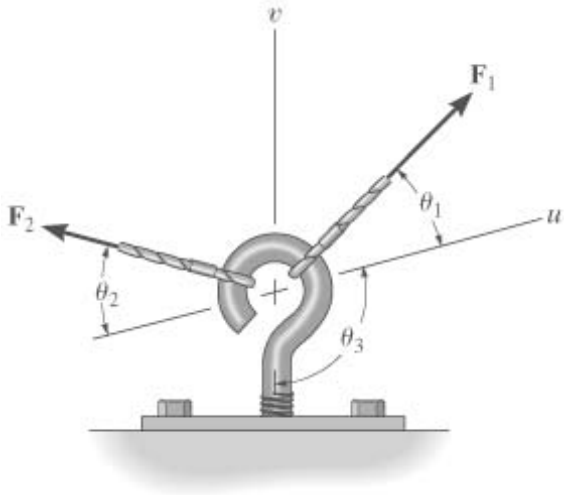
$$\frac{\sin(\theta)}{F_R} = \frac{\sin(90 - \theta_1)}{T}$$

$$\theta = \text{asin}\left(\sin(90\ \text{deg} - \theta_1)\,\frac{F_1}{T}\right) \qquad \theta = 23.8\,\text{deg}$$

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Problem 2-15

Resolve the force $\mathbf{F_1}$ into components acting along the *u* and *v* axes and determine the magnitudes of the components.



Given:

$F_1 = 250\ \text{N}$

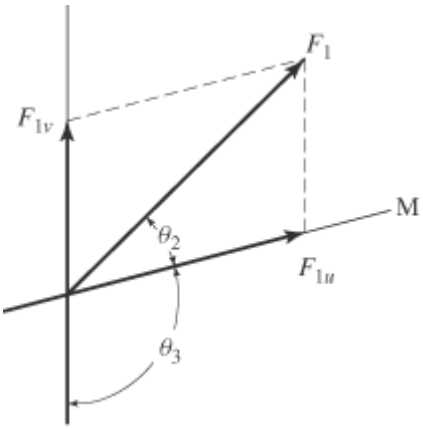
$F_2 = 150\ \text{N}$

$\theta_1 = 30\ \text{deg}$

$\theta_2 = 30\ \text{deg}$

$\theta_3 = 105\ \text{deg}$

Solution:



$$\frac{F_{1v}}{\sin(\theta_1)} = \frac{F_1}{\sin(\theta_3)}$$

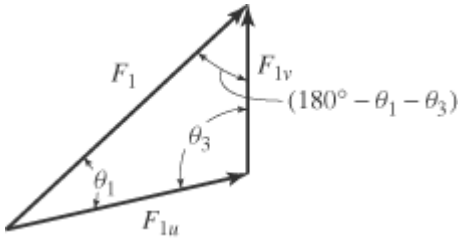
$$F_{1v} = F_1\left(\frac{\sin(\theta_1)}{\sin(\theta_3)}\right)$$

$F_{1v} = 129\,\text{N}$

$$\frac{F_{1u}}{\sin(180\ \text{deg} - \theta_1 - \theta_3)} = \frac{F_1}{\sin(\theta_3)}$$

$$F_{1u} = F_1\left(\frac{\sin(180\ \text{deg} - \theta_1 - \theta_3)}{\sin(\theta_3)}\right)$$

$F_{1u} = 183\ \text{N}$



Problem 2-16

Resolve the force $\mathbf{F_2}$ into components acting along the *u* and *v* axes and determine the magnitudes of the components.

Given:

$F_1 = 250\ \text{N}$

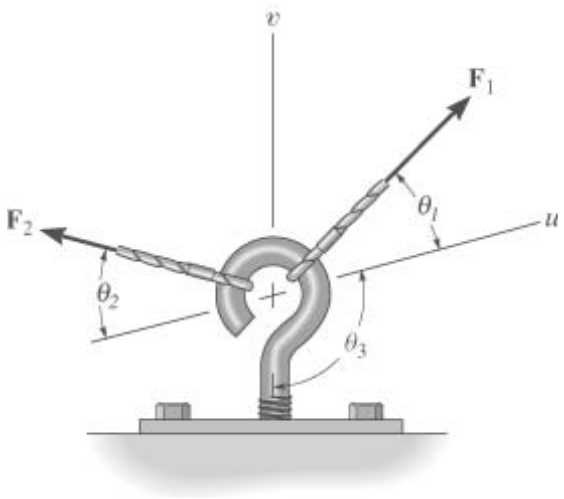
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$F_2 = 150\ \text{N}$

$\theta_1 = 30\ \text{deg}$

$\theta_2 = 30\ \text{deg}$

$\theta_3 = 105\ \text{deg}$



Solution:

$$\frac{F_{1v}}{\sin(\theta_1)} = \frac{F_2}{\sin(180\ \text{deg} - \theta_3)}$$

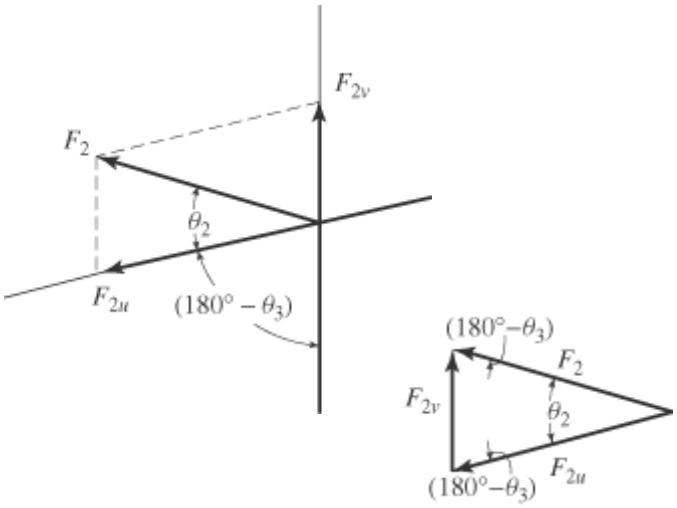
$$F_{1v} = F_2 \left(\frac{\sin(\theta_1)}{\sin(180\ \text{deg} - \theta_3)} \right)$$

$F_{1v} = 77.6\,\text{N}$

$$\frac{F_{2u}}{\sin(180\ \text{deg} - \theta_3)} = \frac{F_2}{\sin(180\ \text{deg} - \theta_3)}$$

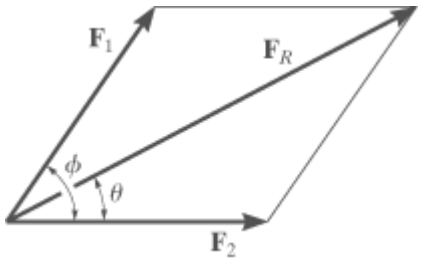
$$F_{2u} = F_2 \left(\frac{\sin(180\ \text{deg} - \theta_3)}{\sin(180\ \text{deg} - \theta_3)} \right)$$

$F_{2u} = 150\ \text{N}$



Problem 2-17

Determine the magnitude and direction of the resultant force $\mathbf{F_R}$. Express the result in terms of the magnitudes of the components $\mathbf{F_1}$ and $\mathbf{F_2}$ and the angle ϕ.



Solution:

$$F_R^2 = F_1^2 + F_2^2 - 2F_1\,F_2\cos(180\ \text{deg} - \phi)$$

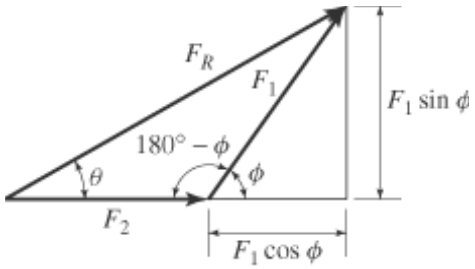
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Since $\cos(180\,\text{deg} - \phi) = -\cos(\phi)$,

$$F_R = \sqrt{F_1^2 + F_2^2 + 2\,F_1\,F_2\,\cos(\phi)}$$

From the figure,

$$\tan(\theta) = \frac{F_1\,\sin(\phi)}{F_2 + F_1\,\cos(\phi)}$$



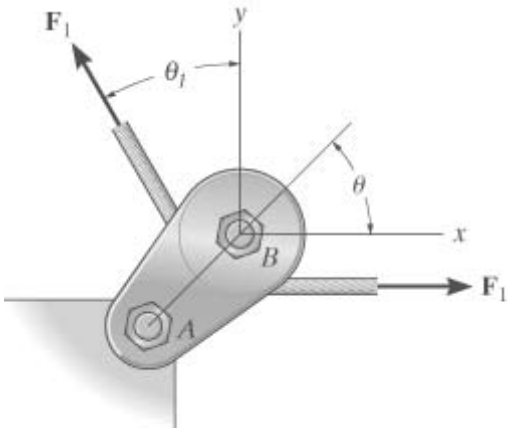
Problem 2-18

If the tension in the cable is $\mathbf{F_1}$, determine the magnitude and direction of the resultant force acting on the pulley. This angle defines the same angle θ of line *AB* on the tailboard block.

Given:

$F_1 = 400$ N

$\theta_1 = 30$ deg



Solution:

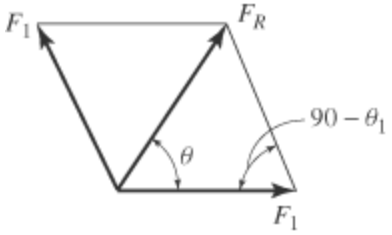
$$F_R = \sqrt{F_1^2 + F_1^2 - 2F_1F_1\cos(90\ \text{deg} - \theta_1)}$$

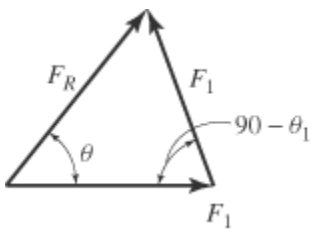
$F_R = 400$ N

$$\frac{\sin(90\ \text{deg} - \theta)}{F_R} = \frac{\sin(\theta_1)}{F_1}$$

$$\theta = 90\ \text{deg} - \text{asin}\left(\frac{F_R}{F_1}\sin(\theta_1)\right)$$

$\theta = 60\,\text{deg}$





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Problem 2-19

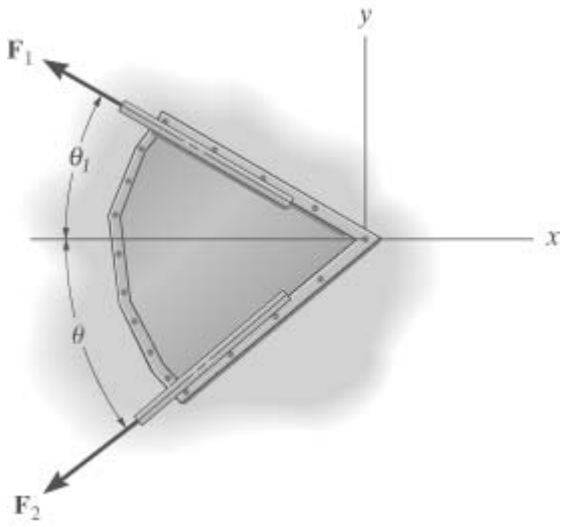
The riveted bracket supports two forces. Determine the angle θ so that the resultant force is directed along the negative *x* axis. What is the magnitude of this resultant force?

Given:

$F_1 = 60$ lb

$F_2 = 70$ lb

$\theta_1 = 30$ deg

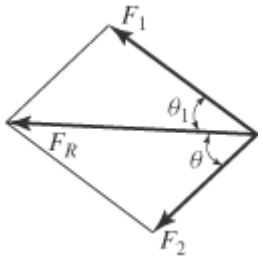


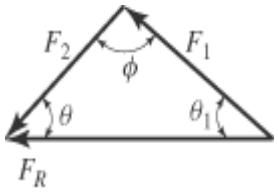
Solution:

$$\frac{\sin(\theta)}{F_1} = \frac{\sin(\theta_1)}{F_2}$$

$$\theta = \operatorname{asin}\left(\sin(\theta_1)\,\frac{F_1}{F_2}\right)$$

$\theta = 25.4\,\text{deg}$





$\phi = 180\ \text{deg} - \theta - \theta_1$

$\phi = 124.6\,\text{deg}$

$$R = \sqrt{F_1^{\,2} + F_2^{\,2} - 2F_1F_2\cos(\phi)}$$

$R = 115\,\text{lb}$

Problem 2-20

The plate is subjected to the forces acting on members *A* and *B* as shown. Determine the magnitude of the resultant of these forces and its direction measured clockwise from the positive *x* axis.

Given:

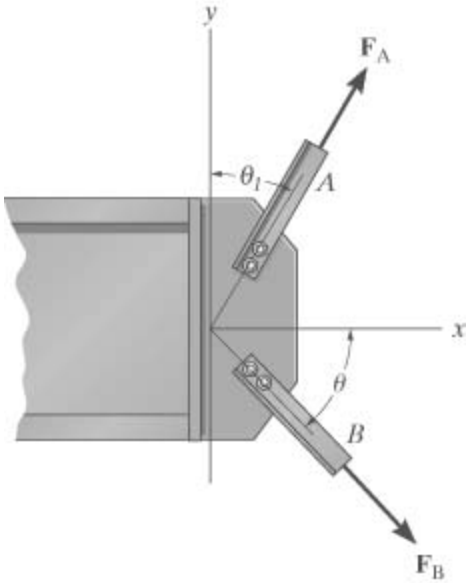
$F_A = 400$ lb

$F_B = 500$ lb

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$\theta_1 = 30 \text{ deg}$

$\theta = 60 \text{ deg}$



Solution:

Cosine law:

$$F_R = \sqrt{F_B{}^2 + F_A{}^2 - 2F_B F_A \cos\left(90 \text{ deg} - \theta + \theta_1\right)}$$

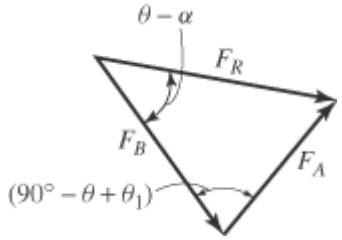
$F_R = 458 \text{ lb}$

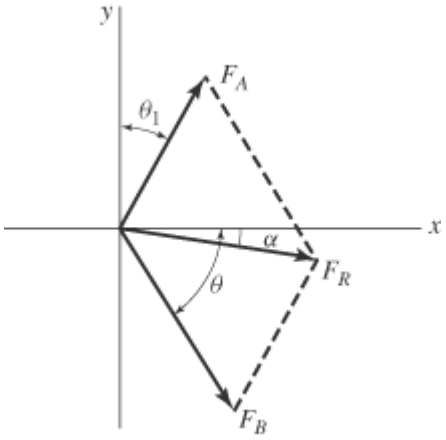
Sine law:

$$\frac{\sin\left(90 \text{ deg} - \theta + \theta_1\right)}{F_R} = \frac{\sin\left(\theta - \alpha\right)}{F_A}$$

$$\alpha = \theta - \text{asin}\left(\sin\left(90 \text{ deg} - \theta + \theta_1\right)\frac{F_A}{F_R}\right)$$

$\alpha = 10.9 \text{ deg}$





Problem 2-21

Determine the angle θ for connecting member B to the plate so that the resultant of $\mathbf{F_A}$ and $\mathbf{F_B}$ is directed along the positive x axis. What is the magnitude of the resultant force?

Given:

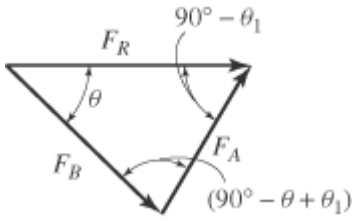
$F_A = 400 \text{ lb}$

$F_B = 500 \text{ lb}$

$\theta_1 = 30 \text{ deg}$

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Solution:



Sine law:

$$\frac{\sin(\theta)}{F_A} = \frac{\sin(90\ \text{deg} - \theta_1)}{F_B}$$

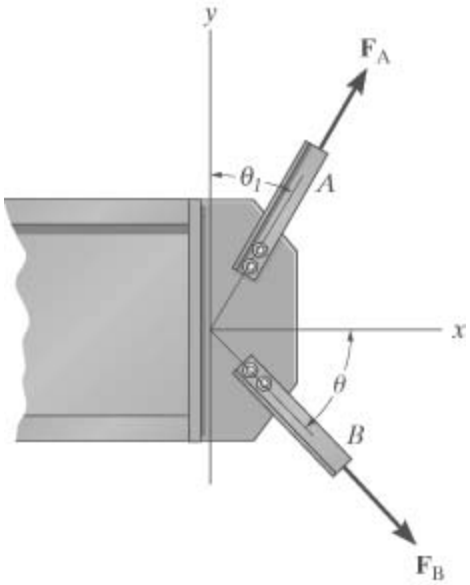
$$\theta = \text{asin}\left(\sin(90\ \text{deg} - \theta_1)\ \frac{F_A}{F_B}\right)$$

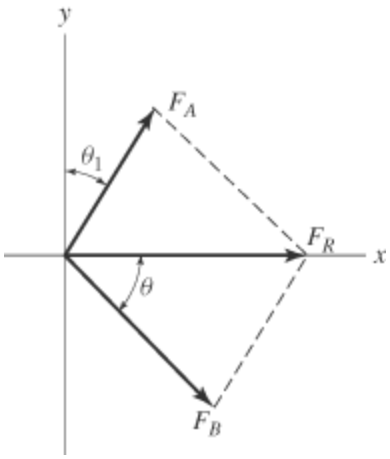
$\theta = 43.9\,\text{deg}$

$$\frac{F_R}{\sin(90\ \text{deg} + \theta_1 - \theta)} = \frac{F_A}{\sin(\theta)}$$

$$F_R = F_A\ \frac{\sin(90\ \text{deg} - \theta + \theta_1)}{\sin(\theta)}$$

$F_R = 561\,\text{lb}$





Problem 2-22

Determine the magnitude and direction of the resultant $\mathbf{F_R} = \mathbf{F_1} + \mathbf{F_2} + \mathbf{F_3}$ of the three forces by first finding the resultant $\mathbf{F'} = \mathbf{F_1} + \mathbf{F_2}$ and then forming $\mathbf{F_R} = \mathbf{F'} + \mathbf{F_3}$.

Given:

$F_1 = 30$ N

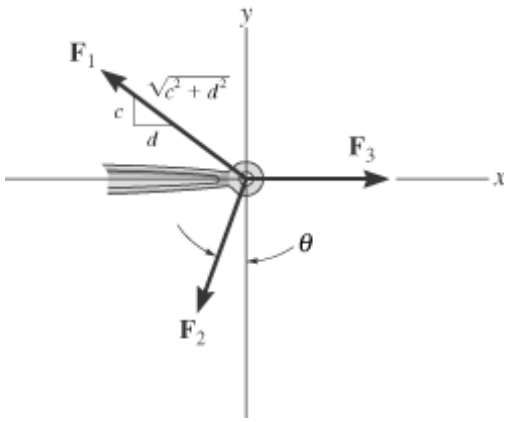
$F_2 = 20$ N

$F_3 = 50$ N

$\theta = 20$ deg

$c = 3$

$d = 4$



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Solution:

$$\alpha = \operatorname{atan}\left(\frac{c}{d}\right)$$

$$F' = \sqrt{F_1{}^2 + F_2{}^2 - 2F_1 F_2 \cos(90\ \text{deg} + \theta - \alpha)}$$

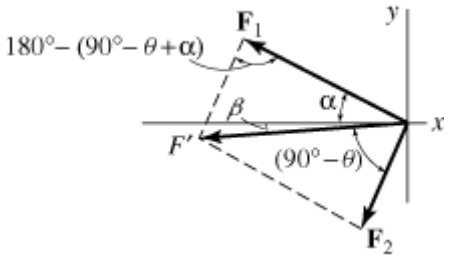
$$F' = 30.9\ \text{N}$$

$$\frac{F'}{\sin((90\ \text{deg} - \theta + \alpha))} = \frac{F_1}{\sin(90\ \text{deg} - \theta - \beta)}$$

$$\beta = 90\ \text{deg} - \theta - \operatorname{asin}\left(F_1\ \frac{\sin(90\ \text{deg} - \theta + \alpha)}{F'}\right)$$

$$\beta = 1.5\ \text{deg}$$

Now add in force F_3.



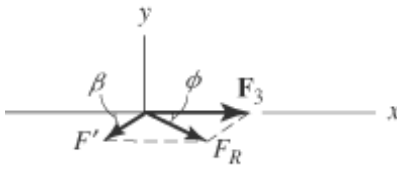
$$F_R = \sqrt{F'^2 + F_3{}^2 - 2F'\ F_3 \cos(\beta)}$$

$$F_R = 19.2\ \text{N}$$

$$\frac{F_R}{\sin(\beta)} = \frac{F'}{\sin(\phi)}$$

$$\phi = \operatorname{asin}\left(F'\ \frac{\sin(\beta)}{F_R}\right)$$

$$\phi = 2.4\ \text{deg}$$



Problem 2-23

Determine the magnitude and direction of the resultant $\mathbf{F_R} = \mathbf{F_1} + \mathbf{F_2} + \mathbf{F_3}$ of the three forces by first finding the resultant $\mathbf{F'} = \mathbf{F_2} + \mathbf{F_3}$ and then forming $\mathbf{F_R} = \mathbf{F'} + \mathbf{F_1}$.

Given:

$$F_1 = 30\ \text{N}$$

$$F_2 = 20\ \text{N}$$

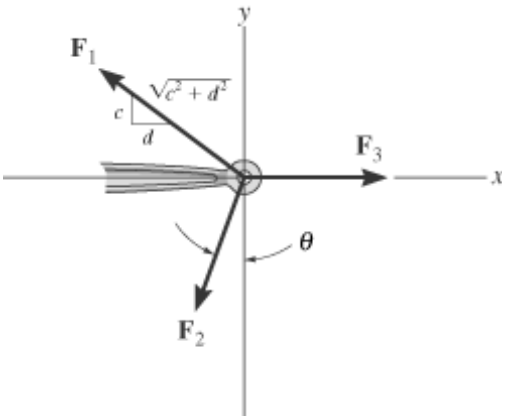
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$F_3 = 50\ \text{N}$

$\theta = 20\ \text{deg}$

$c = 3$

$d = 4$

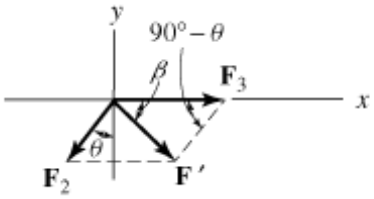


Solution:

$$F' = \sqrt{F_2^2 + F_3^2 - 2F_2F_3\cos((90\ \text{deg} - \theta))}$$

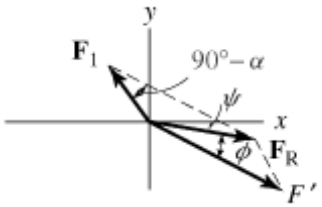
$F' = 47.07\ \text{N}$

$$\frac{F_2}{\sin(\beta)} = \frac{F'}{\sin(90\ \text{deg} - \theta)}$$



$$\beta = \text{asin}\left(F_2\ \frac{\sin(90\ \text{deg} - \theta)}{F'}\right)$$

$\beta = 23.53\,\text{deg}$



$$\alpha = \text{atan}\left(\frac{c}{d}\right)$$

$\gamma = \alpha - \beta$

$$F_R = \sqrt{F'^2 + F_1^2 - 2F'F_1\cos(\gamma)}$$

$F_R = 19.2\ \text{N}$

$$\frac{F_R}{\sin(\gamma)} = \frac{F_1}{\sin(\phi)}$$

$$\phi = \text{asin}\left(F_1\ \frac{\sin(\gamma)}{F_R}\right)$$

$\phi = 21.16\,\text{deg}$

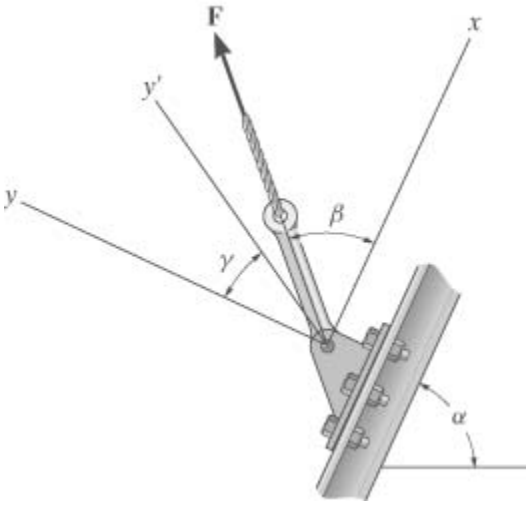
$\psi = \beta - \phi$

$\psi = 2.37\,\text{deg}$

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Problem 2-24

Resolve the force **F** into components acting along (a) the *x* and *y* axes, and (b) the *x* and *y'* axes.



Given:

$F = 50\ \text{lb}$

$\alpha = 65\ \text{deg}$

$\beta = 45\ \text{deg}$

$\gamma = 30\ \text{deg}$

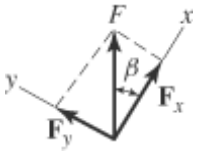
Solution:

(a) $F_x = F\cos(\beta)$

$F_x = 35.4\,\text{lb}$

$F_y = F\sin(\beta)$

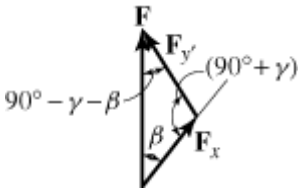
$F_y = 35.4\,\text{lb}$



(b) $$\frac{F_x}{\sin(90\ \text{deg} - \beta - \gamma)} = \frac{F}{\sin(90\ \text{deg} + \gamma)}$$

$$F_x = F\,\frac{\sin(90\ \text{deg} - \beta - \gamma)}{\sin(90\ \text{deg} + \gamma)}$$

$F_x = 14.9\,\text{lb}$



$$\frac{F_{y'}}{\sin(\beta)} = \frac{F}{\sin(90\ \text{deg} + \gamma)}$$

$$F_{y'} = F\,\frac{\sin(\beta)}{\sin(90\ \text{deg} + \gamma)}$$

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$F_{y'} = 40.8\,\text{lb}$

Problem 2-25

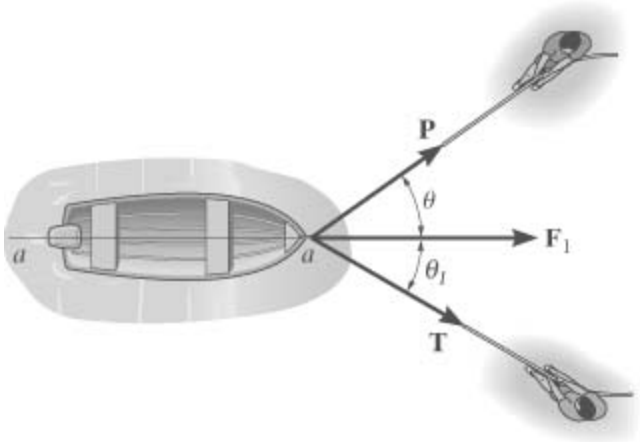
The boat is to be pulled onto the shore using two ropes. Determine the magnitudes of forces **T** and **P** acting in each rope in order to develop a resultant force $\mathbf{F_1}$, directed along the keel axis *aa* as shown.

Given:

$\theta = 40\ \text{deg}$

$\theta_1 = 30\ \text{deg}$

$F_1 = 80\ \text{lb}$

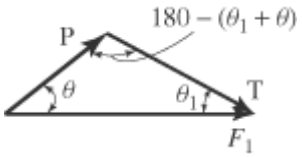


Solution:

$$\frac{F_1}{\sin\left[180\ \text{deg} - \left(\theta + \theta_1\right)\right]} = \frac{T}{\sin\left(\theta\right)}$$

$$T = F_1 \frac{\sin\left(\theta\right)}{\sin\left(180\ \text{deg} - \theta - \theta_1\right)}$$

$T = 54.7\,\text{lb}$



$$\frac{F_1}{\sin\left[180\ \text{deg} - \left(\theta + \theta_1\right)\right]} = \frac{P}{\sin\left(\theta_1\right)}$$

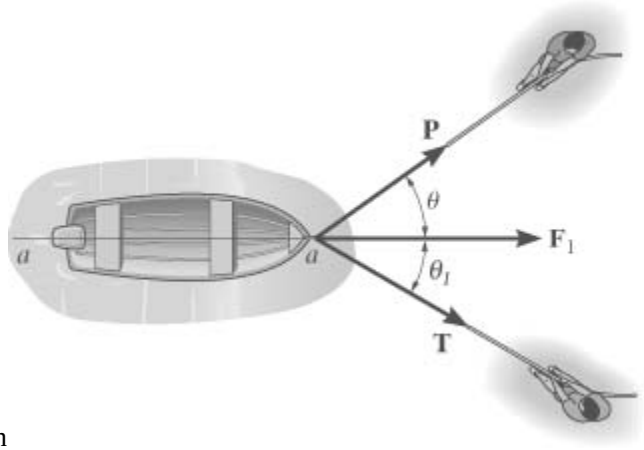
$$P = \sin\left(\theta_1\right) \frac{F_1}{\sin\left[180\ \text{deg} - \left(\theta + \theta_1\right)\right]}$$

$P = 42.6\,\text{lb}$

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Problem 2-26

The boat is to be pulled onto the shore using two ropes. If the resultant force is to be $\mathbf{F_1}$, directed along the keel *aa* as shown, determine the magnitudes of forces **T** and **P** acting in each rope and the angle θ of **P** so that the magnitude of **P** is a *minimum*. **T** acts at θ from the keel as shown.



Given:

$\theta_1 = 30 \text{ deg}$

$F_1 = 80 \text{ lb}$

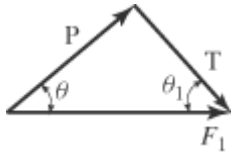
Solution:

From the figure, P is minimum when

$\theta + \theta_1 = 90 \text{ deg}$

$\theta = 90 \text{ deg} - \theta_1$

$\theta = 60 \text{ deg}$



$$\frac{P}{\sin(\theta_1)} = \frac{F_1}{\sin(90 \text{ deg})}$$

$$P = \frac{F_1 \sin(\theta_1)}{\sin(90 \text{ deg})}$$

$P = 40 \text{ lb}$

$$\frac{T}{\sin(\theta)} = \frac{F_1}{\sin(90 \text{ deg})}$$

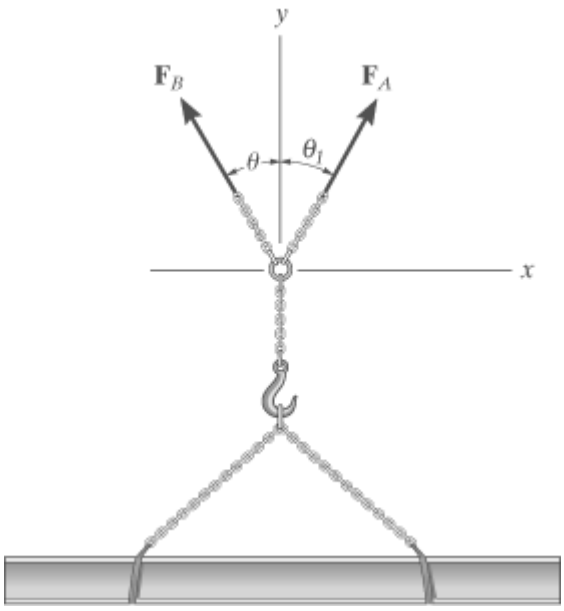
$$T = F_1 \frac{\sin(\theta)}{\sin(90 \text{ deg})}$$

$T = 69.3 \text{ lb}$

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Problem 2-27

The beam is to be hoisted using two chains. Determine the magnitudes of forces $\mathbf{F_A}$ and $\mathbf{F_B}$ acting on each chain in order to develop a resultant force $\mathbf{T}$ directed along the positive *y* axis.



Given:

$T = 600\ \text{N}$

$\theta_1 = 30\ \text{deg}$

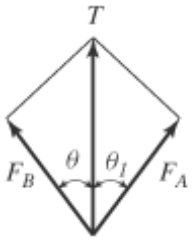
$\theta = 45\ \text{deg}$

Solution:

$$\frac{F_A}{\sin(\theta)} = \frac{T}{\sin\left[180\ \text{deg} - (\theta + \theta_1)\right]}$$

$$F_A = \frac{T\sin(\theta)}{\sin\left[180\ \text{deg} - (\theta + \theta_1)\right]}$$

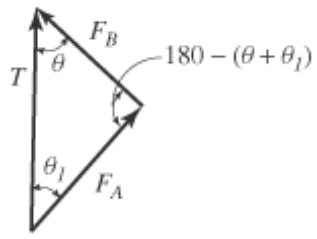
$F_A = 439\,\text{N}$



$$\frac{F_B}{\sin(\theta_1)} = \frac{T}{\sin\left[180\ \text{deg} - (\theta + \theta_1)\right]}$$

$$F_B = T\,\frac{\sin(\theta_1)}{\sin\left[180\ \text{deg} - (\theta + \theta_1)\right]}$$

$F_B = 311\ \text{N}$



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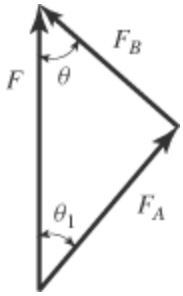
Problem 2-28

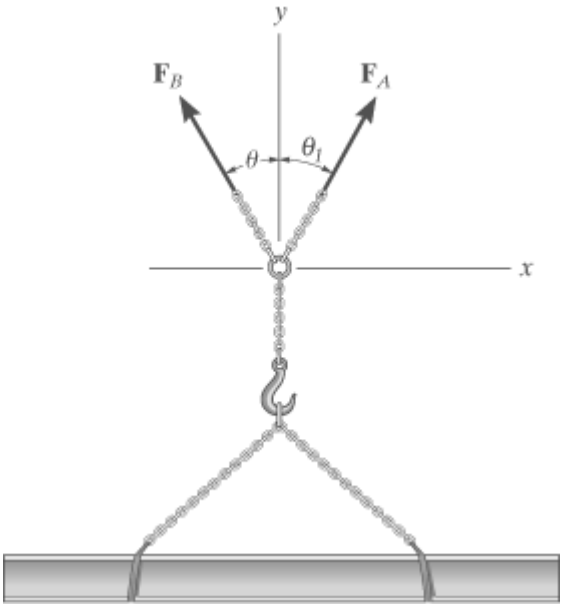
The beam is to be hoisted using two chains. If the resultant force is to be $\mathbf{F}$, directed along the positive *y* axis, determine the magnitudes of forces $\mathbf{F_A}$ and $\mathbf{F_B}$ acting on each chain and the orientation θ of $\mathbf{F_B}$ so that the magnitude of $\mathbf{F_B}$ is a *minimum*.

Given:

$F = 600$ N

$\theta_1 = 30$ deg





Solution:

For minimum $\mathbf{F_B}$, require

$\theta = 90 \text{ deg} - \theta_1$

$\theta = 60 \text{ deg}$

$F_A = F \cos(\theta_1)$

$F_A = 520$ N

$F_B = F \sin(\theta_1)$

$F_B = 300$ N

Problem 2-29

Three chains act on the bracket such that they create a resultant force having magnitude F_R. If two of the chains are subjected to known forces, as shown, determine the orientation θ of the third chain,measured clockwise from the positive *x* axis, so that the magnitude of force $\mathbf{F}$ in this chain is a *minimum*. All forces lie in the *x-y* plane.What is the magnitude of $\mathbf{F}$? *Hint*: First find the resultant of the two known forces. Force $\mathbf{F}$ acts in this direction.

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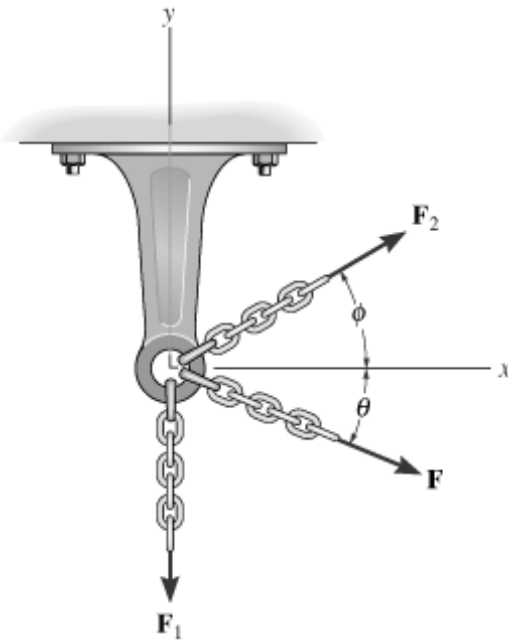
Given:

$F_R = 500$ lb

$F_1 = 200$ lb

$F_2 = 300$ lb

$\phi = 30$ deg



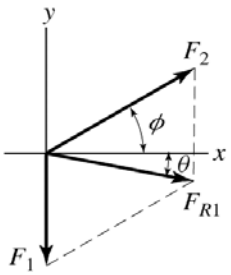
Solution:

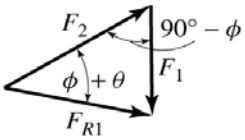
Cosine Law:

$$F_{R1} = \sqrt{F_1^2 + F_2^2 - 2\,F_1\,F_2\cos(90\text{ deg} - \phi)}$$

$F_{R1} = 264.6$ lb

Sine Law: Make **F** parallel to $\mathbf{F}_{R1}$

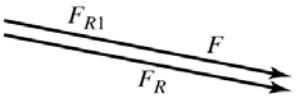




$$\frac{\sin(\phi + \theta)}{F_1} = \frac{\sin(90\text{ deg} - \phi)}{F_{R1}}$$

$$\theta = -\phi + \operatorname{asin}\left(\sin(90\text{ deg} - \phi)\,\frac{F_1}{F_{R1}}\right)$$

$\theta = 10.9$ deg



When **F** is directed along $\mathbf{F}_{R1}$, **F** will be minimum to create the resultant forces.

$F = F_R - F_{R1}$

$F = 235$ lb

Problem 2-30

Three cables pull on the pipe such that they create a resultant force having magnitude F_R. If two of the cables are subjected to known forces, as shown in the figure, determine the direction θ of the third cable so that the magnitude of force **F** in this cable is a *minimum*. All forces lie in the

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x–y plane.What is the magnitude of **F**? *Hint*: First find the resultant of the two known forces.

Given:

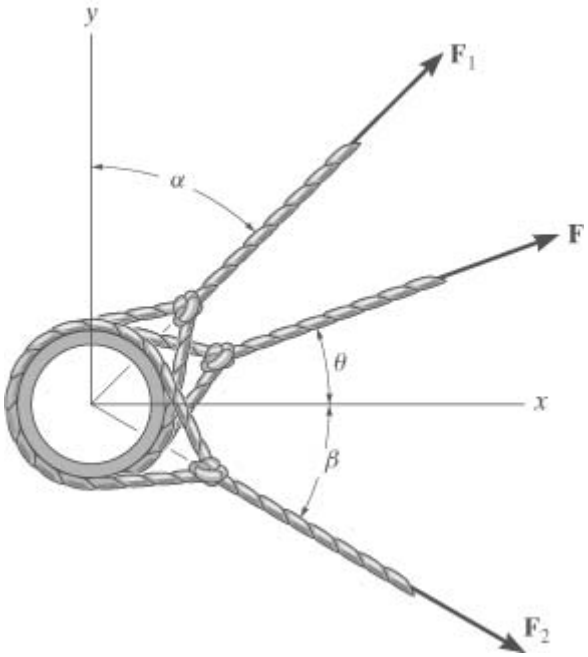
$F_R = 900$ lb

$F_1 = 600$ lb

$F_2 = 400$ lb

$\alpha = 45$ deg

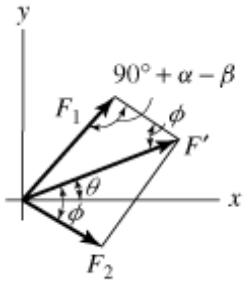
$\beta = 30$ deg



Solution:

$$F' = \sqrt{F_1{}^2 + F_2{}^2 - 2F_1F_2\cos(90\ \text{deg} + \alpha - \beta)}$$

$F' = 802.64$ lb

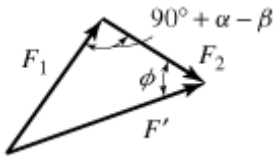


$F = F_R - F'$

$F = 97.4$ lb

$$\frac{\sin(\phi)}{F_1} = \frac{\sin(90\ \text{deg} + \alpha - \beta)}{F'}$$

$$\phi = \text{asin}\left(\sin(90\ \text{deg} + \alpha - \beta)\ \frac{F_1}{F'}\right)$$



$\phi = 46.22$ deg

$\theta = \phi - \beta$

$\theta = 16.2$ deg

Problem 2-31

Determine the *x* and *y* components of the force **F**.

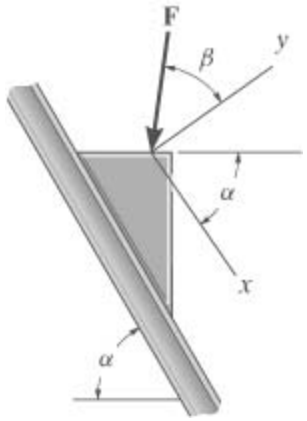
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Given:

$F = 800$ lb

$\alpha = 60$ deg

$\beta = 40$ deg



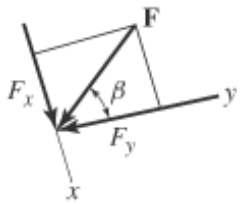
Solution:

$F_x = F\sin(\beta)$

$F_y = -F\cos(\beta)$

$F_x = 514.2$ lb

$F_y = -612.8$ lb



Problem 2-32

Determine the magnitude of the resultant force and its direction, measured clockwise from the positive *x* axis.

Given:

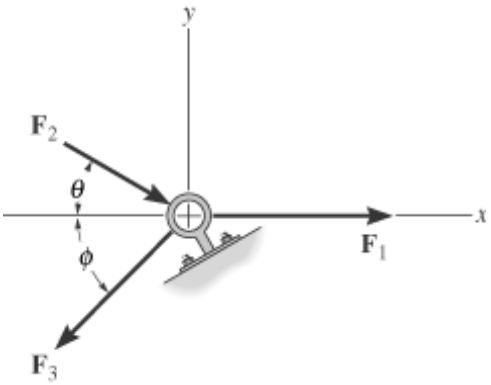
$F_1 = 70$ N

$F_2 = 50$ N

$F_3 = 65$ N

$\theta = 30$ deg

$\phi = 45$ deg



Solution:

$\xrightarrow{+}$ $F_{Rx} = \Sigma F_x;$ $F_{RX} = F_1 + F_2\cos(\theta) - F_3\cos(\phi)$

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$+\uparrow \quad F_{Ry} = \Sigma F_y; \qquad F_{RY} = -F_2 \sin(\theta) - F_3 \sin(\phi)$

$F_R = \sqrt{F_{RX}{}^2 + F_{RY}{}^2}$

$\theta = \operatorname{atan}\left(\frac{|F_{RY}|}{|F_{RX}|}\right)$

$F_R = 97.8 \text{ N}$

$\theta = 46.5 \text{ deg}$

Problem 2-33

Determine the magnitude of the resultant force and its direction measured counterclockwise from the positive *x* axis.

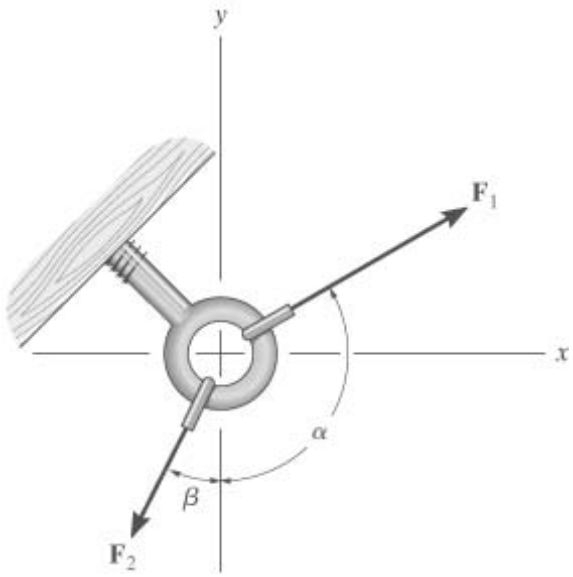
Given:

$F_1 = 50 \text{ lb}$

$F_2 = 35 \text{ lb}$

$\alpha = 120 \text{ deg}$

$\beta = 25 \text{ deg}$



Solution:

$\xrightarrow{+} F_{Rx} = \Sigma F_x; \qquad F_{Rx} = F_1 \sin(\alpha) - F_2 \sin(\beta)$

$F_{Rx} = 28.5 \text{ lb}$

$+\uparrow \quad F_{Ry} = \Sigma F_y; \qquad F_{Ry} = -F_1 \cos(\alpha) - F_2 \cos(\beta)$

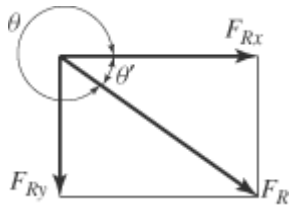
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$F_{Ry} = -6.7\,\text{lb}$

$F_R = \sqrt{F_{Rx}^2 + F_{Ry}^2}$

$F_R = 29.3\,\text{lb}$

$\theta' = \text{atan}\left(\frac{|F_{Ry}|}{|F_{Rx}|}\right)$



$\theta' = 13.3\,\text{deg}$

$\theta = 360\ \text{deg} - \theta'$

$\theta = 347\,\text{deg}$

Problem 2-34

Determine the magnitude of the resultant force and its direction , measured counterclockwise from the positive *x* axis.

Given:

$F_1 = 850\ \text{N}$

$F_2 = 625\ \text{N}$

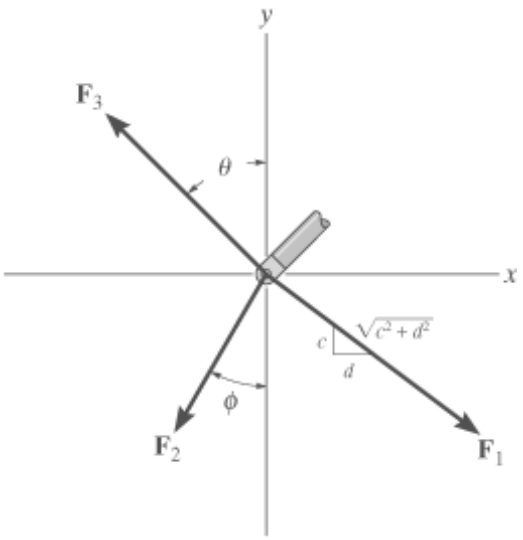
$F_3 = 750\ \text{N}$

$\theta = 45\ \text{deg}$

$\phi = 30\ \text{deg}$

$c = 3$

$d = 4$



Solution:

$\xrightarrow{+}\ F_{Rx} = SF_x; \qquad F_{RX} = F_1\frac{d}{\sqrt{c^2 + d^2}} - F_2\sin(\phi) - F_3\sin(\theta)$

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$+\uparrow \; F_{Ry} = SF_y; \qquad F_{RY} = -F_1 \frac{c}{\sqrt{c^2 + d^2}} - F_2 \cos(\phi) + F_3 \cos(\theta)$

$F_{RX} = -162.8 \text{ N} \qquad F_{RY} = -520.9 \text{ N}$

$F_R = \sqrt{F_{RX}^2 + F_{RY}^2} \qquad F_R = 546 \text{ N}$

$\alpha = \text{atan}\left(\frac{F_{RY}}{F_{RX}}\right)$

$\alpha = 72.64 \text{ deg}$

$\beta = \alpha + 180 \text{ deg}$

$\beta = 252.6 \text{ deg}$

Problem 2-35

Three forces act on the bracket. Determine the magnitude and direction θ of $\mathbf{F_1}$ so that the resultant force is directed along the positive x' axis and has a magnitude of $\mathbf{F_R}$.

Units Used:

$\text{kN} = 10^3 \text{ N}$

Given:

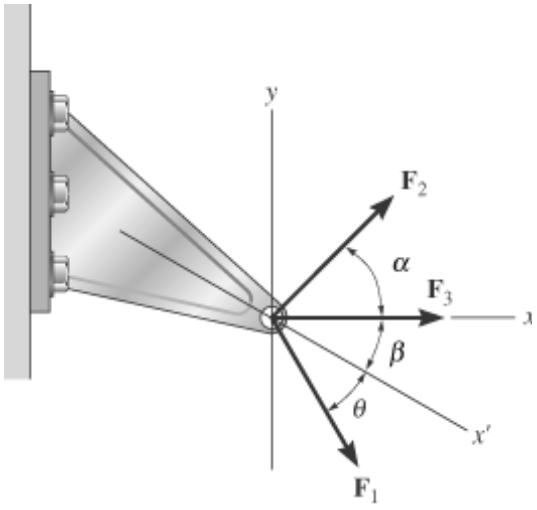
$F_R = 1 \text{ kN}$

$F_2 = 450 \text{ N}$

$F_3 = 200 \text{ N}$

$\alpha = 45 \text{ deg}$

$\beta = 30 \text{ deg}$



Solution:

$\xrightarrow{+} \; F_{Rx} = SF_x; \quad F_R \cos(\beta) = F_3 + F_2 \cos(\alpha) + F_1 \cos(\theta + \beta)$

$+\uparrow \; F_{Ry} = SF_y; \quad -F_R \sin(\beta) = F_2 \sin(\alpha) - F_1 \sin(\theta + \beta)$

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$$F_1 \cos(\theta + \beta) = F_R \cos(\beta) - F_3 - F_2 \cos(\alpha)$$

$$F_1 \sin(\theta + \beta) = F_2 \sin(\alpha) + F_R \sin(\beta)$$

$$\theta = \operatorname{atan}\left(\frac{F_2 \sin(\alpha) + F_R \sin(\beta)}{F_R \cos(\beta) - F_3 - F_2 \cos(\alpha)}\right) - \beta$$

$$\theta = 37\,\text{deg}$$

$$F_1 = \sqrt{\left(F_R \cos(\beta) - F_3 - F_2 \cos(\alpha)\right)^2 + \left(F_2 \sin(\alpha) + F_R \sin(\beta)\right)^2}$$

$$F_1 = 889\ \text{N}$$

Problem 2-36

Determine the magnitude and direction, measured counterclockwise from the x' axis, of the resultant force of the three forces acting on the bracket.

Given:

$F_1 = 300\ \text{N}$

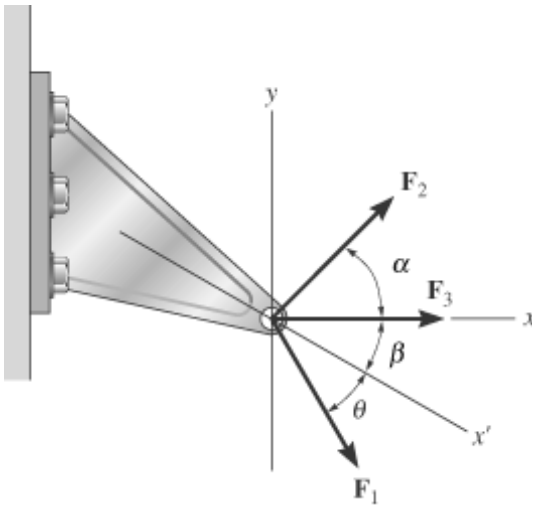
$F_2 = 450\ \text{N}$

$F_3 = 200\ \text{N}$

$\alpha = 45\ \text{deg}$

$\beta = 30\ \text{deg}$

$\theta = 20\ \text{deg}$



Solution:

$$F_{Rx} = F_1 \cos(\theta + \beta) + F_3 + F_2 \cos(\alpha) \qquad F_{Rx} = 711.03\ \text{N}$$

$$F_{Ry} = -F_1 \sin(\theta + \beta) + F_2 \sin(\alpha) \qquad F_{Ry} = 88.38\ \text{N}$$

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$F_R = \sqrt{F_{Rx}^2 + F_{Ry}^2}$ $F_R = 717$ N

ϕ (angle from x axis)

$\phi = \text{atan}\left(\frac{F_{Ry}}{F_{Rx}}\right)$ $\phi = 7.1$ deg

ϕ' (angle from x' axis)

$\phi' = \beta + \phi$ $\phi' = 37.1$ deg

Problem 2-37

Determine the magnitude of the resultant force and its direction, measured counterclockwise from the positive x axis.

Given:

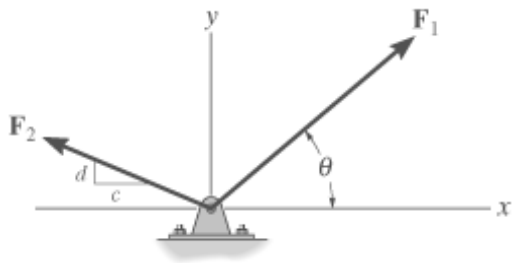
$F_1 = 800$ N

$F_2 = 600$ N

$\theta = 40$ deg

$c = 12$

$d = 5$



Solution:

$\xrightarrow{+} F_{Rx} = \Sigma F_x;$ $F_{Rx} = F_1 \cos(\theta) - F_2\left(\frac{c}{\sqrt{c^2 + d^2}}\right)$ $F_{Rx} = 59$ N

$+\uparrow F_{Ry} = \Sigma F_y;$ $F_{Ry} = F_1 \sin(\theta) + F_2\left(\frac{d}{\sqrt{c^2 + d^2}}\right)$ $F_{Ry} = 745$ N

$F_R = \sqrt{F_{Rx}^2 + F_{Ry}^2}$ $F_R = 747$ N

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$$\theta = \operatorname{atan}\left(\frac{F_{Ry}}{F_{Rx}}\right) \qquad \theta = 85.5\,\text{deg}$$

Problem 2-38

Determine the magnitude of the resultant force and its direction, measured counterclockwise from the positive *x* axis.

Units Used:

$\text{kN} = 10^3\ \text{N}$

Given:

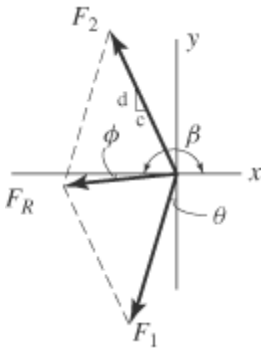
$F_1 = 30\ \text{kN}$

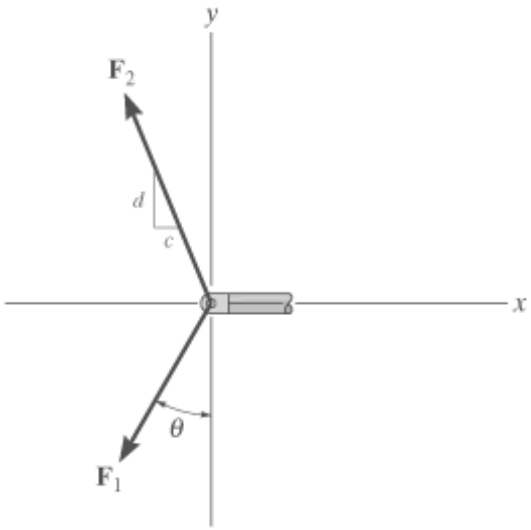
$F_2 = 26\ \text{kN}$

$\theta = 30\ \text{deg}$

$c = 5$

$d = 12$





Solution:

$\xrightarrow{+} F_{Rx} = \Sigma F_x;$ $\quad F_{Rx} = -F_1 \sin(\theta) - \left(\frac{c}{\sqrt{c^2 + d^2}}\right) F_2 \qquad F_{Rx} = -25\,\text{kN}$

$+\uparrow F_{Ry} = \Sigma F_y;$ $\quad F_{Ry} = -F_1 \cos(\theta) + \left(\frac{d}{\sqrt{c^2 + d^2}}\right) F_2 \qquad F_{Ry} = -2\,\text{kN}$

$F_R = \sqrt{F_{Rx}^2 + F_{Ry}^2} \qquad F_R = 25.1\,\text{kN}$

$\phi = \operatorname{atan}\left(\frac{F_{Ry}}{F_{Rx}}\right) \qquad \phi = 4.5\,\text{deg}$

$\beta = 180\ \text{deg} + \phi \qquad \beta = 184.5\,\text{deg}$

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Problem 2-39

Determine the magnitude of the resultant force and its direction measured counterclockwise from the positive *x* axis.

Given:

$F_1 = 60$ lb

$F_2 = 70$ lb

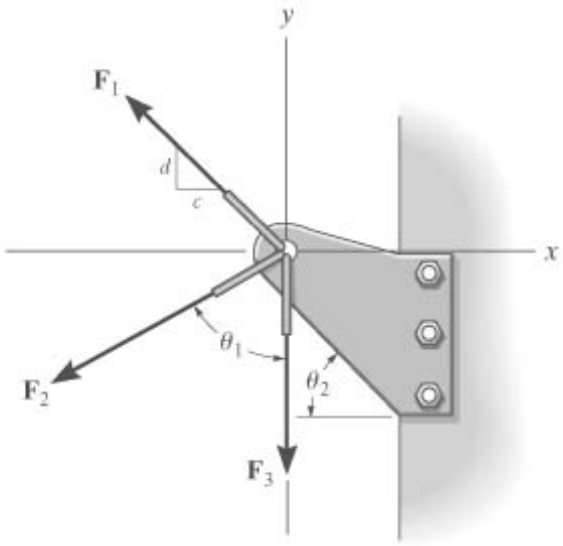
$F_3 = 50$ lb

$\theta_1 = 60$ deg

$\theta_2 = 45$ deg

$c = 1$

$d = 1$



Solution:

$$\theta_3 = \operatorname{atan}\left(\frac{d}{c}\right)$$

$$F_{Rx} = -F_1\cos(\theta_3) - F_2\sin(\theta_1) \qquad F_{Rx} = -103\,\text{lb}$$

$$F_{Ry} = F_1\sin(\theta_3) - F_2\cos(\theta_1) - F_3 \qquad F_{Ry} = -42.6\,\text{lb}$$

$$F_R = \sqrt{F_{Rx}^2 + F_{Ry}^2} \qquad F_R = 111.5\,\text{lb}$$

$$\theta = 180\ \text{deg} + \operatorname{atan}\left(\frac{F_{Ry}}{F_{Rx}}\right) \qquad \theta = 202\,\text{deg}$$

Problem 2-40

Determine the magnitude of the resultant force $\mathbf{F_R = F_1 + F_2}$ and its direction, measured counterclockwise from the positive *x* axis by summing the rectangular or *x*, *y* components of the forces to obtain the resultant force.

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Given:

$F_1 = 600$ N

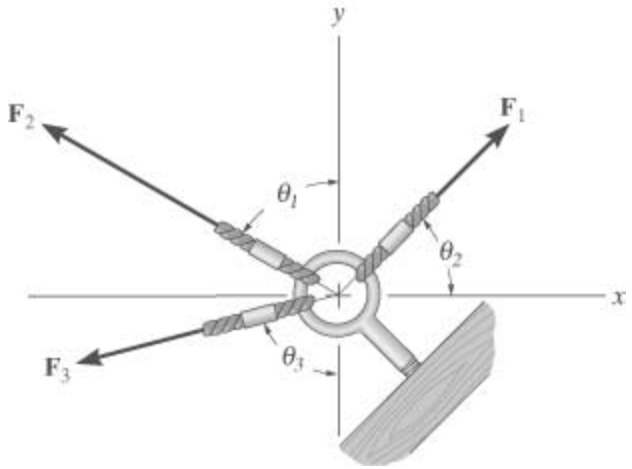
$F_2 = 800$ N

$F_3 = 450$ N

$\theta_1 = 60$ deg

$\theta_2 = 45$ deg

$\theta_3 = 75$ deg



Solution:

$\overset{+}{\rightarrow} F_{Rx} = \Sigma F_x;$ $\quad F_{Rx} = F_1 \cos(\theta_2) - F_2 \sin(\theta_1)$ $\quad F_{Rx} = -268.556$ N

$+\uparrow F_{Ry} = \Sigma F_y;$ $\quad F_{Ry} = F_1 \sin(\theta_2) + F_2 \cos(\theta_1)$ $\quad F_{Ry} = 824.264$ N

$$F_R = \sqrt{F_{Rx}^2 + F_{Ry}^2} \qquad F_R = 867 \text{ N}$$

$$\theta = 180 \text{ deg} - \operatorname{atan}\left(\frac{|F_{Ry}|}{|F_{Rx}|}\right) \qquad \theta = 108 \text{ deg}$$

Problem 2-41

Determine the magnitude and direction of the resultant $\mathbf{F_R} = \mathbf{F_1} + \mathbf{F_2} + \mathbf{F_3}$ of the three forces by summing the rectangular or *x, y* components of the forces to obtain the resultant force.

Given:

$F_1 = 30$ N

$F_2 = 20$ N

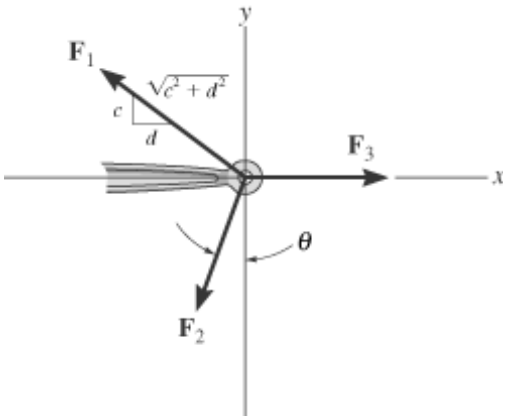
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$F_3 = 50$ N

$\theta = 20$ deg

$c = 3$

$d = 4$



Solution:

$$F_{Rx} = -F_1\left(\frac{d}{\sqrt{c^2 + d^2}}\right) - F_2(\sin(\theta)) + F_3 \qquad F_{Rx} = 19.2 \text{ N}$$

$$F_{Ry} = F_1\left(\frac{c}{\sqrt{c^2 + d^2}}\right) - F_2\cos(\theta) \qquad F_{Ry} = -0.8 \text{ N}$$

$$F_R = \sqrt{F_{Rx}^2 + F_{Ry}^2} \qquad F_R = 19.2 \text{ N}$$

$$\theta = \text{atan}\left(\frac{-F_{Ry}}{F_{Rx}}\right) \qquad \theta = 2.4 \text{ deg}$$

Problem 2-42

Determine the magnitude and orientation, measured counterclockwise from the positive *y* axis, of the resultant force acting on the bracket.

Given:

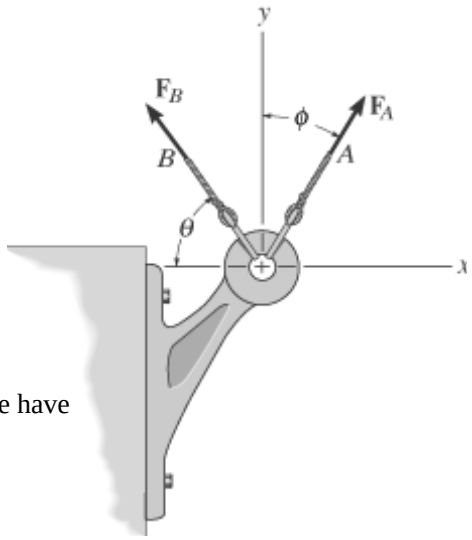
$F_A = 700$ N

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$F_B = 600$ N

$\theta = 20$ deg

$\phi = 30$ deg



Solution:

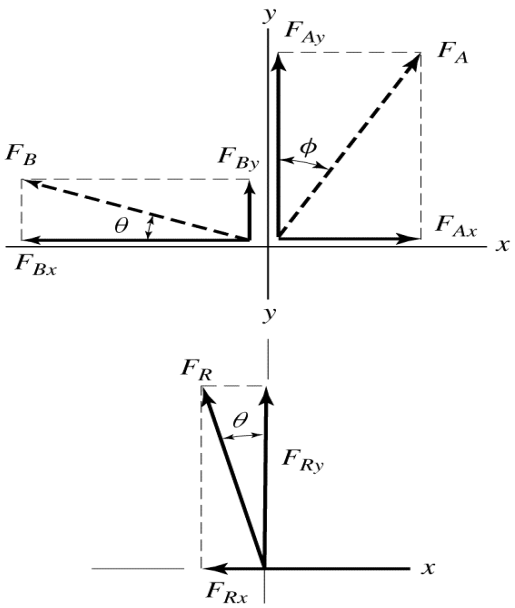
Scalar Notation: Suming the force components algebraically, we have

$F_{Rx} = \Sigma F_x; \quad F_{Rx} = F_A \sin(\phi) - F_B \cos(\theta)$

$F_{Rx} = -213.8$ N

$F_{Ry} = \Sigma F_y; \quad F_{Ry} = F_A \cos(\phi) + F_B \sin(\theta)$

$F_{Ry} = 811.4$ N



The magnitude of the resultant force F_R is

$$F_R = \sqrt{F_{Rx}^2 + F_{Ry}^2}$$

$F_R = 839$ N

The directional angle θ measured counterclockwise from the positive x axis is

$$\theta = \text{atan}\left(\frac{|F_{Rx}|}{F_{Ry}}\right)$$

$\theta = 14.8$ deg

Problem 2-43

Determine the magnitude and direction, measured counterclockwise from the positive x' axis, of the resultant force of the three forces acting on the bracket.

Given:

$F_1 = 300$ N

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$F_2 = 200$ N

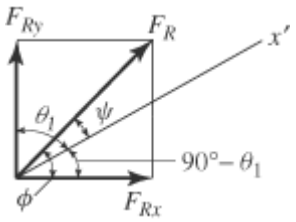
$F_3 = 180$ N

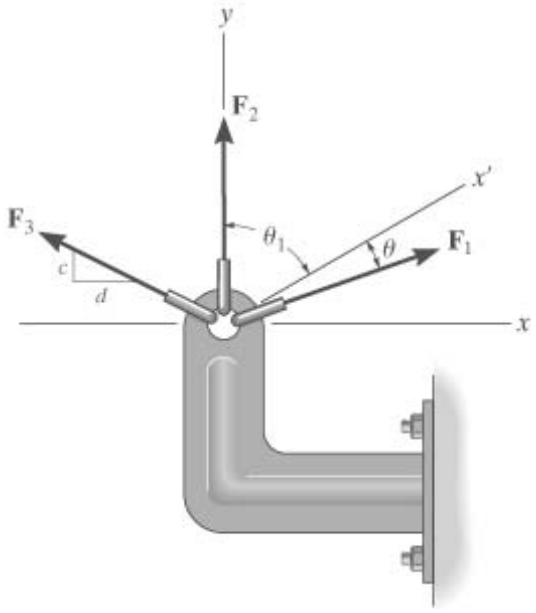
$\theta = 10$ deg

$\theta_1 = 60$ deg

$c = 5$

$d = 12$





Solution:

$\xrightarrow{+} F_{Rx} = \Sigma F_x;$ $\quad F_{Rx} = F_1 \sin(\theta_1 + \theta) - \left(\frac{d}{\sqrt{c^2 + d^2}}\right) F_3$ $\quad F_{Rx} = 115.8$ N

$+\uparrow F_{Ry} = \Sigma F_y;$ $\quad F_{Ry} = F_1 \cos(\theta_1 + \theta) + F_2 + \left(\frac{c}{\sqrt{c^2 + d^2}}\right) F_3$ $\quad F_{Ry} = 371.8$ N

$F_R = \sqrt{F_{Rx}^2 + F_{Ry}^2}$ $\quad F_R = 389$ N

$\phi = \operatorname{atan}\left(\frac{F_{Ry}}{F_{Rx}}\right)$ $\quad \phi = 72.7$ deg

$\psi = \phi - (90 \text{ deg} - \theta_1)$ $\quad \psi = 42.7$ deg

---

Problem 2-44

Determine the x and y components of $\mathbf{F_1}$ and $\mathbf{F_2}$.

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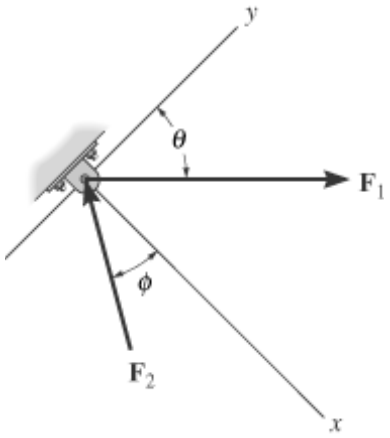
Given:

$F_1 = 200$ N

$F_2 = 150$ N

$\theta = 45$ deg

$\phi = 30$ deg



Solution:

$F_{1x} = F_1 \sin(\theta)$

$F_{1x} = 141.4$ N

$F_{1y} = F_1 \cos(\theta)$

$F_{1y} = 141.4$ N

$F_{2x} = -F_2 \cos(\phi)$

$F_{2x} = -129.9$ N

$F_{2y} = F_2 \sin(\phi)$

$F_{2y} = 75$ N

Problem 2-45

Determine the magnitude of the resultant force and its direction, measured counterclockwise from the positive *x* axis.

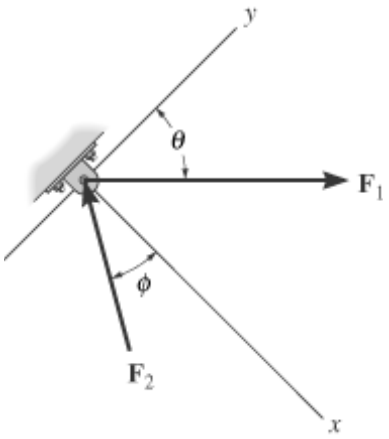
Given:

$F_1 = 200$ N

$F_2 = 150$ N

$\theta = 45$ deg

$\phi = 30$ deg



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Solution:

$\xrightarrow{+} F_{Rx} = \Sigma F_x;$ $\quad F_{Rx} = F_1 \sin(\theta) - F_2 \cos(\phi)$ $\quad F_{Rx} = 11.5 \text{ N}$

$+\uparrow F_{Ry} = \Sigma F_y;$ $\quad F_{Ry} = F_1 \cos(\theta) + F_2 \sin(\phi)$ $\quad F_{Ry} = 216.4 \text{ N}$

$$F = \sqrt{F_{Rx}^2 + F_{Ry}^2} \qquad F = 217 \text{ N}$$

$$\beta = \operatorname{atan}\left(\frac{F_{Ry}}{F_{Rx}}\right) \qquad \beta = 87 \text{ deg}$$

Problem 2-46

Determine the *x* and *y* components of each force acting on the *gusset plate* of the bridge truss.

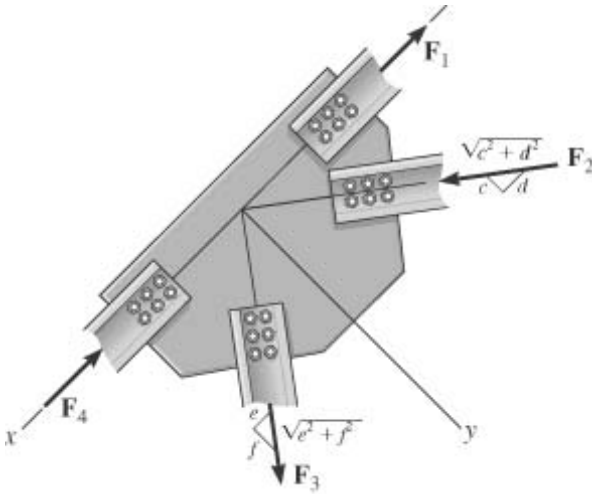
Given:

$F_1 = 200 \text{ lb}$ $\quad c = 3$

$F_2 = 400 \text{ lb}$ $\quad d = 4$

$F_3 = 300 \text{ lb}$ $\quad e = 3$

$F_4 = 300 \text{ lb}$ $\quad f = 4$



Solution:

$F_{1x} = -F_1$

$F_{1x} = -200 \text{ lb}$

$F_{1y} = 0 \text{ lb}$

$$F_{2x} = F_2\left(\frac{d}{\sqrt{c^2 + d^2}}\right)$$

$F_{2x} = 320 \text{ lb}$

$$F_{2y} = -F_2\left(\frac{c}{\sqrt{c^2 + d^2}}\right)$$

$F_{2y} = -240 \text{ lb}$

$$F_{3x} = F_3\left(\frac{e}{\sqrt{e^2 + f^2}}\right)$$

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$F_{3x} = 180\,\text{lb}$

$F_{3y} = F_3\left(\frac{f}{\sqrt{e^2 + f^2}}\right)$

$F_{3y} = 240\,\text{lb}$

$F_{4x} = -F_4$

$F_{4x} = -300\,\text{lb}$

$F_{4y} = 0\ \text{lb}$

Problem 2-47

Determine the magnitude of the resultant force and its direction measured clockwise from the positive x axis.

Units Used:

$\text{kN} = 10^3\ \text{N}$

Given:

$F_1 = 20\ \text{kN}$

$F_2 = 40\ \text{kN}$

$F_3 = 50\ \text{kN}$

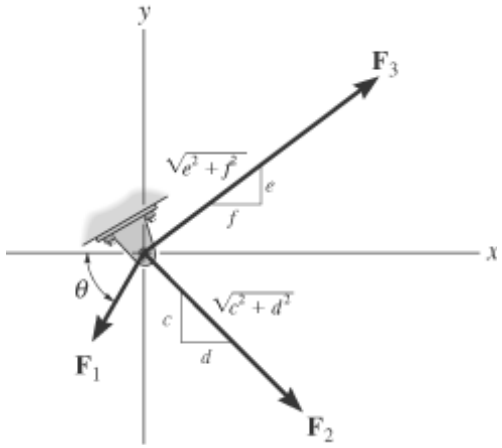
$\theta = 60\ \text{deg}$

$c = 1$

$d = 1$

$e = 3$

$f = 4$



Solution:

$\xrightarrow{+} F_{Rx} = \Sigma F_x; \qquad F_{Rx} = F_3\left(\frac{f}{\sqrt{e^2 + f^2}}\right) + F_2\left(\frac{d}{\sqrt{c^2 + d^2}}\right) - F_1\cos(\theta)$

$F_{Rx} = 58.28\,\text{kN}$

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$+\uparrow \quad F_{Ry} = \Sigma F_y; \qquad F_{Ry} = F_3\left(\frac{e}{\sqrt{e^2 + f^2}}\right) - F_2\left(\frac{c}{\sqrt{c^2 + d^2}}\right) - F_1 \sin(\theta)$

$F_{Ry} = -15.6\,\text{kN}$

$F = \sqrt{F_{Rx}{}^2 + F_{Ry}{}^2}$

$F = 60.3\,\text{kN}$

$\theta = \text{atan}\left(\frac{|F_{Ry}|}{F_{Rx}}\right)$

$\theta = 15\,\text{deg}$

Problem 2-48

Three forces act on the bracket. Determine the magnitude and direction θ of $\mathbf{F_1}$ so that the resultant force is directed along the positive x' axis and has magnitude $\mathbf{F_R}$.

Given:

$F_2 = 200\ \text{N}$

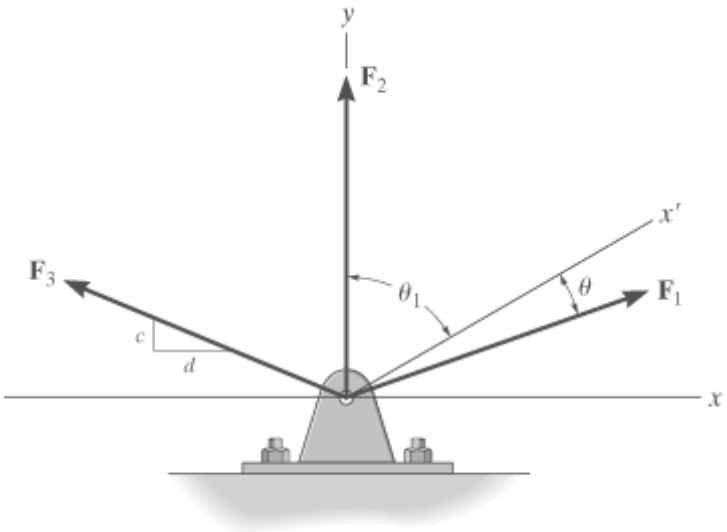
$F_3 = 180\ \text{N}$

$\theta_1 = 60\ \text{deg}$

$F_R = 800\ \text{N}$

$c = 5$

$d = 12$



Solution:

Initial Guesses: $F_1 = 100\ \text{N} \qquad \theta = 10\ \text{deg}$

Given

$\rightarrow \quad F_{Rx} = \Sigma F_x; \qquad F_R \sin(\theta_1) = F_1 \sin(\theta_1 + \theta) - \left(\frac{d}{\sqrt{c^2 + d^2}}\right) F_3$

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$+\uparrow \quad F_{Ry} = \Sigma F_y; \qquad F_R(\cos(\theta_1)) = F_1 \cos(\theta_1 + \theta) + F_2 + \left(\frac{c}{\sqrt{c^2 + d^2}}\right) F_3$

$$\begin{pmatrix} F_1 \\ \theta \end{pmatrix} = \text{Find}(F_1, \theta)$$

$F_1 = 869$ N $\qquad \theta = 21.3$ deg

Problem 2-49

Determine the magnitude and direction, measured counterclockwise from the positive x' axis, of the resultant force acting on the bracket.

Given:

$F_1 = 300$ N

$F_2 = 200$ N

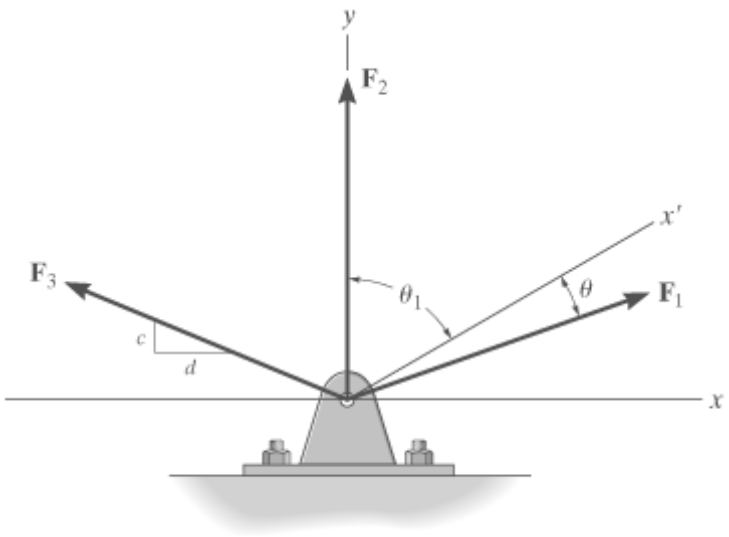
$F_3 = 180$ N

$\theta_1 = 60$ deg

$\theta = 10$ deg

$c = 5$

$d = 12$



Solution:

Guesses $\qquad F_{Rx} = 100$ N $\qquad F_{Ry} = 100$ N

Given

$\rightarrow \quad F_{Rx} = \Sigma F_x; \qquad F_{Rx} = F_1 \sin(\theta_1 + \theta) - \left(\frac{d}{\sqrt{c^2 + d^2}}\right) F_3$

$+\uparrow \quad F_{Ry} = \Sigma F_y; \qquad F_{Ry} = F_1 \cos(\theta_1 + \theta) + F_2 + \left(\frac{c}{\sqrt{c^2 + d^2}}\right)(F_3)$

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$$\begin{pmatrix} F_{Rx} \\ F_{Ry} \end{pmatrix} = \text{Find}(F_{Rx}, F_{Ry}) \qquad \begin{pmatrix} F_{Rx} \\ F_{Ry} \end{pmatrix} = \begin{pmatrix} 115.8 \\ 371.8 \end{pmatrix} \text{N}$$

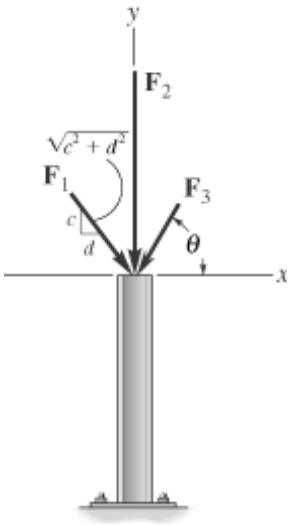
$$F_R = \sqrt{F_{Rx}^2 + F_{Ry}^2} \qquad F_R = 389 \text{ N}$$

$$\phi = \text{atan}\left(\frac{F_{Ry}}{F_{Rx}}\right) \qquad \phi = 72.7 \text{ deg}$$

$$\phi' = \left[\phi - (90 \text{ deg} - \theta_1)\right] \qquad \phi' = 42.7 \text{ deg}$$

Problem 2-50

Express each of the three forces acting on the column in Cartesian vector form and compute the magnitude of the resultant force.



Given:

$F_1 = 150$ lb $\quad \theta = 60$ deg

$F_2 = 275$ lb $\quad c = 4$

$F_3 = 75$ lb $\quad d = 3$

Solution:

Find the components of each force.

$$F_{1x} = F_1\left(\frac{d}{\sqrt{c^2 + d^2}}\right) \qquad F_{1y} = F_1\left(\frac{-c}{\sqrt{c^2 + d^2}}\right)$$

$$\mathbf{F_{1v}} = \begin{pmatrix} F_{1x} \\ F_{1y} \end{pmatrix} \qquad \mathbf{F_{1v}} = \begin{pmatrix} 90 \\ -120 \end{pmatrix} \text{lb}$$

$$F_{2x} = 0 \text{ lb} \qquad F_{2y} = -F_2$$

$$\mathbf{F_{2v}} = \begin{pmatrix} F_{2x} \\ F_{2y} \end{pmatrix} \qquad \mathbf{F_{2v}} = \begin{pmatrix} 0 \\ -275 \end{pmatrix} \text{lb}$$

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$F_{3x} = -F_3 \cos(\theta)$ $F_{3y} = -F_3 \sin(\theta)$

$\mathbf{F_{3v}} = \begin{pmatrix} F_{3x} \\ F_{3y} \end{pmatrix}$ $\mathbf{F_{3v}} = \begin{pmatrix} -37.5 \\ -65 \end{pmatrix}$ lb

Now find the magnitude of the resultant force.

$F_R = \left|\mathbf{F_{1v}} + \mathbf{F_{2v}} + \mathbf{F_{3v}}\right|$ $F_R = 462.9$ lb

---

Problem 2-51

Determine the magnitude of force $\mathbf{F}$ so that the resultant $\mathbf{F_R}$ of the three forces is as small as possible. What is the minimum magnitude of $\mathbf{F_R}$?

Units Used:

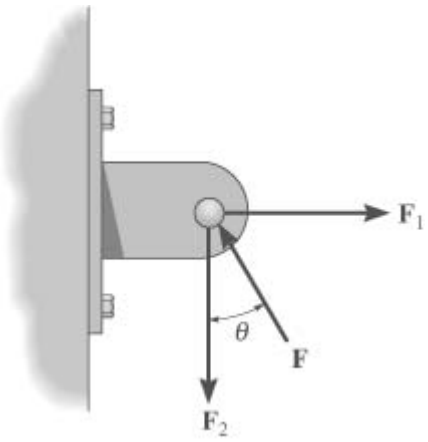
kN = 1000 N

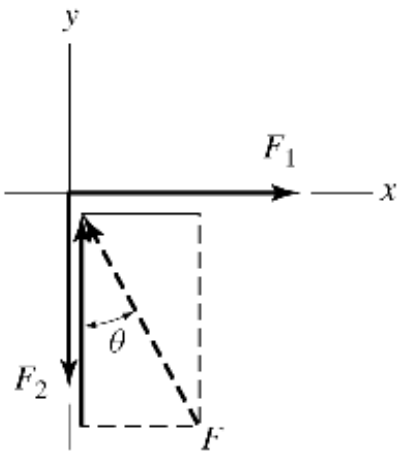
Given:

$F_1 = 5$ kN

$F_2 = 4$ kN

$\theta = 30$ deg



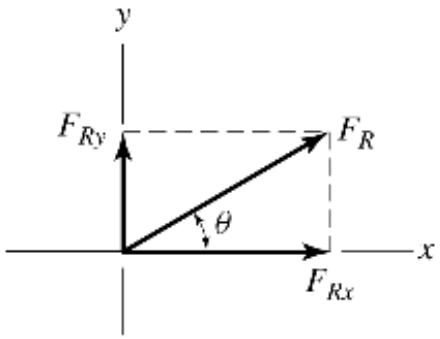


Solution:

Scalar Notation: Suming the force components algebrically, we have

$\overset{+}{\rightarrow}$ $F_{Rx} = \Sigma F_x; \quad F_{Rx} = F_1 - F\sin(\theta)$

$+\uparrow$ $F_{Ry} = \Sigma F_y; \quad F_{Ry} = F\cos(\theta) - F_2$



The magnitude of the resultant force $\mathbf{F_R}$ is

$$F_R = \sqrt{F_{Rx}^2 + F_{Ry}^2} = \sqrt{\left(F_1 - F\sin(\theta)\right)^2 + \left(F\cos(\theta) - F_2\right)^2}$$

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$$F_R^2 = F_1^2 + F_2^2 + F^2 - 2FF_1\sin(\theta) - 2F_2F\cos(\theta)$$

$$2F_R\frac{dF_R}{dF} = 2F - 2F_1\sin(\theta) - 2F_2\cos(\theta)$$

If F is a minimum, then $\left(\frac{dF_R}{dF} = 0\right)$ $F = F_1\sin(\theta) + F_2\cos(\theta)$ $F = 5.96\,\text{kN}$

$$F_R = \sqrt{\left(F_1 - F\sin(\theta)\right)^2 + \left(F\cos(\theta) - F_2\right)^2} \qquad F_R = 2.3\,\text{kN}$$

Problem 2-52

Express each of the three forces acting on the bracket in Cartesian vector form with respect to the x and y axes. Determine the magnitude and direction θ of $\mathbf{F_1}$ so that the resultant force is directed along the positive x' axis and has magnitude $\mathbf{F_R}$.

Units Used:

kN = 1000 N

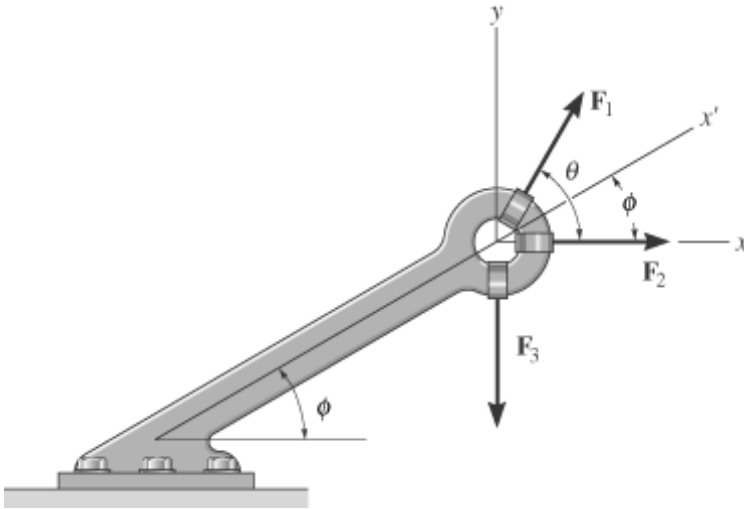
Given:

$F_R = 600$ N

$F_2 = 350$ N

$F_3 = 100$ N

$\phi = 30$ deg



Solution:

$$\mathbf{F_{2v}} = \begin{pmatrix} F_2 \\ 0 \end{pmatrix} \qquad \mathbf{F_{3v}} = \begin{pmatrix} 0 \\ -F_3 \end{pmatrix}$$

$$\mathbf{F_{1v}} = \begin{pmatrix} F_1\cos(\theta) \\ F_1\sin(\theta) \end{pmatrix} \qquad \mathbf{F_{2v}} = \begin{pmatrix} 350 \\ 0 \end{pmatrix}\text{ N} \qquad \mathbf{F_{3v}} = \begin{pmatrix} 0 \\ -100 \end{pmatrix}\text{ N}$$

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The initial guesses**:** $F_1 = 20\ \text{N} \quad \theta = 10\ \text{deg}$

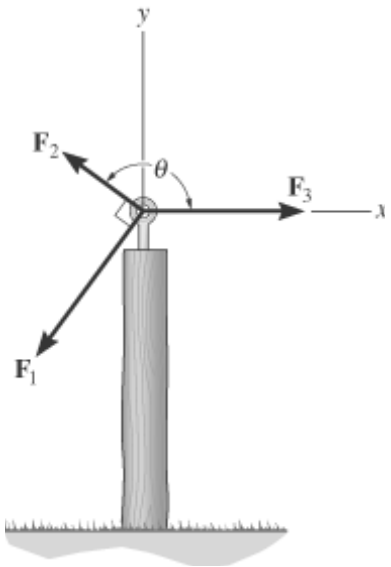
Given

$$\begin{pmatrix} F_1\cos(\theta) \\ F_1\sin(\theta) \end{pmatrix} + \mathbf{F_{2v}} + \mathbf{F_{3v}} = \begin{pmatrix} F_R\cos(\phi) \\ F_R\sin(\phi) \end{pmatrix}$$

$$\begin{pmatrix} F_1 \\ \theta \end{pmatrix} = \text{Find}(F_1, \theta) \qquad F_1 = 434.5\ \text{N} \qquad \theta = 67\,\text{deg}$$

Problem 2-53

The three concurrent forces acting on the post produce a resultant force $\mathbf{F_R} = 0$. If $\mathbf{F_2} = (1/2)\mathbf{F_1}$, and $\mathbf{F_1}$ is to be 90° from $\mathbf{F_2}$ as shown, determine the required magnitude $\mathbf{F_3}$ expressed in terms of $\mathbf{F_1}$ and the angle θ.



Solution: Use the primed coordiates.

$\Sigma F_{Rx} = 0 \qquad F_3\cos(\theta - 90\ \text{deg}) - F_1 = 0$

$\Sigma F_{Ry} = 0 \qquad -F_3\sin(\theta - 90\ \text{deg}) + F_2 = 0$

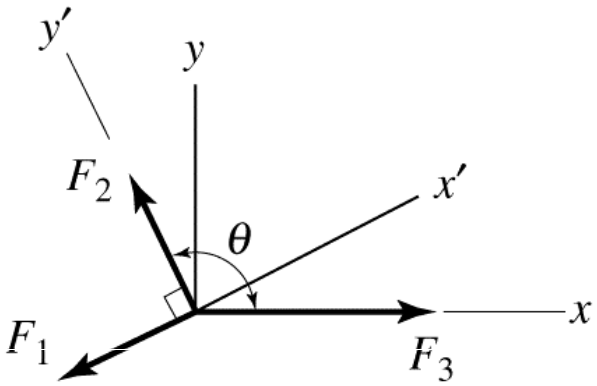
$$\tan(\theta - 90\ \text{deg}) = \frac{F_2}{F_1} = \frac{1}{2}$$

$$\theta = 90\ \text{deg} + \text{atan}\left(\frac{1}{2}\right)$$

$\theta = 117\,\text{deg}$

$$k = \frac{1}{\cos(\theta - 90\ \text{deg})}$$

$k = 1.1 \qquad F_3 = kF_1$



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Problem 2-54

Three forces act on the bracket. Determine the magnitude and orientation θ of $\mathbf{F}_2$ so that the resultant force is directed along the positive u axis and has magnitude F_R.

Given:

$F_R = 50$ lb

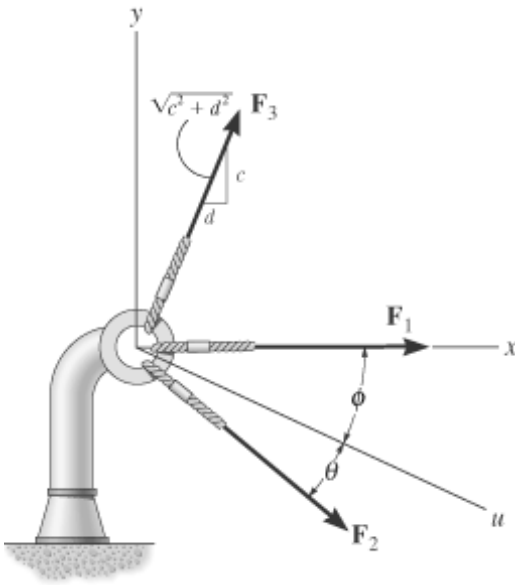
$F_1 = 80$ lb

$F_3 = 52$ lb

$\phi = 25$ deg

$c = 12$

$d = 5$



Solution:

Guesses

$F_2 = 1$ lb $\quad \theta = 120$ deg

Given

$$F_R \cos(\phi) = F_1 + F_2 \cos(\phi + \theta) + \left(\frac{d}{\sqrt{c^2 + d^2}}\right) F_3$$

$$-F_R \sin(\phi) = -F_2 \sin(\phi + \theta) + \left(\frac{c}{\sqrt{c^2 + d^2}}\right) F_3$$

$$\begin{pmatrix} F_2 \\ \theta \end{pmatrix} = \text{Find}(F_2, \theta) \qquad F_2 = 88.1\,\text{lb} \qquad \theta = 103.3\,\text{deg}$$

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Problem 2-55

Determine the magnitude and orientation, measured clockwise from the positive x axis, of the resultant force of the three forces acting on the bracket.

Given:

$F_1 = 80$ lb

$F_2 = 150$ lb

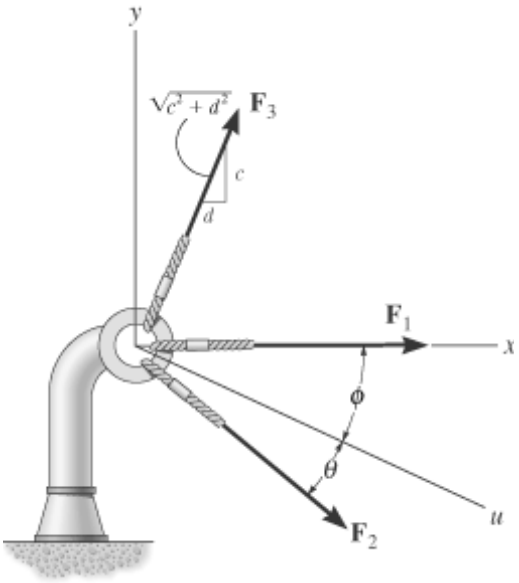
$F_3 = 52$lb

$\theta = 55$ deg

$\phi = 25$ deg

$c = 12$ m

$d = 5$ m



Solution:

$$F_{Rx} = F_1 + F_3\left(\frac{d}{\sqrt{c^2 + d^2}}\right) + F_2\cos(\theta + \phi) \qquad F_{Rx} = 126.05\,\text{lb}$$

$$F_{Ry} = F_3\left(\frac{c}{\sqrt{c^2 + d^2}}\right) - F_2\sin(\theta + \phi) \qquad F_{Ry} = -99.7\,\text{lb}$$

$$F_R = \sqrt{F_{Rx}^2 + F_{Ry}^2} \qquad F_R = 161\,\text{lb}$$

$$\beta = \text{atan}\left(\frac{|F_{Ry}|}{|F_{Rx}|}\right) \qquad \beta = 38.3\,\text{deg}$$

Problem 2-56

Three forces act on the ring. Determine the range of values for the magnitude of **P** so that the magnitude of the resultant force does not exceed F. Force **P** is always directed to the right.

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Units Used:

$\text{kN} = 10^3 \text{ N}$

Given:

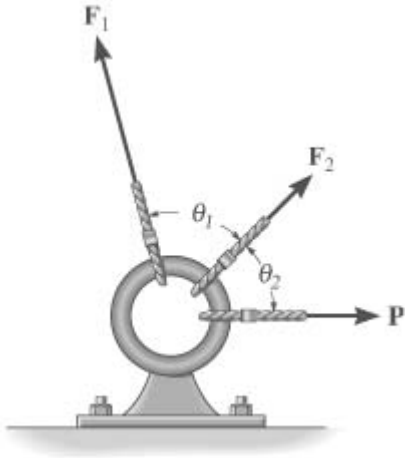
$F = 2500 \text{ N}$

$F_1 = 1500 \text{ N}$

$F_2 = 600 \text{ N}$

$\theta_1 = 60 \text{ deg}$

$\theta_2 = 45 \text{ deg}$



Solution:

Initial Guesses: $F_{Rx} = 100 \text{ N}$ $F_{Ry} = 100 \text{ N}$ $P = 100 \text{ N}$

Given

$\xrightarrow{+}$ $F_{Rx} = \Sigma F_x;$ $F_{Rx} = P + F_2\cos(\theta_2) + F_1\cos(\theta_1 + \theta_2)$

$+\uparrow$ $F_{Ry} = \Sigma F_y;$ $F_{Ry} = F_2\sin(\theta_2) + F_1\sin(\theta_1 + \theta_2)$

$F = \sqrt{F_{Rx}^2 + F_{Ry}^2}$

$\begin{pmatrix} F_{Rx} \\ F_{Ry} \\ P_{max} \end{pmatrix} = \text{Find}(F_{Rx}, F_{Ry}, P)$ $P_{max} = 1.6 \text{ kN}$

Initial Guesses: $F_{Rx} = -100 \text{ N}$ $F_{Ry} = 100 \text{ N}$ $P = -2000 \text{ N}$

Given

$\xrightarrow{+}$ $F_{Rx} = \Sigma F_x;$ $F_{Rx} = P + F_2\cos(\theta_2) + F_1\cos(\theta_1 + \theta_2)$

$+\uparrow$ $F_{Ry} = \Sigma F_y;$ $F_{Ry} = F_2\sin(\theta_2) + F_1\sin(\theta_1 + \theta_2)$

$F = \sqrt{F_{Rx}^2 + F_{Ry}^2}$

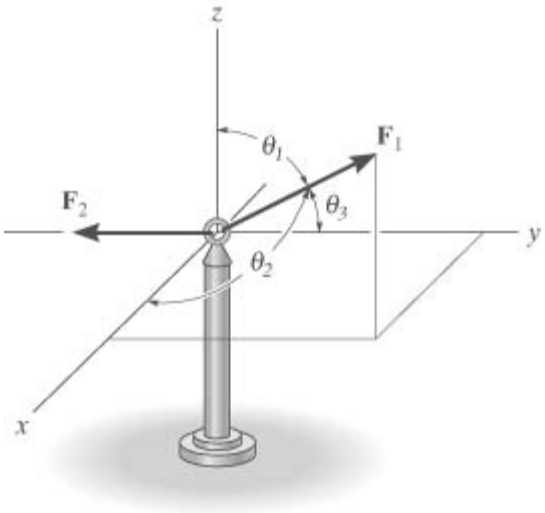
$\begin{pmatrix} F_{Rx} \\ F_{Ry} \\ P_{min} \end{pmatrix} = \text{Find}(F_{Rx}, F_{Ry}, P)$ $P_{min} = -1.7 \text{ kN}$

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Since $P > 0$ we conclude that $0 <= P <= P_{max} = 1.6\,\text{kN}$

Problem 2-57

Determine the magnitude and coordinate direction angles of $\mathbf{F_1}$ and $\mathbf{F_2}$. Sketch each force on an *x, y, z* reference.



Given:

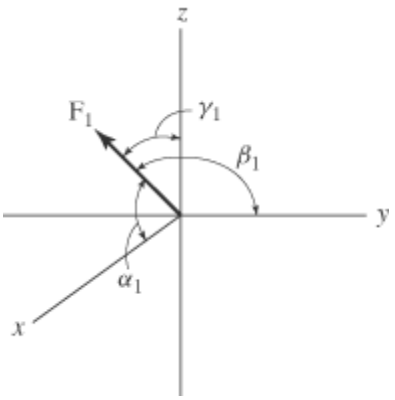
$$\mathbf{F_1} = \begin{pmatrix} 60 \\ -50 \\ 40 \end{pmatrix} \text{N}$$

$$\mathbf{F_2} = \begin{pmatrix} -40 \\ -85 \\ 30 \end{pmatrix} \text{N}$$

Solution:

$$\left|\mathbf{F_1}\right| = 87.7\ \text{N}$$

$$\begin{pmatrix} \alpha_1 \\ \beta_1 \\ \gamma_1 \end{pmatrix} = \text{acos}\left(\frac{\mathbf{F_1}}{\left|\mathbf{F_1}\right|}\right)$$



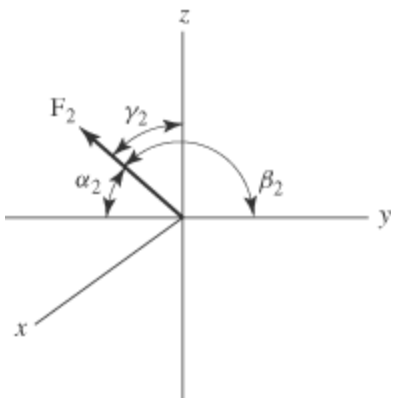
$$\begin{pmatrix} \alpha_1 \\ \beta_1 \\ \gamma_1 \end{pmatrix} = \begin{pmatrix} 46.9 \\ 124.7 \\ 62.9 \end{pmatrix} \text{deg}$$

$$\left|\mathbf{F_2}\right| = 98.6\ \text{N}$$

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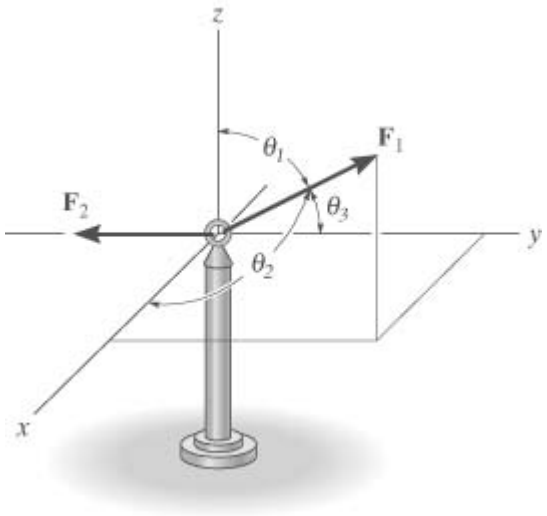
$$\begin{pmatrix}\alpha_2\\ \beta_2\\ \gamma_2\end{pmatrix} = \operatorname{acos}\left(\frac{\mathbf{F_2}}{|\mathbf{F_2}|}\right)$$

$$\begin{pmatrix}\alpha_2\\ \beta_2\\ \gamma_2\end{pmatrix} = \begin{pmatrix}113.9\\ 149.5\\ 72.3\end{pmatrix}\text{deg}$$



Problem 2-58

Express each force in Cartesian vector form.



Units Used:

$\text{kN} = 10^3\ \text{N}$

Given:

$F_1 = 5\ \text{kN}$

$F_2 = 2\ \text{kN}$

$\theta_1 = 60\ \text{deg}$

$\theta_2 = 60\ \text{deg}$

$\theta_3 = 45\ \text{deg}$

Solution:

$$\mathbf{F_{1v}} = F_1\begin{pmatrix}\cos(\theta_2)\\ \cos(\theta_3)\\ \cos(\theta_1)\end{pmatrix} \qquad \mathbf{F_{1v}} = \begin{pmatrix}2.5\\ 3.5\\ 2.5\end{pmatrix}\text{kN}$$

$$\mathbf{F_{2v}} = F_2\begin{pmatrix}0\\ -1\\ 0\end{pmatrix} \qquad \mathbf{F_{2v}} = \begin{pmatrix}0\\ -2\\ 0\end{pmatrix}\text{kN}$$

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Problem 2-59

Determine the magnitude and coordinate direction angles of the force **F** acting on the stake.

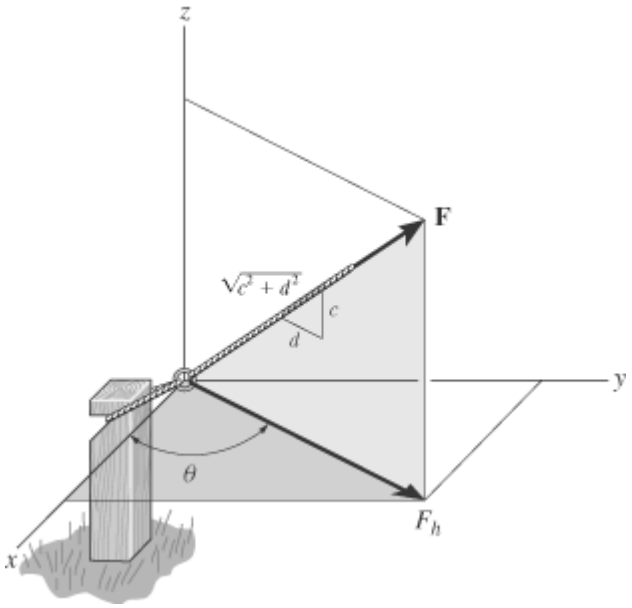
Given:

$F_h = 40\ \text{N}$

$\theta = 70\ \text{deg}$

$c = 3$

$d = 4$



Solution:

$$F = F_h\left(\frac{\sqrt{c^2+d^2}}{d}\right)$$

$F = 50\ \text{N}$

$$F_x = F_h\cos(\theta) \qquad F_y = F_h\sin(\theta) \qquad F_z = \left(\frac{c}{\sqrt{c^2+d^2}}\right)F$$

$F_x = 13.7\ \text{N}$ $F_y = 37.6\ \text{N}$ $F_z = 30\ \text{N}$

$$\alpha = \text{acos}\left(\frac{F_x}{F}\right) \qquad \beta = \text{acos}\left(\frac{F_y}{F}\right) \qquad \gamma = \text{acos}\left(\frac{F_z}{F}\right)$$

$\alpha = 74.1\,\text{deg}$ $\beta = 41.3\,\text{deg}$ $\gamma = 53.1\,\text{deg}$

Problem 2-60

Express each force in Cartesian vector form.

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Given:

$F_1 = 400$ lb

$F_2 = 600$ lb

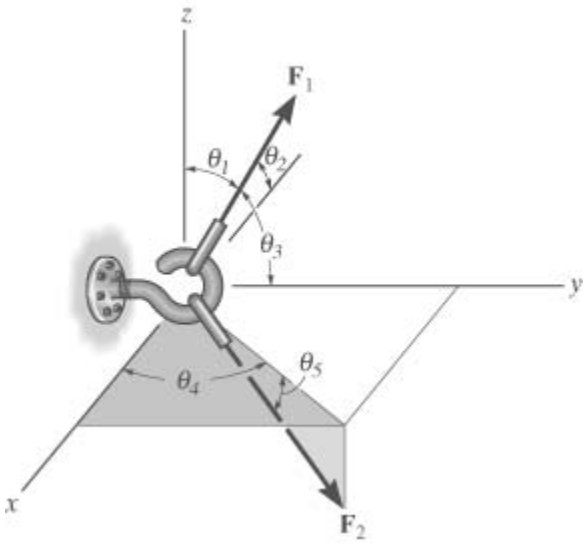
$\theta_1 = 45$ deg

$\theta_2 = 60$ deg

$\theta_3 = 60$ deg

$\theta_4 = 45$ deg

$\theta_5 = 30$ deg



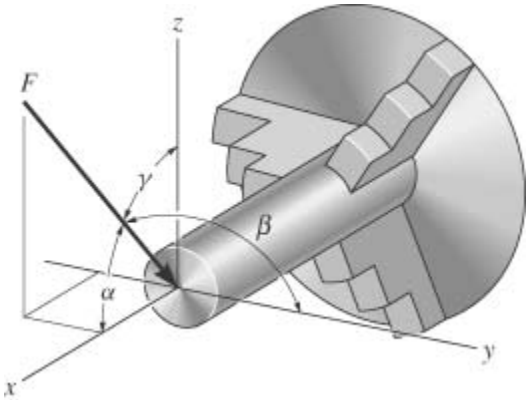
Solution:

$$\mathbf{F_{1v}} = F_1 \begin{pmatrix} -\cos(\theta_2) \\ \cos(\theta_3) \\ \cos(\theta_1) \end{pmatrix} \qquad \mathbf{F_{1v}} = \begin{pmatrix} -200 \\ 200 \\ 282.8 \end{pmatrix} \text{lb}$$

$$\mathbf{F_{2v}} = F_2 \begin{pmatrix} \cos(\theta_5)\cos(\theta_4) \\ \cos(\theta_5)\sin(\theta_4) \\ -\sin(\theta_5) \end{pmatrix} \qquad \mathbf{F_{2v}} = \begin{pmatrix} 367.4 \\ 367.4 \\ -300 \end{pmatrix} \text{lb}$$

Problem 2-61

The stock *S* mounted on the lathe is subjected to a force **F**, which is caused by the die *D*. Determine the coordinate direction angle β and express the force as a Cartesian vector.



Given:

$F = 60$ N

$\alpha = 60$ deg

$\gamma = 30$ deg

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Solution:

$$\cos^2(\alpha) + \cos^2(\beta) + \cos^2(\gamma) = 1$$

$$\beta = \operatorname{acos}\left(1 - \cos(\alpha)^2 - \cos(\gamma)^2\right) \qquad \beta = 90\,\text{deg}$$

$$\mathbf{F_v} = F\begin{pmatrix} -\cos(\alpha) \\ -\cos(\beta) \\ -\cos(\gamma) \end{pmatrix} \qquad \mathbf{F_v} = \begin{pmatrix} -30 \\ 0 \\ -52 \end{pmatrix}\,\text{N}$$

Problem 2-62

Determine the magnitude and coordinate direction angles of the resultant force.

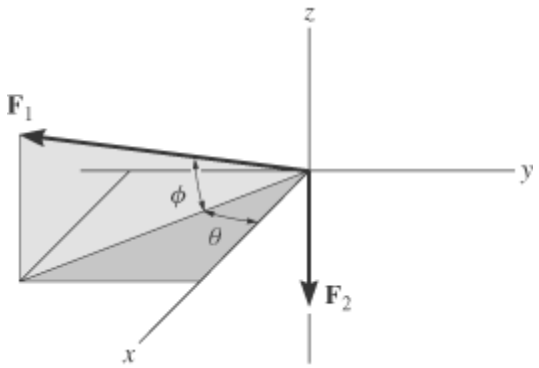
Given:

$F_1 = 80$ lb

$F_2 = 130$ lb

$\theta = 40$ deg

$\phi = 30$ deg



Solution:

$$\mathbf{F_{1v}} = F_1\begin{pmatrix} \cos(\phi)\cos(\theta) \\ -\cos(\phi)\sin(\theta) \\ \sin(\phi) \end{pmatrix} \qquad \mathbf{F_{1v}} = \begin{pmatrix} 53.1 \\ -44.5 \\ 40 \end{pmatrix}\,\text{lb}$$

$$\mathbf{F_{2v}} = F_2\begin{pmatrix} 0 \\ 0 \\ -1 \end{pmatrix} \qquad \mathbf{F_{2v}} = \begin{pmatrix} 0 \\ 0 \\ -130 \end{pmatrix}\,\text{lb}$$

$$\mathbf{F_R} = \mathbf{F_{1v}} + \mathbf{F_{2v}} \qquad \left|\mathbf{F_R}\right| = 113.6\,\text{lb}$$

$$\begin{pmatrix} \alpha \\ \beta \\ \gamma \end{pmatrix} = \operatorname{acos}\left(\frac{\mathbf{F_R}}{\left|\mathbf{F_R}\right|}\right) \qquad \begin{pmatrix} \alpha \\ \beta \\ \gamma \end{pmatrix} = \begin{pmatrix} 62.1 \\ 113.1 \\ 142.4 \end{pmatrix}\,\text{deg}$$

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Problem 2-63

Specify the coordinate direction angles of $\mathbf{F_1}$ and $\mathbf{F_2}$ and express each force as a cartesian vector.

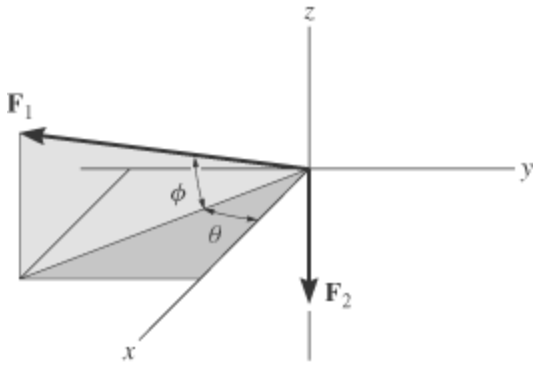
Given:

$F_1 = 80$ lb

$F_2 = 130$ lb

$\phi = 30$ deg

$\theta = 40$ deg



Solution:

$$\mathbf{F_{1v}} = F_1 \begin{pmatrix} \cos(\phi)\cos(\theta) \\ -\cos(\phi)\sin(\theta) \\ \sin(\phi) \end{pmatrix} \qquad \mathbf{F_{1v}} = \begin{pmatrix} 53.1 \\ -44.5 \\ 40 \end{pmatrix} \text{lb}$$

$$\begin{pmatrix} \alpha_1 \\ \beta_1 \\ \gamma_1 \end{pmatrix} = \operatorname{acos}\left(\frac{\mathbf{F_{1v}}}{F_1}\right) \qquad \begin{pmatrix} \alpha_1 \\ \beta_1 \\ \gamma_1 \end{pmatrix} = \begin{pmatrix} 48.4 \\ 123.8 \\ 60 \end{pmatrix} \text{deg}$$

$$\mathbf{F_{2v}} = F_2 \begin{pmatrix} 0 \\ 0 \\ -1 \end{pmatrix} \qquad \mathbf{F_{2v}} = \begin{pmatrix} 0 \\ 0 \\ -130 \end{pmatrix} \text{lb}$$

$$\begin{pmatrix} \alpha_2 \\ \beta_2 \\ \gamma_2 \end{pmatrix} = \operatorname{acos}\left(\frac{\mathbf{F_{2v}}}{F_2}\right) \qquad \begin{pmatrix} \alpha_2 \\ \beta_2 \\ \gamma_2 \end{pmatrix} = \begin{pmatrix} 90 \\ 90 \\ 180 \end{pmatrix} \text{deg}$$

Problem 2-64

The mast is subjected to the three forces shown. Determine the coordinate angles $\alpha_1, \beta_1, \gamma_1$ of $\mathbf{F_1}$ so that the resultant force acting on the mast is $F_R\mathbf{i}$.

Given:

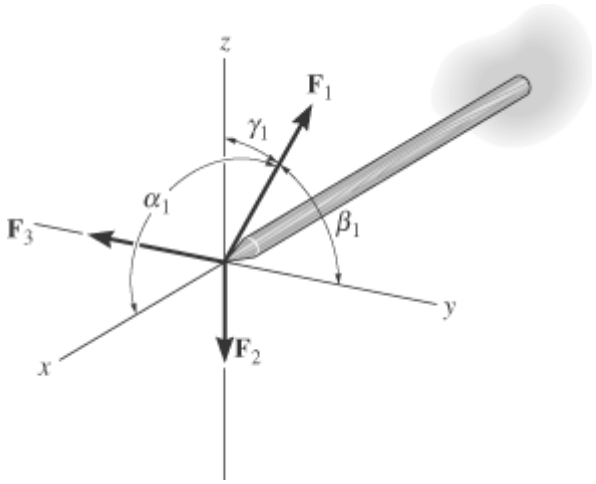
$F_R = 350$ N

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$F_1 = 500$ N

$F_2 = 200$ N

$F_3 = 300$ N



Solution:

Guesses $\alpha = 20$ deg

$\beta = 20$ deg

$\gamma = 20$ deg

Given $$F_1\begin{pmatrix}\cos(\alpha)\\ \cos(\beta)\\ \cos(\gamma)\end{pmatrix} + F_2\begin{pmatrix}0\\ 0\\ -1\end{pmatrix} + F_3\begin{pmatrix}0\\ -1\\ 0\end{pmatrix} = F_R\begin{pmatrix}1\\ 0\\ 0\end{pmatrix}$$

$$\begin{pmatrix}\alpha\\ \beta\\ \gamma\end{pmatrix} = \text{Find}(\alpha,\beta,\gamma) \qquad \begin{pmatrix}\alpha\\ \beta\\ \gamma\end{pmatrix} = \begin{pmatrix}45.6\\ 53.1\\ 66.4\end{pmatrix}\text{deg}$$

Problem 2-65

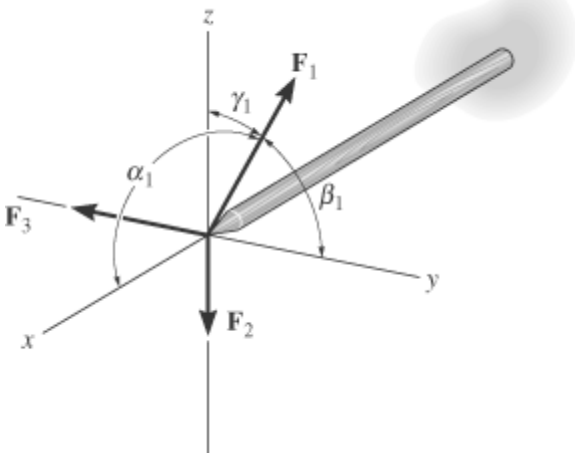
The mast is subjected to the three forces shown. Determine the coordinate angles $\alpha_1, \beta_1, \gamma_1$ of $\mathbf{F_1}$ so that the resultant force acting on the mast is zero.

Given:

$F_1 = 500$ N

$F_2 = 200$ N

$F_3 = 300$ N



Solution:

Guesses $\alpha = 20$ deg

$\beta = 20$ deg

$\gamma = 20$ deg

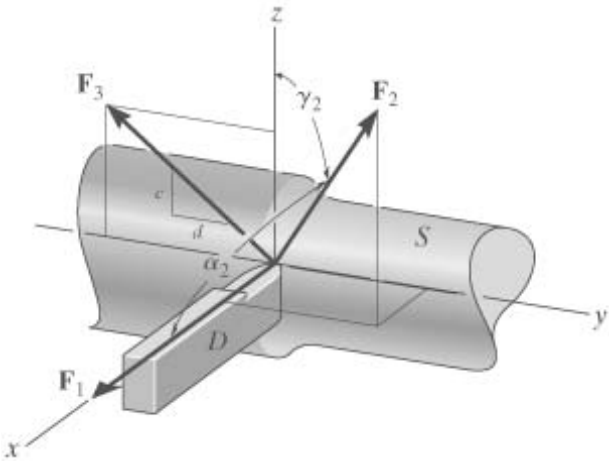
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Given $F_1\begin{pmatrix}\cos(\alpha)\\ \cos(\beta)\\ \cos(\gamma)\end{pmatrix} + F_2\begin{pmatrix}0\\ 0\\ -1\end{pmatrix} + F_3\begin{pmatrix}0\\ -1\\ 0\end{pmatrix} = 0$

$\begin{pmatrix}\alpha\\ \beta\\ \gamma\end{pmatrix} = \text{Find}(\alpha,\beta,\gamma)$ $\begin{pmatrix}\alpha\\ \beta\\ \gamma\end{pmatrix} = \begin{pmatrix}90\\ 53.1\\ 66.4\end{pmatrix}\text{deg}$

Problem 2-66

The shaft *S* exerts three force components on the die *D*. Find the magnitude and direction of the resultant force. Force $\mathbf{F_2}$ acts within the octant shown.



Given:

$F_1 = 400$ N

$F_2 = 300$ N

$F_3 = 200$ N

$\alpha_2 = 60$ deg

$\gamma_2 = 60$ deg

$c = 3$

$d = 4$

Solution:

$\cos(\alpha_2)^2 + \cos(\beta_2)^2 + \cos(\gamma_2)^2 = 1$

Solving for the positive root,

$\beta_2 = \text{acos}\left(\sqrt{1 - \cos(\alpha_2)^2 - \cos(\gamma_2)^2}\right)$ $\beta_2 = 45\,\text{deg}$

$\mathbf{F_{1v}} = \begin{pmatrix}F_1\\ 0\\ 0\end{pmatrix}$ $\mathbf{F_{2v}} = F_2\begin{pmatrix}\cos(\alpha_2)\\ \cos(\beta_2)\\ \cos(\gamma_2)\end{pmatrix}$ $\mathbf{F_{3v}} = \frac{F_3}{\sqrt{c^2 + d^2}}\begin{pmatrix}0\\ -d\\ c\end{pmatrix}$

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$\mathbf{F_R} = \mathbf{F_{1v}} + \mathbf{F_{2v}} + \mathbf{F_{3v}}$ $\quad |\mathbf{F_R}| = 615 \text{ N}$

$\begin{pmatrix} \alpha \\ \beta \\ \gamma \end{pmatrix} = \text{acos}\left(\frac{\mathbf{F_R}}{|\mathbf{F_R}|}\right)$ $\quad \begin{pmatrix} \alpha \\ \beta \\ \gamma \end{pmatrix} = \begin{pmatrix} 26.6 \\ 85.1 \\ 64.0 \end{pmatrix} \text{deg}$

Problem 2-67

The beam is subjected to the two forces shown. Express each force in Cartesian vector form and determine the magnitude and coordinate direction angles of the resultant force.

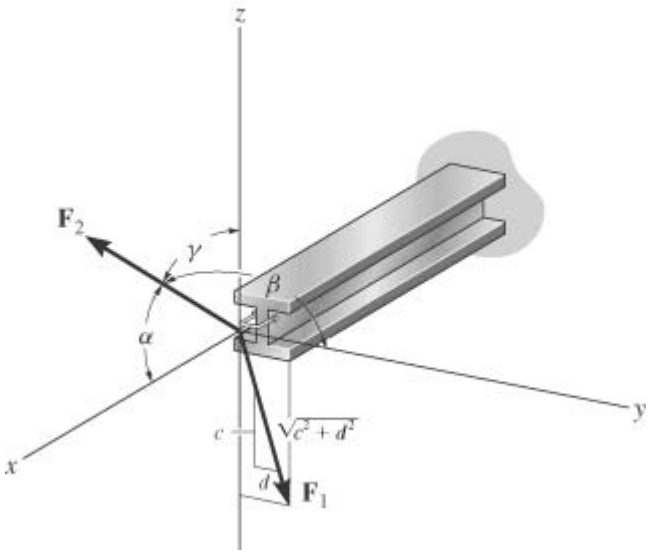
Given:

$F_1 = 630 \text{ lb} \quad \alpha = 60 \text{ deg}$

$F_2 = 250 \text{ lb} \quad \beta = 135 \text{ deg}$

$c = 24 \quad \gamma = 60 \text{ deg}$

$d = 7$



Solution:

$\mathbf{F_{1v}} = \frac{F_1}{\sqrt{c^2 + d^2}} \begin{pmatrix} 0 \\ d \\ -c \end{pmatrix}$ $\quad \mathbf{F_{1v}} = \begin{pmatrix} 0 \\ 176.4 \\ -604.8 \end{pmatrix} \text{lb}$

$\mathbf{F_{2v}} = F_2 \begin{pmatrix} \cos(\alpha) \\ \cos(\beta) \\ \cos(\gamma) \end{pmatrix}$ $\quad \mathbf{F_{2v}} = \begin{pmatrix} 125 \\ -176.8 \\ 125 \end{pmatrix} \text{lb}$

$\mathbf{F_R} = \mathbf{F_{1v}} + \mathbf{F_{2v}}$ $\quad |\mathbf{F_R}| = 495.8 \text{ lb}$

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$$\begin{pmatrix}\alpha_R\\ \beta_R\\ \gamma_R\end{pmatrix} = \operatorname{acos}\left(\frac{\mathbf{F_R}}{|\mathbf{F_R}|}\right) \qquad \begin{pmatrix}\alpha_R\\ \beta_R\\ \gamma_R\end{pmatrix} = \begin{pmatrix}75.4\\ 90\\ 165.4\end{pmatrix}\text{deg}$$

Problem 2-68

Determine the magnitude and coordinate direction angles of the resultant force.

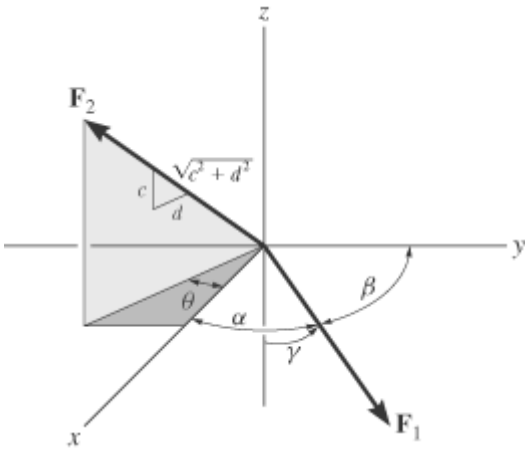
Given:

$F_1 = 350$ N $\quad \alpha = 60$ deg

$F_2 = 250$N $\quad \beta = 60$ deg

$c = 3$ $\quad \gamma = 45$ deg

$d = 4$ $\quad \theta = 30$ deg



Solution:

$$\mathbf{F_{1v}} = F_1\begin{pmatrix}\cos(\alpha)\\ \cos(\beta)\\ -\cos(\gamma)\end{pmatrix} \qquad \mathbf{F_{1v}} = \begin{pmatrix}175\\ 175\\ -247.5\end{pmatrix}\text{N}$$

$$F_{2h} = F_2\left(\frac{d}{\sqrt{c^2+d^2}}\right) \qquad F_{2y} = F_2\left(\frac{c}{\sqrt{c^2+d^2}}\right)$$

$$\mathbf{F_{2v}} = \begin{pmatrix}F_{2h}\cos(\theta)\\ -F_{2h}\sin(\theta)\\ F_{2y}\end{pmatrix} \qquad \mathbf{F_{2v}} = \begin{pmatrix}173.2\\ -100\\ 150\end{pmatrix}\text{N}$$

$$\mathbf{F_R} = \mathbf{F_{1v}} + \mathbf{F_{2v}} \qquad |\mathbf{F_R}| = 369.3\text{ N}$$

$$\begin{pmatrix}\alpha_R\\ \beta_R\\ \gamma_R\end{pmatrix} = \operatorname{acos}\left(\frac{\mathbf{F_R}}{|\mathbf{F_R}|}\right) \qquad \begin{pmatrix}\alpha_R\\ \beta_R\\ \gamma_R\end{pmatrix} = \begin{pmatrix}19.5\\ 78.3\\ 105.3\end{pmatrix}\text{deg}$$

Problem 2-69

Determine the magnitude and coordinate direction angles of $\mathbf{F_3}$ so that the resultant of the three

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forces acts along the positive y axis and has magnitude F.

Given:

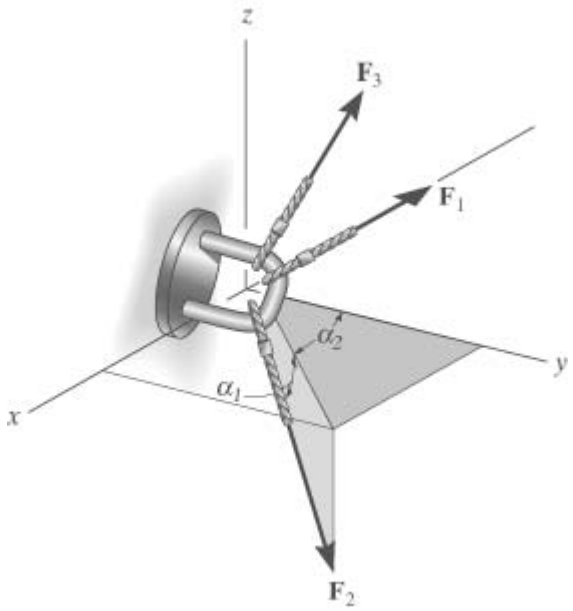
$F = 600$ lb

$F_1 = 180$ lb

$F_2 = 300$ lb

$\alpha_1 = 30$ deg

$\alpha_2 = 40$ deg



Solution:

Initial guesses:

$\alpha = 40$ deg $\quad \gamma = 50$ deg

$\beta = 50$ deg $\quad F_3 = 45$ lb

Given

$F_{Rx} = \Sigma F_x; \quad 0 = -F_1 + F_2\cos(\alpha_1)\sin(\alpha_2) + F_3\cos(\alpha)$

$F_{Ry} = \Sigma F_y; \quad F = F_2\cos(\alpha_1)\cos(\alpha_2) + F_3\cos(\beta)$

$F_{Rz} = \Sigma F_z; \quad 0 = -F_2\sin(\alpha_1) + F_3\cos(\gamma)$

$\cos(\alpha)^2 + \cos(\beta)^2 + \cos(\gamma)^2 = 1$

$$\begin{pmatrix} F_3 \\ \alpha \\ \beta \\ \gamma \end{pmatrix} = \text{Find}(F_3, \alpha, \beta, \gamma) \qquad F_3 = 428\,\text{lb} \qquad \begin{pmatrix} \alpha \\ \beta \\ \gamma \end{pmatrix} = \begin{pmatrix} 88.3 \\ 20.6 \\ 69.5 \end{pmatrix} \text{deg}$$

Problem 2-70

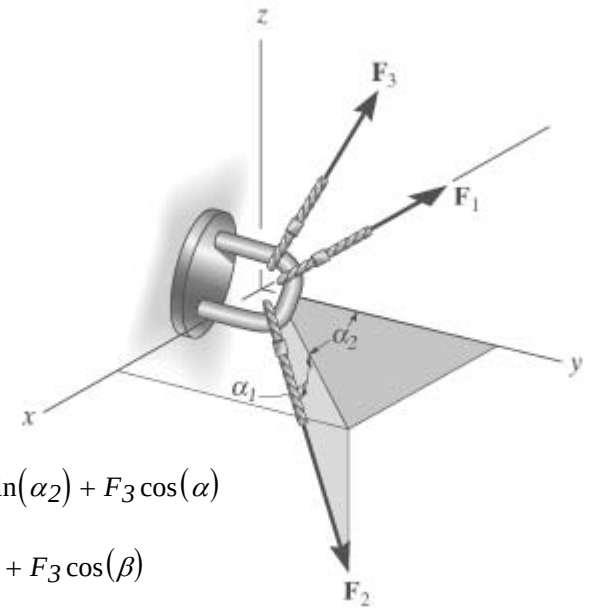
Determine the magnitude and coordinate direction angles of $\mathbf{F_3}$ so that the resultant of the three forces is zero.

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Given:

$F_1 = 180$ lb $\quad \alpha_1 = 30$ deg

$F_2 = 300$ lb $\quad \alpha_2 = 40$ deg



Solution:

Initial guesses:

$\alpha = 40$ deg $\quad \gamma = 50$ deg

$\beta = 50$ deg $\quad F_3 = 45$ lb

Given

$F_{Rx} = \Sigma F_x; \quad 0 = -F_1 + F_2\cos(\alpha_1)\sin(\alpha_2) + F_3\cos(\alpha)$

$F_{Ry} = \Sigma F_y; \quad 0 = F_2\cos(\alpha_1)\cos(\alpha_2) + F_3\cos(\beta)$

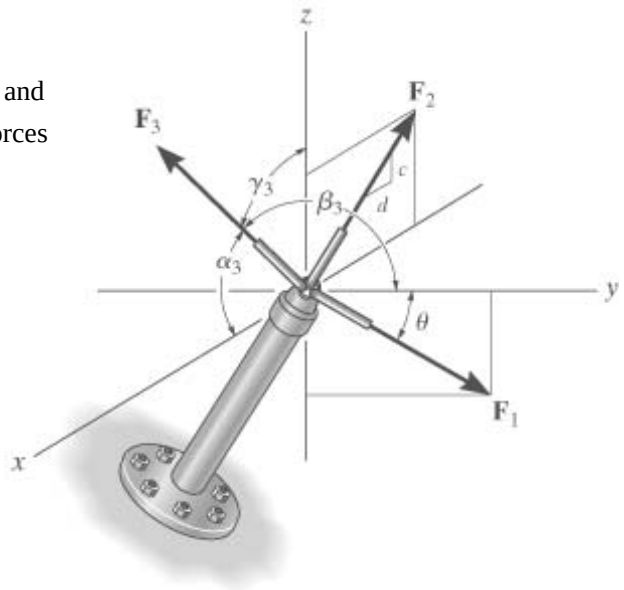
$F_{Rz} = \Sigma F_z; \quad 0 = -F_2\sin(\alpha_1) + F_3\cos(\gamma)$

$\cos(\alpha)^2 + \cos(\beta)^2 + \cos(\gamma)^2 = 1$

$\begin{pmatrix} F_3 \\ \alpha \\ \beta \\ \gamma \end{pmatrix} = \text{Find}(F_3, \alpha, \beta, \gamma) \qquad F_3 = 250\,\text{lb} \qquad \begin{pmatrix} \alpha \\ \beta \\ \gamma \end{pmatrix} = \begin{pmatrix} 87.0 \\ 142.9 \\ 53.1 \end{pmatrix} \text{deg}$

Problem 2-71

Specify the magnitude F_3 and directions α_3, β_3, and γ_3 of $\mathbf{F_3}$ so that the resultant force of the three forces is $\mathbf{F_R}$.



Units Used:

kN $= 10^3$ N

Given:

$F_1 = 12$ kN $\quad c = 5$

$F_2 = 10$ kN $\quad d = 12$

$\theta = 30$ deg

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$$\mathbf{F_R} = \begin{pmatrix} 0 \\ 9 \\ 0 \end{pmatrix} \text{ kN}$$

Solution:

Initial Guesses: $F_{3x} = 1 \text{ kN}$ $F_{3y} = 1 \text{ kN}$ $F_{3z} = 1 \text{ kN}$

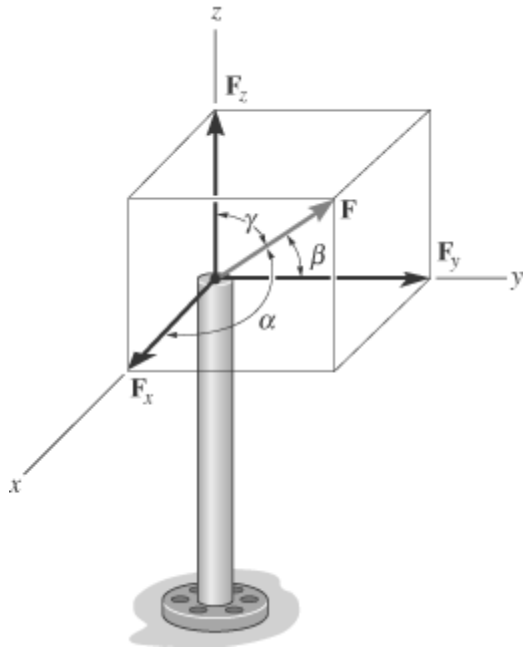
Given $$\mathbf{F_R} = \begin{pmatrix} F_{3x} \\ F_{3y} \\ F_{3z} \end{pmatrix} + F_1 \begin{pmatrix} 0 \\ \cos(\theta) \\ -\sin(\theta) \end{pmatrix} + \frac{F_2}{\sqrt{c^2 + d^2}} \begin{pmatrix} -d \\ 0 \\ c \end{pmatrix}$$

$$\begin{pmatrix} F_{3x} \\ F_{3y} \\ F_{3z} \end{pmatrix} = \text{Find}(F_{3x}, F_{3y}, F_{3z}) \qquad \mathbf{F_3} = \begin{pmatrix} F_{3x} \\ F_{3y} \\ F_{3z} \end{pmatrix} \qquad \mathbf{F_3} = \begin{pmatrix} 9.2 \\ -1.4 \\ 2.2 \end{pmatrix} \text{ kN} \qquad |\mathbf{F_3}| = 9.6 \text{ kN}$$

$$\begin{pmatrix} \alpha_3 \\ \beta_3 \\ \gamma_3 \end{pmatrix} = \text{acos}\left(\frac{\mathbf{F_3}}{|\mathbf{F_3}|}\right) \qquad \begin{pmatrix} \alpha_3 \\ \beta_3 \\ \gamma_3 \end{pmatrix} = \begin{pmatrix} 15.5 \\ 98.4 \\ 77.0 \end{pmatrix} \text{ deg}$$

Problem 2-72

The pole is subjected to the force **F**, which has components acting along the *x,y,z* axes as shown. Given β and γ, determine the magnitude of the three components of **F**.



Units Used:

$\text{kN} = 1000 \text{ N}$

Given:

$F = 3 \text{ kN}$

$\beta = 30 \text{ deg}$

$\gamma = 75 \text{ deg}$

Solution:

$$\cos(\alpha)^2 + \cos(\beta)^2 + \cos(\gamma)^2 = 1$$

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$\alpha = \text{acos}\left(\sqrt{-\cos(\beta)^2 - \cos(\gamma)^2 + 1}\right)$

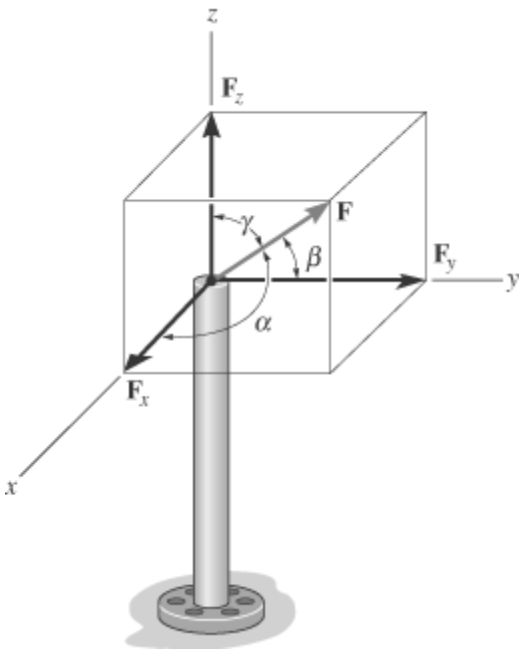
$\alpha = 64.67\,\text{deg}$

$F_x = F\cos(\alpha)$ $F_y = F\cos(\beta)$ $F_z = F\cos(\gamma)$

$F_x = 1.28\,\text{kN}$ $F_y = 2.60\,\text{kN}$ $F_z = 0.8\,\text{kN}$

Problem 2-73

The pole is subjected to the force **F** which has components $\mathbf{F_x}$ and $\mathbf{F_z}$. Determine the magnitudes of **F** and $\mathbf{F_y}$.



Units Used:

kN = 1000 N

Given:

$F_x = 1.5$ kN

$F_z = 1.25$ kN

$\beta = 75$ deg

Solution:

$\cos(\alpha)^2 + \cos(\beta)^2 + \cos(\gamma)^2 = 1$

$\left(\frac{F_x}{F}\right)^2 + \cos(\beta)^2 + \left(\frac{F_z}{F}\right)^2 = 1$

$F = \sqrt{\frac{F_x^{\,2} + F_z^{\,2}}{1 - \cos(\beta)^2}}$ $F = 2.02\,\text{kN}$

$F_y = F\cos(\beta)$ $F_y = 0.5\,\text{kN}$

Problem 2-74

The eye bolt is subjected to the cable force **F** which has a component F_x along the *x* axis, a component F_z along the *z* axis, and a coordinate direction angle β. Determine the magnitude of **F**.

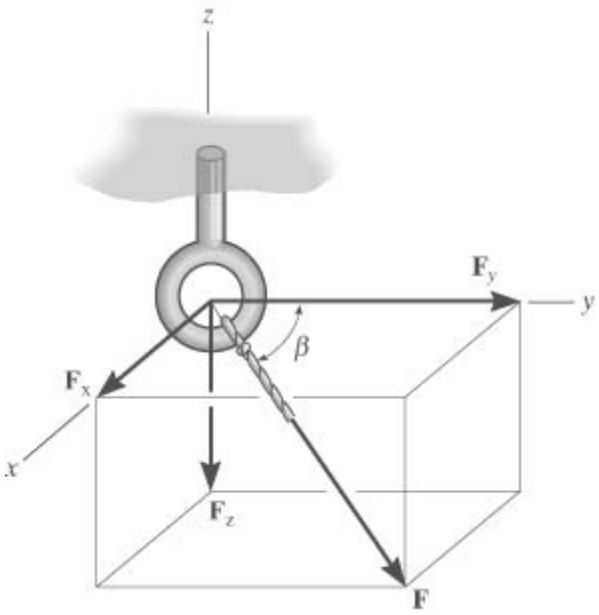
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Given:

$F_x = 60$ N

$F_z = -80$ N

$\beta = 80$ deg



Solution:

$F_y = F\cos(\beta)$

$F_y = \sqrt{F_x^2 + F_z^2 + F_y^2}\cos(\beta)$

$F_y = \sqrt{\frac{F_x^2 + F_z^2}{1 - \cos(\beta)^2}}\cos(\beta)$ $F_y = 17.6$ N

$F = \sqrt{F_x^2 + F_y^2 + F_z^2}$ $F = 102$ N

Problem 2-75

Three forces act on the hook. If the resultant force $\mathbf{F_R}$ has a magnitude and direction as shown, determine the magnitude and the coordinate direction angles of force $\mathbf{F_3}$.

Given:

$F_R = 120$ N

$F_1 = 80$ N

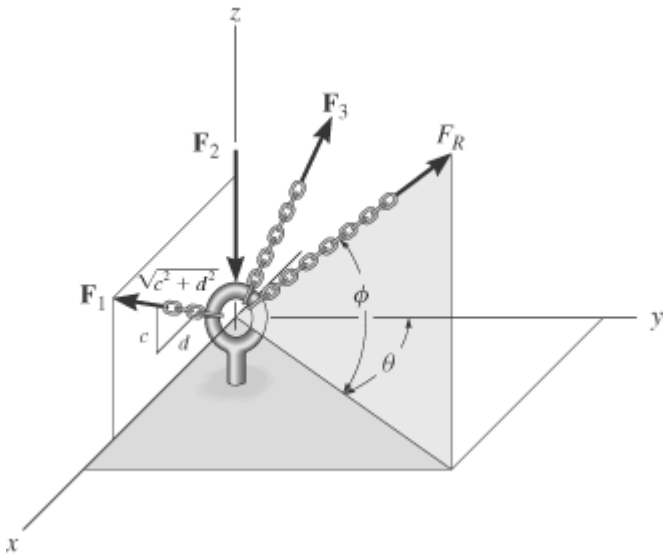
$F_2 = 110$ N

$c = 3$

$d = 4$

$\theta = 30$ deg

$\phi = 45$ deg



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Solution:

$$\mathbf{F_{1v}} = \left(\frac{F_1}{\sqrt{c^2 + d^2}}\right)\begin{pmatrix} d \\ 0 \\ c \end{pmatrix} \qquad \mathbf{F_{1v}} = \begin{pmatrix} 64 \\ 0 \\ 48 \end{pmatrix} \text{ N}$$

$$\mathbf{F_{2v}} = F_2\begin{pmatrix} 0 \\ 0 \\ -1 \end{pmatrix} \qquad \mathbf{F_{2v}} = \begin{pmatrix} 0 \\ 0 \\ -110 \end{pmatrix} \text{ N}$$

$$\mathbf{F_{Rv}} = F_R\begin{pmatrix} \cos(\phi)\sin(\theta) \\ \cos(\phi)\cos(\theta) \\ \sin(\phi) \end{pmatrix} \qquad \mathbf{F_{Rv}} = \begin{pmatrix} 42.4 \\ 73.5 \\ 84.9 \end{pmatrix} \text{ N}$$

$$\mathbf{F_{3v}} = \mathbf{F_{Rv}} - \mathbf{F_{1v}} - \mathbf{F_{2v}} \qquad \mathbf{F_{3v}} = \begin{pmatrix} -21.6 \\ 73.5 \\ 146.9 \end{pmatrix} \text{ N} \qquad \left|\mathbf{F_{3v}}\right| = 165.6 \text{ N}$$

$$\begin{pmatrix} \alpha \\ \beta \\ \gamma \end{pmatrix} = \text{acos}\left(\frac{\mathbf{F_{3v}}}{\left|\mathbf{F_{3v}}\right|}\right) \qquad \begin{pmatrix} \alpha \\ \beta \\ \gamma \end{pmatrix} = \begin{pmatrix} 97.5 \\ 63.7 \\ 27.5 \end{pmatrix} \text{deg}$$

Problem 2-76

Determine the coordinate direction angles of $\mathbf{F_1}$ and $\mathbf{F_R}$.

Given:

$F_R = 120$ N

$F_1 = 80$ N

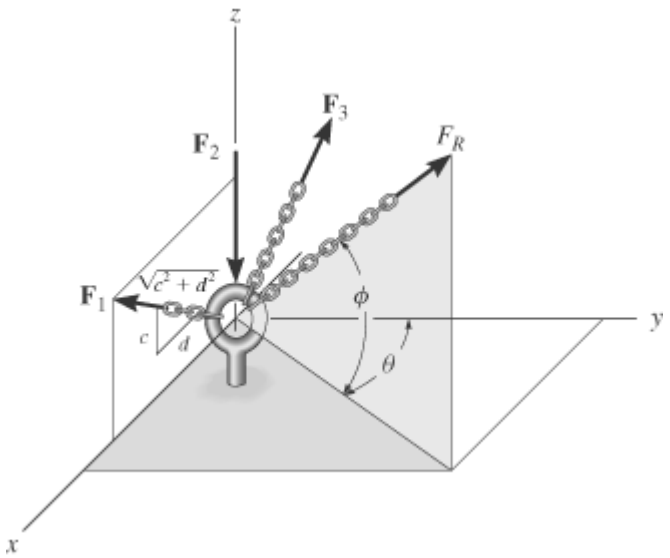
$F_2 = 110$ N

$c = 3$

$d = 4$

$\theta = 30$ deg

$\phi = 45$ deg



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Solution:

$$\mathbf{F_{1v}} = \left(\frac{F_1}{\sqrt{c^2 + d^2}}\right)\begin{pmatrix} d \\ 0 \\ c \end{pmatrix} \qquad \mathbf{F_{1v}} = \begin{pmatrix} 64 \\ 0 \\ 48 \end{pmatrix} \text{ N}$$

$$\begin{pmatrix} \alpha_1 \\ \beta_1 \\ \gamma_1 \end{pmatrix} = \text{acos}\left(\frac{\mathbf{F_{1v}}}{|\mathbf{F_{1v}}|}\right) \qquad \begin{pmatrix} \alpha_1 \\ \beta_1 \\ \gamma_1 \end{pmatrix} = \begin{pmatrix} 36.9 \\ 90 \\ 53.1 \end{pmatrix} \text{deg}$$

$$\mathbf{F_{Rv}} = F_R\begin{pmatrix} \cos(\phi)\sin(\theta) \\ \cos(\phi)\cos(\theta) \\ \sin(\phi) \end{pmatrix} \qquad \mathbf{F_{Rv}} = \begin{pmatrix} 42.4 \\ 73.5 \\ 84.9 \end{pmatrix} \text{ N}$$

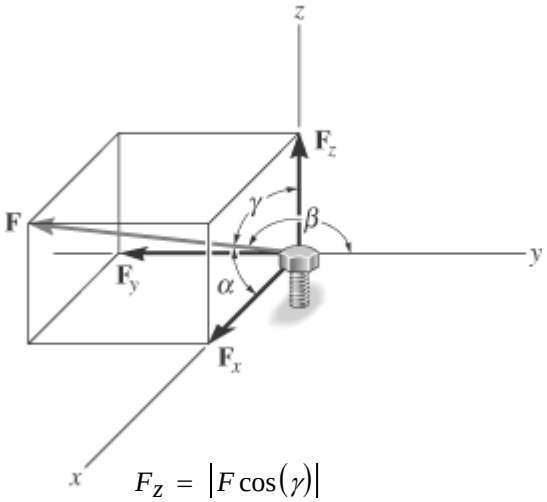
$$\begin{pmatrix} \alpha_R \\ \beta_R \\ \gamma_R \end{pmatrix} = \text{acos}\left(\frac{\mathbf{F_{Rv}}}{|\mathbf{F_{Rv}}|}\right) \qquad \begin{pmatrix} \alpha_R \\ \beta_R \\ \gamma_R \end{pmatrix} = \begin{pmatrix} 69.3 \\ 52.2 \\ 45 \end{pmatrix} \text{deg}$$

Problem 2-77

The pole is subjected to the force **F**, which has components acting along the *x*, *y*, *z* axes as shown. Given the magnitude of **F** and the angles α and γ, determine the magnitudes of the components of **F**.

Given:

$F = 80 \text{ N} \quad \alpha = 60 \text{ deg} \quad \gamma = 45 \text{ deg}$



Solution:

$$\beta = \text{acos}\left(-\sqrt{1 - \cos(\alpha)^2 - \cos(\gamma)^2}\right)$$

$\beta = 120 \text{ deg}$

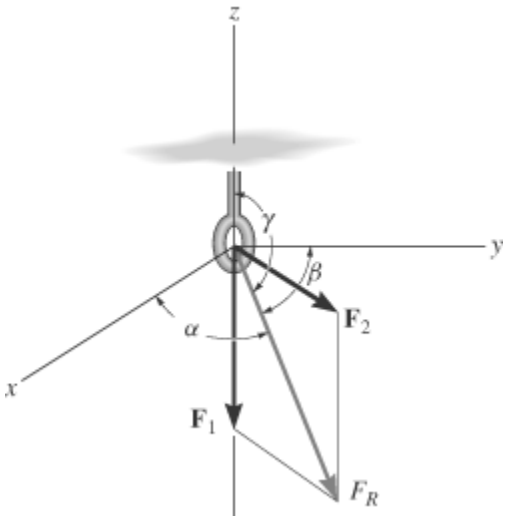
$$F_x = |F\cos(\alpha)| \qquad F_y = |F\cos(\beta)| \qquad F_z = |F\cos(\gamma)|$$

$F_x = 40 \text{ N}$ $F_y = 40 \text{ N}$ $F_z = 56.6 \text{ N}$

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Problem 2-78

Two forces $\mathbf{F_1}$ and $\mathbf{F_2}$ act on the bolt. If the resultant force $\mathbf{F_R}$ has magnitude F_R and coordinate direction angles α and β, as shown, determine the magnitude of $\mathbf{F_2}$ and its coordinate direction angles.



Given:

$F_1 = 20$ lb

$F_R = 50$ lb

$\alpha = 110$ deg

$\beta = 80$ deg

Solution:

$$\cos(\alpha)^2 + \cos(\beta)^2 + \cos(\gamma)^2 = 1$$

$$\gamma = \operatorname{acos}\left(-\sqrt{1 - \cos(\alpha)^2 - \cos(\beta)^2}\right) \qquad \gamma = 157.44\,\text{deg}$$

Initial Guesses $F_{2x} = 1$ lb $F_{2y} = 1$ lb $F_{2z} = 1$ lb

Given $$F_R\begin{pmatrix}\cos(\alpha)\\ \cos(\beta)\\ \cos(\gamma)\end{pmatrix} = F_1\begin{pmatrix}0\\ 0\\ -1\end{pmatrix} + \begin{pmatrix}F_{2x}\\ F_{2y}\\ F_{2z}\end{pmatrix} \qquad \begin{pmatrix}F_{2x}\\ F_{2y}\\ F_{2z}\end{pmatrix} = \text{Find}\left(F_{2x}, F_{2y}, F_{2z}\right)$$

$$\mathbf{F_2} = \begin{pmatrix}F_{2x}\\ F_{2y}\\ F_{2z}\end{pmatrix} \qquad \mathbf{F_2} = \begin{pmatrix}-17.1\\ 8.7\\ -26.2\end{pmatrix}\text{ lb} \qquad \left|\mathbf{F_2}\right| = 32.4\,\text{lb}$$

$$\begin{pmatrix}\alpha_2\\ \beta_2\\ \gamma_2\end{pmatrix} = \operatorname{acos}\left(\frac{\mathbf{F_2}}{\left|\mathbf{F_2}\right|}\right) \qquad \begin{pmatrix}\alpha_2\\ \beta_2\\ \gamma_2\end{pmatrix} = \begin{pmatrix}121.8\\ 74.5\\ 143.8\end{pmatrix}\text{ deg}$$

Problem 2-79

Given $\mathbf{r_1}$, $\mathbf{r_2}$, and $\mathbf{r_3}$, determine the magnitude and direction of $\mathrm{r} = 2\mathbf{r_1} - \mathbf{r_2} + 3\mathbf{r_3}$.

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Given:

$$\mathbf{r_1} = \begin{pmatrix} 3 \\ -4 \\ 3 \end{pmatrix} \text{ m} \qquad \mathbf{r_2} = \begin{pmatrix} 4 \\ 0 \\ -5 \end{pmatrix} \text{ m} \qquad \mathbf{r_3} = \begin{pmatrix} 3 \\ -2 \\ 5 \end{pmatrix} \text{ m}$$

Solution:

$$\mathbf{r} = 2\mathbf{r_1} - \mathbf{r_2} + 3\mathbf{r_3}$$

$$\mathbf{r} = \begin{pmatrix} 11 \\ -14 \\ 26 \end{pmatrix} \text{ m} \qquad |\mathbf{r}| = 31.5 \text{ m}$$

$$\begin{pmatrix} \alpha \\ \beta \\ \gamma \end{pmatrix} = \text{acos}\left(\frac{\mathbf{r}}{|\mathbf{r}|}\right) \qquad \begin{pmatrix} \alpha \\ \beta \\ \gamma \end{pmatrix} = \begin{pmatrix} 69.6 \\ 116.4 \\ 34.4 \end{pmatrix} \text{deg}$$

Problem 2-80

Represent the position vector **r** acting from point $A(a, b, c)$ to point $B(d, e, f)$ in Cartesian vector form. Determine its coordinate direction angles and find the distance between points A and B.

Given:

$a = 3$ m

$b = 5$ m

$c = 6$ m

$d = 5$ m

$e = -2$ m

$f = 1$ m

Solution:

$$\mathbf{r} = \begin{pmatrix} d - a \\ e - b \\ f - c \end{pmatrix} \qquad \mathbf{r} = \begin{pmatrix} 2 \\ -7 \\ -5 \end{pmatrix} \text{ m} \qquad |\mathbf{r}| = 8.8 \text{ m}$$

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$$\alpha = \operatorname{acos}\left(\frac{d-a}{|\mathbf{r}|}\right) \qquad \alpha = 76.9\,\text{deg}$$

$$\beta = \operatorname{acos}\left(\frac{e-b}{|\mathbf{r}|}\right) \qquad \beta = 142\,\text{deg}$$

$$\gamma = \operatorname{acos}\left(\frac{f-c}{|\mathbf{r}|}\right) \qquad \gamma = 124\,\text{deg}$$

Problem 2-81

A position vector extends from the origin to point $A(a, b, c)$. Determine the angles α, β, γ which the tail of the vector makes with the x, y, z axes, respectively.

Given:

$a = 2$ m $\quad b = 3$ m $\quad c = 6$ m

Solution:

$$\mathbf{r} = \begin{pmatrix} a \\ b \\ c \end{pmatrix} \qquad \mathbf{r} = \begin{pmatrix} 2 \\ 3 \\ 6 \end{pmatrix} \text{m}$$

$$\begin{pmatrix} \alpha \\ \beta \\ \gamma \end{pmatrix} = \operatorname{acos}\left(\frac{\mathbf{r}}{|\mathbf{r}|}\right) \qquad \begin{pmatrix} \alpha \\ \beta \\ \gamma \end{pmatrix} = \begin{pmatrix} 73.4 \\ 64.6 \\ 31.0 \end{pmatrix} \text{deg}$$

Problem 2-82

Express the position vector $\mathbf{r}$ in Cartesian vector form; then determine its magnitude and coordinate direction angles.

Given:

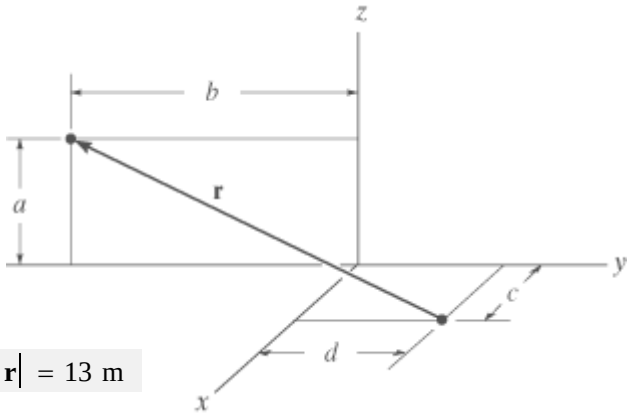
$a = 4$ m

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$b = 8$ m

$c = 3$ m

$d = 4$ m



Solution:

$$\mathbf{r} = \begin{pmatrix} -c \\ -d - b \\ a \end{pmatrix} \qquad \mathbf{r} = \begin{pmatrix} -3 \\ -12 \\ 4 \end{pmatrix} \text{ m} \qquad |\mathbf{r}| = 13 \text{ m}$$

$$\begin{pmatrix} \alpha \\ \beta \\ \gamma \end{pmatrix} = \operatorname{acos}\left(\frac{\mathbf{r}}{|\mathbf{r}|}\right) \qquad \begin{pmatrix} \alpha \\ \beta \\ \gamma \end{pmatrix} = \begin{pmatrix} 103.3 \\ 157.4 \\ 72.1 \end{pmatrix} \text{deg}$$

Problem 2-83

Express the position vector **r** in Cartesian vector form; then determine its magnitude and coordinate direction angles.

Given:

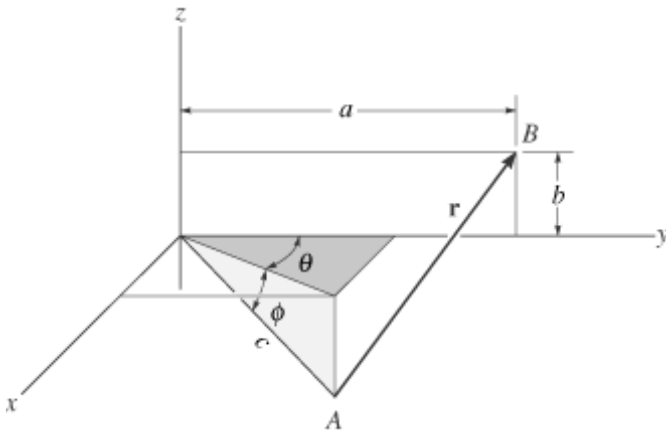
$a = 8$ ft

$b = 2$ ft

$c = 5$ ft

$\theta = 30$ deg

$\phi = 20$ deg



Solution:

$$\mathbf{r} = \begin{pmatrix} -c\cos(\phi)\sin(\theta) \\ a - c\cos(\phi)\cos(\theta) \\ b + c\sin(\phi) \end{pmatrix} \qquad \mathbf{r} = \begin{pmatrix} -2.35 \\ 3.93 \\ 3.71 \end{pmatrix} \text{ ft} \qquad |\mathbf{r}| = 5.89 \text{ ft}$$

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$$\begin{pmatrix}\alpha\\ \beta\\ \gamma\end{pmatrix} = \operatorname{acos}\left(\frac{\mathbf{r}}{|\mathbf{r}|}\right) \qquad \begin{pmatrix}\alpha\\ \beta\\ \gamma\end{pmatrix} = \begin{pmatrix}113.5\\ 48.2\\ 51\end{pmatrix}\text{deg}$$

Problem 2-84

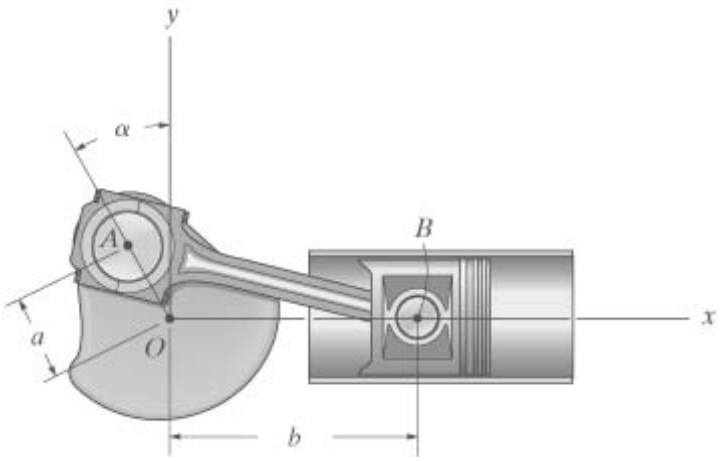
Determine the length of the connecting rod *AB* by first formulating a Cartesian position vector from *A* to *B* and then determining its magnitude.

Given:

$b = 16$ in

$a = 5$ in

$\alpha = 30$ deg



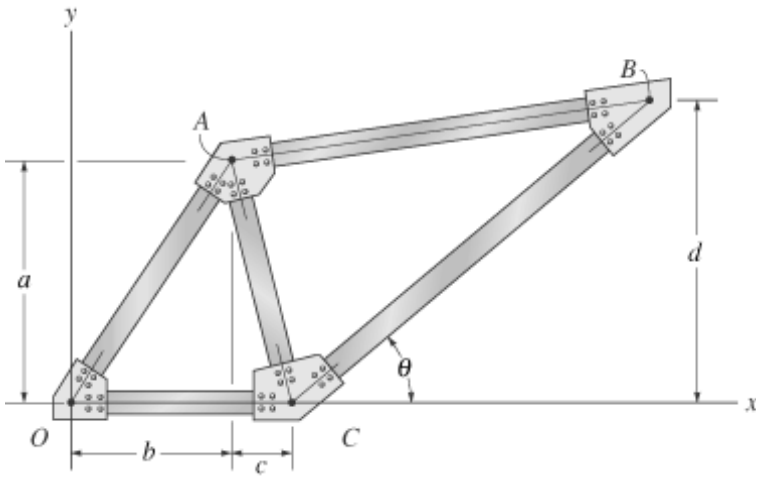
Solution:

$$\mathbf{r} = \begin{pmatrix} a\sin(\alpha) + b \\ -a\cos(\alpha)\end{pmatrix} \qquad \mathbf{r} = \begin{pmatrix}18.5\\ -4.3\end{pmatrix}\text{in} \qquad |\mathbf{r}| = 19\,\text{in}$$

Problem 2-85

Determine the length of member *AB* of the truss by first establishing a Cartesian position vector from *A* to *B* and then determining its magnitude.

Given:



$a = 1.2$ m

$b = 0.8$ m

$c = 0.3$ m

$d = 1.5$ m

$\theta = 40$ deg

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Solution:

$$\mathbf{r} = \begin{pmatrix} c + d\,\cot(\theta) \\ d - a \end{pmatrix} \qquad \mathbf{r} = \begin{pmatrix} 2.09 \\ 0.3 \end{pmatrix} \text{ m} \qquad |\mathbf{r}| = 2.11 \text{ m}$$

Problem 2- 86

The positions of point *A* on the building and point *B* on the antenna have been measured relative to the electronic distance meter (EDM) at *O*. Determine the distance between *A* and *B*. *Hint:* Formulate a position vector directed from *A* to *B*; then determine its magnitude.

Given:

$a = 460$ m

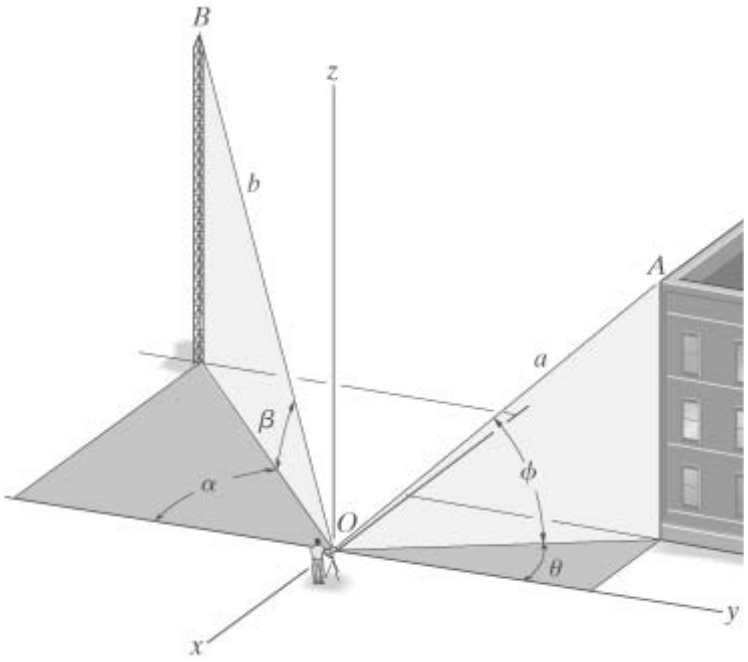
$b = 653$ m

$\alpha = 60$ deg

$\beta = 55$ deg

$\theta = 30$ deg

$\phi = 40$ deg



Solution:

$$\mathbf{r_{OA}} = \begin{pmatrix} -a\cos(\phi)\sin(\theta) \\ a\cos(\phi)\cos(\theta) \\ a\sin(\phi) \end{pmatrix}$$

$$\mathbf{r_{OB}} = \begin{pmatrix} -b\cos(\beta)\sin(\alpha) \\ -b\cos(\beta)\cos(\alpha) \\ b\sin(\beta) \end{pmatrix}$$

$$\mathbf{r_{AB}} = \mathbf{r_{OB}} - \mathbf{r_{OA}} \qquad \mathbf{r_{AB}} = \begin{pmatrix} -148.2 \\ -492.4 \\ 239.2 \end{pmatrix} \text{ m} \qquad |\mathbf{r_{AB}}| = 567.2 \text{ m}$$

Problem 2-87

Determine the lengths of cords *ACB* and *CO*. The knot at *C* is located midway between *A* and *B*.

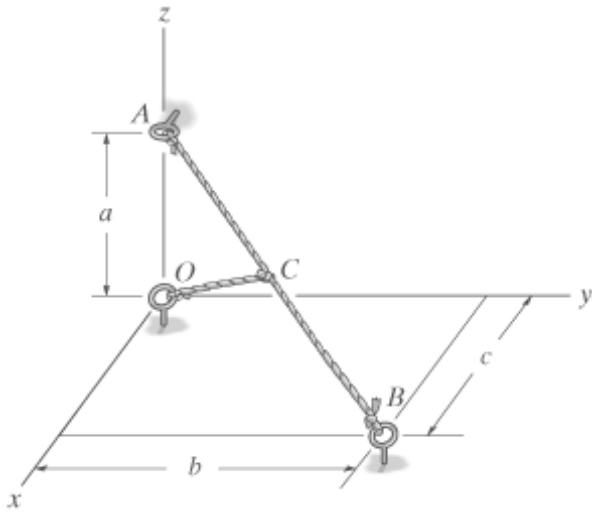
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Given:

$a = 3$ ft

$b = 6$ ft

$c = 4$ ft



Solution:

$$\mathbf{r_{AB}} = \begin{pmatrix} c \\ b \\ -a \end{pmatrix} \qquad \mathbf{r_{OA}} = \begin{pmatrix} 0 \\ 0 \\ a \end{pmatrix}$$

$$\mathbf{r_{AC}} = \frac{\mathbf{r_{AB}}}{2}$$

$$\left|\mathbf{r_{AB}}\right| = 7.8\,\text{ft}$$

$$\mathbf{r_{OC}} = \mathbf{r_{OA}} + \mathbf{r_{AC}}$$

$$\left|\mathbf{r_{OC}}\right| = 3.91\,\text{ft}$$

Problem 2-88

Determine the length of the crankshaft *AB* by first formulating a Cartesian position vector from *A* to *B* and then determining its magnitude.

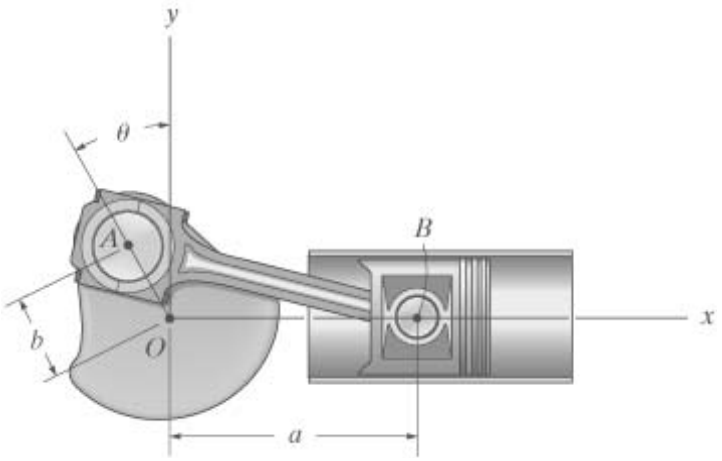
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Given:

$a = 400$

$b = 125$

$\theta = 25$ deg



Solution:

$$\mathbf{r_{AB}} = \begin{bmatrix} a + b\sin(\theta) \\ -(b\cos(\theta)) \\ 0 \end{bmatrix} \text{ mm} \qquad |\mathbf{r_{AB}}| = 467\,\text{mm}$$

Problem 2-89

Determine the length of wires *AD*, *BD*, and *CD*. The ring at *D* is midway between *A* and *B*.

Given:

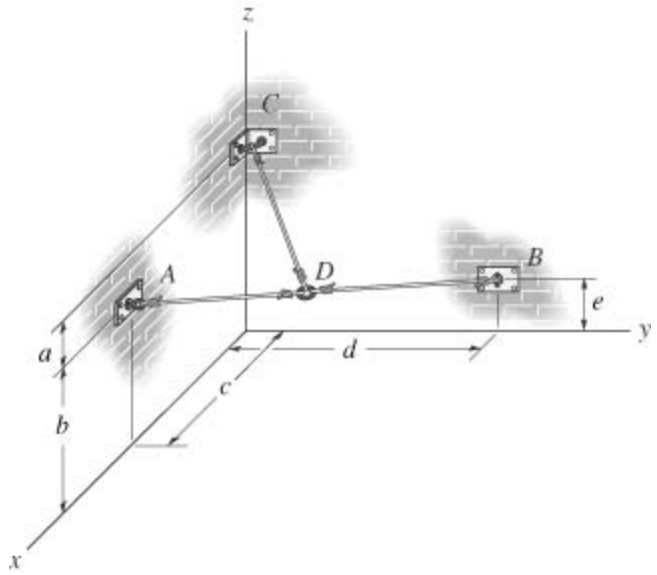
$a = 0.5$ m

$b = 1.5$ m

$c = 2$ m

$d = 2$ m

$e = 0.5$ m



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Solution:

$$\mathbf{r_{AD}} = \begin{pmatrix} \frac{-c}{2} \\ \frac{d}{2} \\ \frac{e}{2} - \frac{b}{2} \end{pmatrix} \qquad \mathbf{r_{AD}} = \begin{pmatrix} -1 \\ 1 \\ -0.5 \end{pmatrix} \text{ m} \qquad \left|\mathbf{r_{AD}}\right| = 1.5 \text{ m}$$

$$\mathbf{r_{BD}} = -\mathbf{r_{AD}}$$

$$\left|\mathbf{r_{BD}}\right| = 1.5 \text{ m}$$

$$\mathbf{r_{CD}} = \begin{pmatrix} \frac{c}{2} \\ \frac{d}{2} \\ a + \frac{b}{2} - \frac{e}{2} \end{pmatrix} \qquad \mathbf{r_{CD}} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \text{ m} \qquad \left|\mathbf{r_{CD}}\right| = 1.7 \text{ m}$$

Problem 2-90

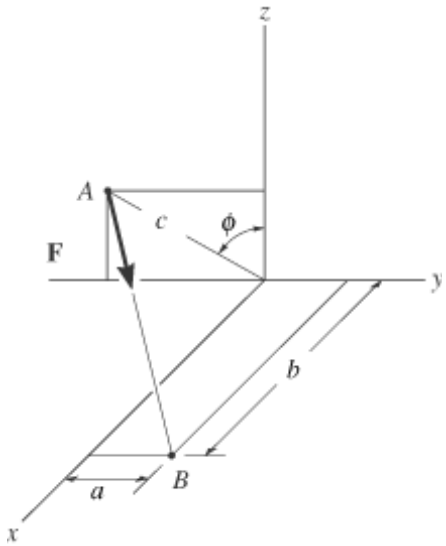
Express force **F** as a Cartesian vector; then determine its coordinate direction angles.

Given:

$F = 600$ lb $\quad c = 3$ ft

$a = 1.5$ ft $\quad \phi = 60$ deg

$b = 5$ ft



Solution:

$$r = b\mathbf{i} + \left(a + c\sin(\phi)\right)\mathbf{j} + \left(0 - c\cos(\phi)\right)\mathbf{k}$$

$$r = \sqrt{b^2 + \left(a + c\sin(\phi)\right)^2 + \left(c\cos(\phi)\right)^2}$$

$r = 2$ m

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$d = F\frac{b}{r}$ $e = F\left(\frac{a + c\sin(\phi)}{r}\right)$ $f = F\left[\frac{-(c\cos(\phi))}{r}\right]$

$d = 452\,\text{lb}$ $e = 370\,\text{lb}$ $f = -136\,\text{lb}$

$F = (d\mathbf{i} + e\mathbf{j} + f\mathbf{k})\ \text{lb}$

$\alpha = \text{acos}\left(\frac{d}{F}\right)$ $\alpha = 41.1\,\text{deg}$

$\beta = \text{acos}\left(\frac{e}{F}\right)$ $\beta = 51.9\,\text{deg}$

$\gamma = \text{acos}\left(\frac{f}{F}\right)$ $\gamma = 103\,\text{deg}$

Problem 2-91

Express force **F** as a Cartesian vector; then determine its coordinate direction angles.

Given:

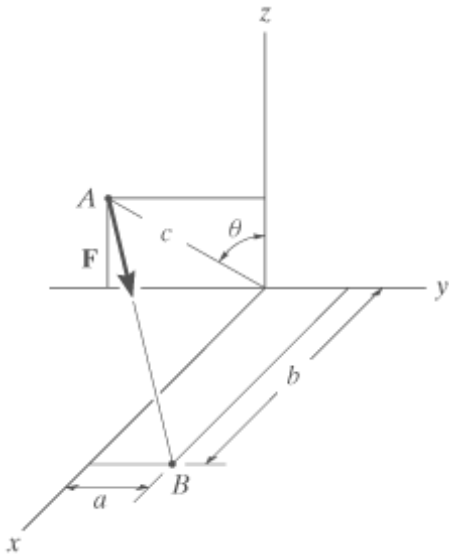
$a = 1.5$ ft

$b = 5$ ft

$c = 3$ ft

$\theta = 60$ deg

$F = 600$ lb



Solution:

$\mathbf{r} = \begin{pmatrix} b \\ a + c\sin(\theta) \\ -c\cos(\theta) \end{pmatrix}$ $\mathbf{F} = F\frac{\mathbf{r}}{|\mathbf{r}|}$ $\mathbf{F} = \begin{pmatrix} 452 \\ 370 \\ -136 \end{pmatrix}\text{lb}$

$\begin{pmatrix} \alpha \\ \beta \\ \gamma \end{pmatrix} = \text{acos}\left(\frac{\mathbf{F}}{|\mathbf{F}|}\right)$ $\begin{pmatrix} \alpha \\ \beta \\ \gamma \end{pmatrix} = \begin{pmatrix} 41.1 \\ 51.9 \\ 103.1 \end{pmatrix}\text{deg}$

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Problem 2-92

Determine the magnitude and coordinate direction angles of the resultant force acting at point A.

Given:

$F_1 = 150$ N

$F_2 = 200$ N

$a = 1.5$ m

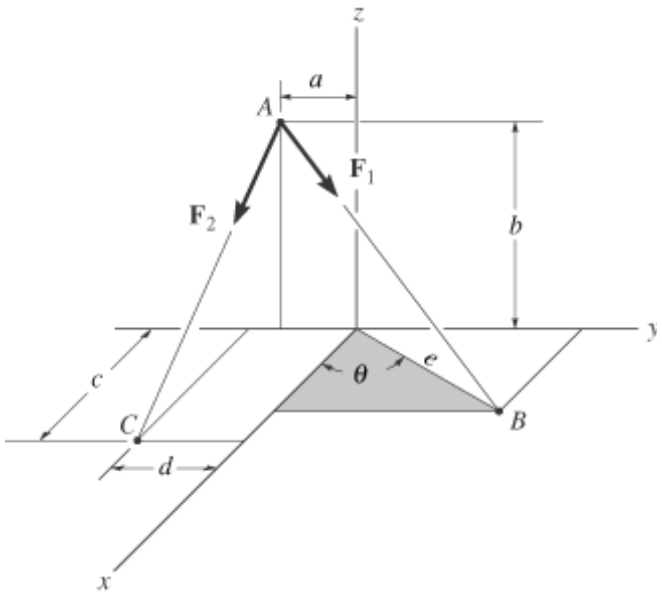
$b = 4$ m

$c = 3$ m

$d = 2$ m

$e = 3$ m

$\theta = 60$ deg



Solution:

Define the position vectors and then the forces

$$\mathbf{r_{AB}} = \begin{pmatrix} e\cos(\theta) \\ a + e\sin(\theta) \\ -b \end{pmatrix} \qquad \mathbf{F_{1v}} = F_1 \frac{\mathbf{r_{AB}}}{|\mathbf{r_{AB}}|} \qquad \mathbf{F_{1v}} = \begin{pmatrix} 38 \\ 103.8 \\ -101.4 \end{pmatrix} \text{ N}$$

$$\mathbf{r_{AC}} = \begin{pmatrix} c \\ a - d \\ -b \end{pmatrix} \qquad \mathbf{F_{2v}} = F_2 \frac{\mathbf{r_{AC}}}{|\mathbf{r_{AC}}|} \qquad \mathbf{F_{2v}} = \begin{pmatrix} 119.4 \\ -19.9 \\ -159.2 \end{pmatrix} \text{ N}$$

Add the forces and find the magnitude of the resultant

$$\mathbf{F_R} = \mathbf{F_{1v}} + \mathbf{F_{2v}} \qquad \mathbf{F_R} = \begin{pmatrix} 157.4 \\ 83.9 \\ -260.6 \end{pmatrix} \text{ N} \qquad |\mathbf{F_R}| = 316 \text{ N}$$

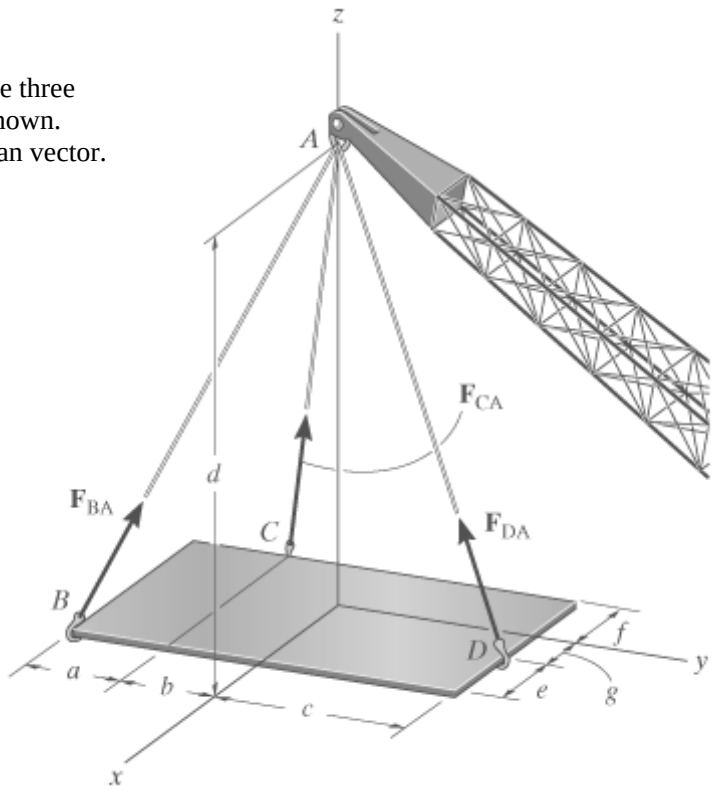
Find the direction cosine angles

$$\begin{pmatrix} \alpha \\ \beta \\ \gamma \end{pmatrix} = \operatorname{acos}\left(\frac{\mathbf{F_R}}{|\mathbf{F_R}|}\right) \qquad \begin{pmatrix} \alpha \\ \beta \\ \gamma \end{pmatrix} = \begin{pmatrix} 60.1 \\ 74.6 \\ 145.6 \end{pmatrix} \text{deg}$$

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Problem 2-93

The plate is suspended using the three cables which exert the forces shown. Express each force as a Cartesian vector.



Given:

$F_{BA} = 350$ lb

$F_{CA} = 500$ lb

$F_{DA} = 400$ lb

$a = 3$ ft

$b = 3$ ft

$c = 6$ ft

$d = 14$ ft

$e = 3$ ft

$f = 3$ ft

$g = 2$ ft

Solution:

$$\mathbf{r_{BA}} = \begin{pmatrix} -e-g \\ a+b \\ d \end{pmatrix} \qquad \mathbf{F_{BAv}} = F_{BA}\frac{\mathbf{r_{BA}}}{|\mathbf{r_{BA}}|} \qquad \mathbf{F_{BAv}} = \begin{pmatrix} -109.2 \\ 131 \\ 305.7 \end{pmatrix} \text{lb}$$

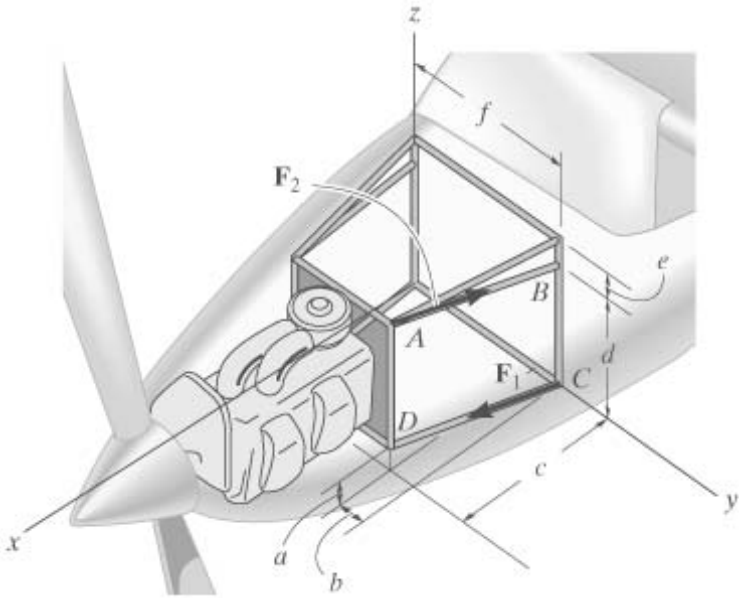
$$\mathbf{r_{CA}} = \begin{pmatrix} f \\ b \\ d \end{pmatrix} \qquad \mathbf{F_{CAv}} = F_{CA}\frac{\mathbf{r_{CA}}}{|\mathbf{r_{CA}}|} \qquad \mathbf{F_{CAv}} = \begin{pmatrix} 102.5 \\ 102.5 \\ 478.5 \end{pmatrix} \text{lb}$$

$$\mathbf{r_{DA}} = \begin{pmatrix} -g \\ -c \\ d \end{pmatrix} \qquad \mathbf{F_{DAv}} = F_{DA}\frac{\mathbf{r_{DA}}}{|\mathbf{r_{DA}}|} \qquad \mathbf{F_{DAv}} = \begin{pmatrix} -52.1 \\ -156.2 \\ 364.5 \end{pmatrix} \text{lb}$$

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Problem 2-94

The engine of the lightweight plane is supported by struts that are connected to the space truss that makes up the structure of the plane. The anticipated loading in two of the struts is shown. Express each of these forces as a Cartesian vector.



Given:

$F_1 = 400$ lb

$F_2 = 600$ lb

$a = 0.5$ ft

$b = 0.5$ ft

$c = 3.0$ ft

$d = 2.0$ ft

$e = 0.5$ ft

$f = 3.0$ ft

Solution:

$$\mathbf{r_{CD}} = \begin{pmatrix} c \\ -b \\ a \end{pmatrix} \qquad \mathbf{F_{1v}} = F_1 \frac{\mathbf{r_{CD}}}{|\mathbf{r_{CD}}|} \qquad \mathbf{F_{1v}} = \begin{pmatrix} 389.3 \\ -64.9 \\ 64.9 \end{pmatrix} \text{lb}$$

$$\mathbf{r_{AB}} = \begin{pmatrix} -c \\ b \\ -e \end{pmatrix} \qquad \mathbf{F_{2v}} = F_2 \frac{\mathbf{r_{AB}}}{|\mathbf{r_{AB}}|} \qquad \mathbf{F_{2v}} = \begin{pmatrix} -584.0 \\ 97.3 \\ -97.3 \end{pmatrix} \text{lb}$$

Problem 2-95

The window is held open by cable *AB*. Determine the length of the cable and express the force **F** acting at *A* along the cable as a Cartesian vector.

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Given:

$a = 300$ mm

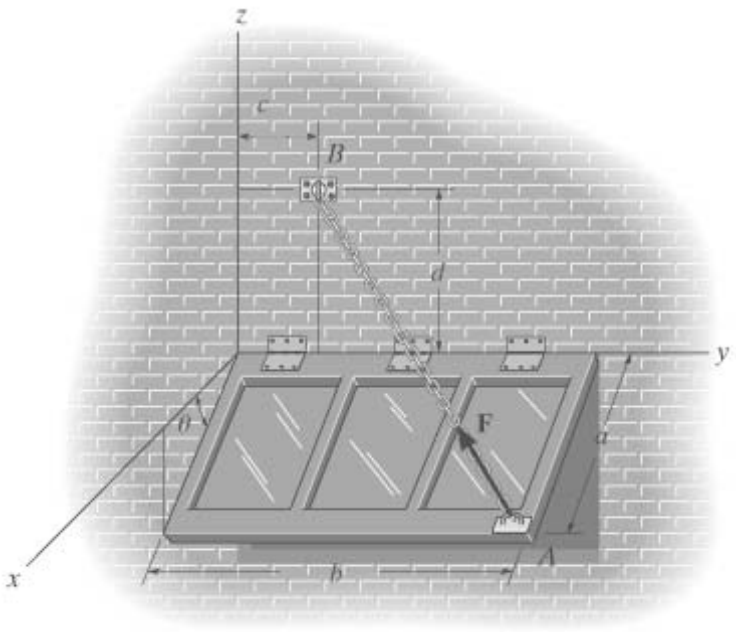
$b = 500$ mm

$c = 150$ mm

$d = 250$ mm

$\theta = 30$ deg

$F = 30$ N



Solution:

$$\mathbf{r_{AB}} = \begin{pmatrix} -a\cos(\theta) \\ c - b \\ d + a\sin(\theta) \end{pmatrix} \qquad \left|\mathbf{r_{AB}}\right| = 591.6\,\text{mm}$$

$$\mathbf{F_V} = F\frac{\mathbf{r_{AB}}}{\left|\mathbf{r_{AB}}\right|} \qquad \mathbf{F_V} = \begin{pmatrix} -13.2 \\ -17.7 \\ 20.3 \end{pmatrix}\text{ N}$$

Problem 2-96

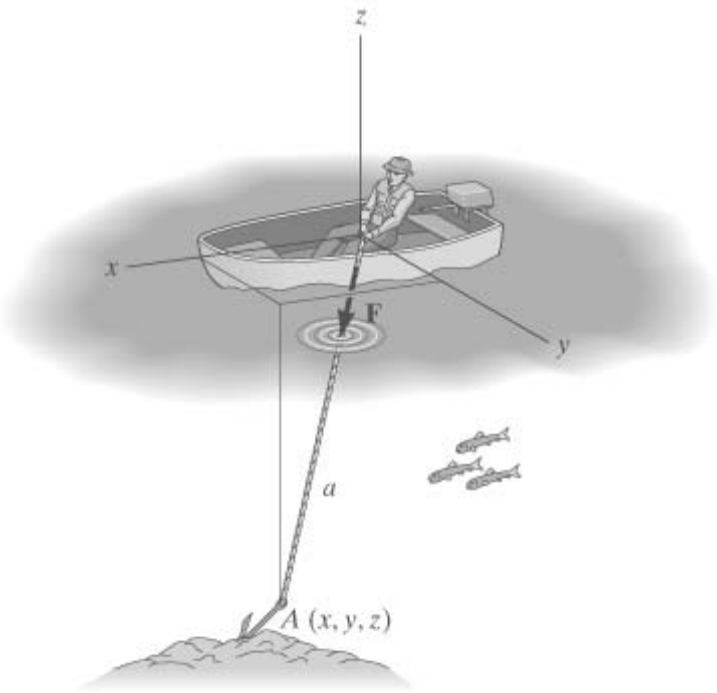
The force acting on the man, caused by his pulling on the anchor cord, is $\mathbf{F}$. If the length of the cord is L, determine the coordinates $A(x, y, z)$ of the anchor.

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Given:

$$\mathbf{F} = \begin{pmatrix} 40 \\ 20 \\ -50 \end{pmatrix} \text{ N}$$

$L = 25$ m



Solution:

$$\mathbf{r} = L\frac{\mathbf{F}}{|\mathbf{F}|}$$

$$\mathbf{r} = \begin{pmatrix} 14.9 \\ 7.5 \\ -18.6 \end{pmatrix} \text{ m}$$

Problem 2-97

Express each of the forces in Cartesian vector form and determine the magnitude and coordinate direction angles of the resultant force.

Given:

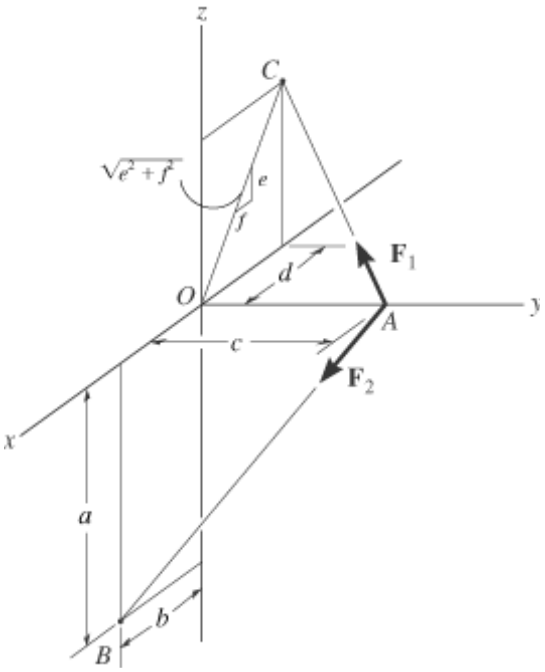
$F_1 = 80$ lb $\quad c = 4$ ft

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$F_2 = 50$ lb $\quad d = 2.5$ ft

$a = 6$ ft $\quad e = 12$

$b = 2$ ft $\quad f = 5$



Solution:

$$\mathbf{r_{AC}} = \begin{pmatrix} -d \\ -c \\ e\,\dfrac{d}{f} \end{pmatrix} \qquad \mathbf{F_{1v}} = F_1\,\frac{\mathbf{r_{AC}}}{|\mathbf{r_{AC}}|} \qquad \mathbf{F_{1v}} = \begin{pmatrix} -26.2 \\ -41.9 \\ 62.9 \end{pmatrix} \text{lb}$$

$$\mathbf{r_{AB}} = \begin{pmatrix} b \\ -c \\ -a \end{pmatrix} \qquad \mathbf{F_{2v}} = F_2\,\frac{\mathbf{r_{AB}}}{|\mathbf{r_{AB}}|} \qquad \mathbf{F_{2v}} = \begin{pmatrix} 6.1 \\ -12.1 \\ -18.2 \end{pmatrix} \text{kg}$$

$$\mathbf{F_R} = \mathbf{F_{1v}} + \mathbf{F_{2v}} \qquad \mathbf{F_R} = \begin{pmatrix} -12.8 \\ -68.7 \\ 22.8 \end{pmatrix} \text{lb} \qquad |\mathbf{F_R}| = 73.5\,\text{lb}$$

$$\begin{pmatrix} \alpha \\ \beta \\ \gamma \end{pmatrix} = \operatorname{acos}\left(\frac{\mathbf{F_R}}{|\mathbf{F_R}|}\right) \qquad \begin{pmatrix} \alpha \\ \beta \\ \gamma \end{pmatrix} = \begin{pmatrix} 1.7 \\ 2.8 \\ 1.3 \end{pmatrix}$$

Problem 2-98

The cable attached to the tractor at B exerts force $\mathbf{F}$ on the framework. Express this force as a Cartesian vector

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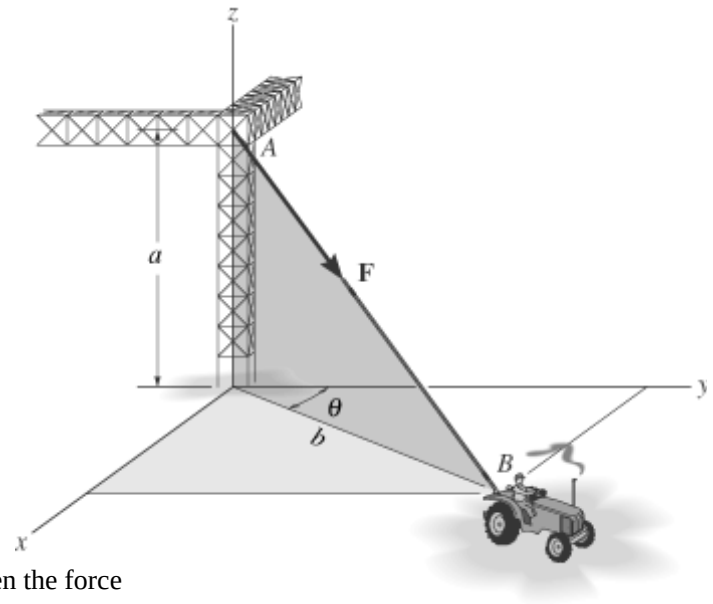
Given:

$F = 350$ lb

$a = 35$ ft

$b = 50$ ft

$\theta = 20$ deg



Solution:

Find the position vector and then the force

$$\mathbf{r_{AB}} = \begin{pmatrix} b\sin(\theta) \\ b\cos(\theta) \\ -a \end{pmatrix} \qquad \mathbf{F_v} = F\frac{\mathbf{r_{AB}}}{|\mathbf{r_{AB}}|} \qquad \mathbf{F_v} = \begin{pmatrix} 98.1 \\ 269.4 \\ -200.7 \end{pmatrix} \text{lb}$$

Problem 2-99

The cable *OA* exerts force **F** on point *O*. If the length of the cable is *L*, what are the coordinates (*x*, *y*, *z*) of point *A*?

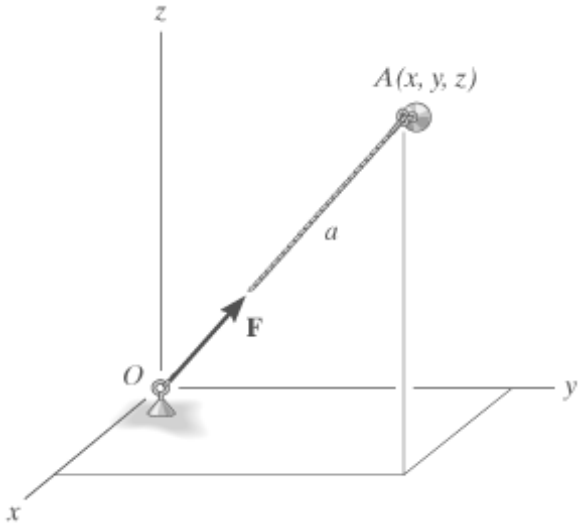
Given:

$$\mathbf{F} = \begin{pmatrix} 40 \\ 60 \\ 70 \end{pmatrix} \text{N}$$

$L = 3$ m

Solution:

$$\mathbf{r} = L\frac{\mathbf{F}}{|\mathbf{F}|}$$



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$$\mathbf{r} = \begin{pmatrix} 1.2 \\ 1.8 \\ 2.1 \end{pmatrix} \text{ m}$$

Problem 2-100

Determine the position $(x, y, 0)$ for fixing cable BA so that the resultant of the forces exerted on the pole is directed along its axis, from B toward O, and has magnitude F_R. Also, what is the magnitude of force $\mathbf{F}_3$?

Given:

$F_1 = 500$ N

$F_2 = 400$ N

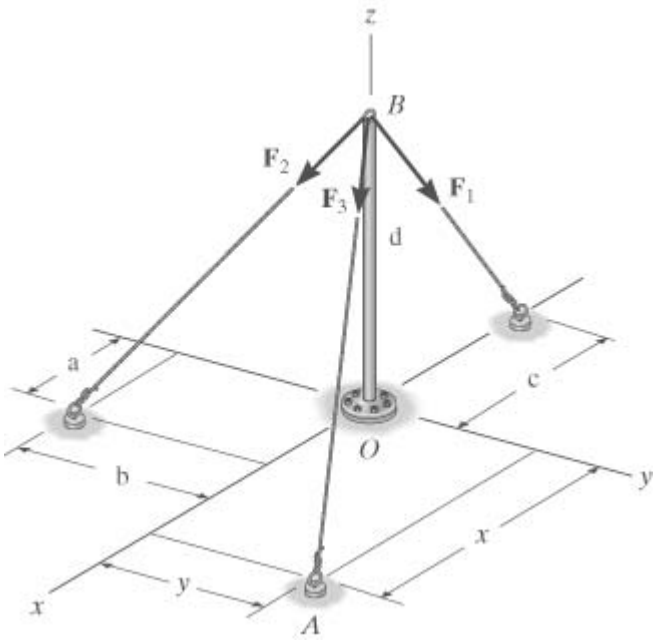
$F_R = 1000$ N

$a = 1$ m

$b = 2$ m

$c = 2$ m

$d = 3$ m



Solution:

Initial Guesses

$F_3 = 1$ N $\quad x = 1$ m $\quad y = 1$ m

Given

$$F_R \begin{pmatrix} 0 \\ 0 \\ -1 \end{pmatrix} = \left(\frac{F_1}{\sqrt{c^2 + d^2}}\right) \begin{pmatrix} -c \\ 0 \\ -d \end{pmatrix} + \left(\frac{F_2}{\sqrt{a^2 + b^2 + d^2}}\right) \begin{pmatrix} a \\ -b \\ -d \end{pmatrix} + \left(\frac{F_3}{\sqrt{x^2 + y^2 + d^2}}\right) \begin{pmatrix} x \\ y \\ -d \end{pmatrix}$$

$$\begin{pmatrix} F_3 \\ x \\ y \end{pmatrix} = \text{Find}(F_3, x, y) \qquad \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1.9 \\ 2.4 \end{pmatrix} \text{ m} \qquad F_3 = 380 \text{ N}$$

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Problem 2-101

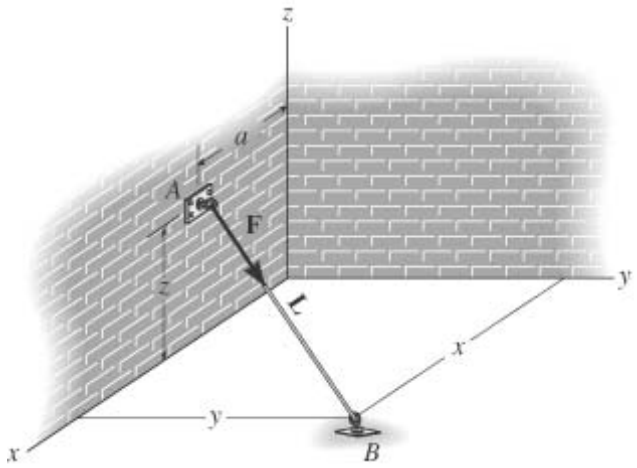
The cord exerts a force **F** on the hook. If the cord is length L, determine the location x, y of the point of attachment B, and the height z of the hook.

Given:

$$\mathbf{F} = \begin{pmatrix} 12 \\ 9 \\ -8 \end{pmatrix} \text{ lb}$$

$L = 8$ ft

$a = 2$ ft



Solution:

Initial guesses $x = 1$ ft $y = 1$ ft $z = 1$ ft

Given $\begin{pmatrix} x - a \\ y \\ -z \end{pmatrix} = L \dfrac{\mathbf{F}}{|\mathbf{F}|}$ $\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \text{Find}(x, y, z)$ $\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 7.65 \\ 4.24 \\ 3.76 \end{pmatrix}$ ft

---

Problem 2-102

The cord exerts a force of magnitude F on the hook. If the cord length L, the distance z, and the x component of the force, F_x, are given, determine the location x, y of the point of attachment B of the cord to the ground.

Given:

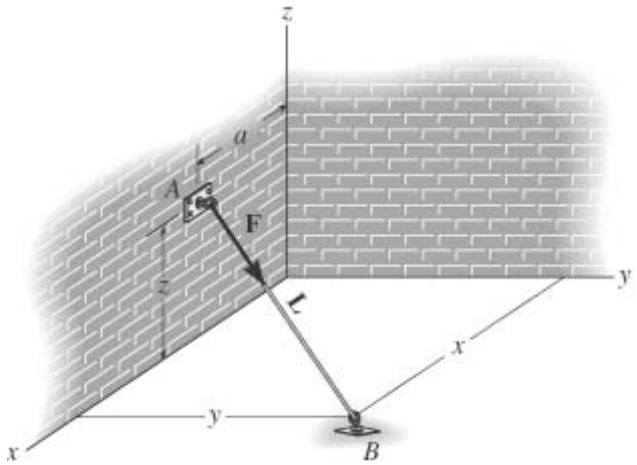
$F = 30$ lb

$L = 8$ ft

$z = 4$ ft

$F_x = 25$ lb

$a = 2$ ft



Solution:

Guesses

$x = 1$ ft

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$y = 1$ ft

Given

$$F_x = \left(\frac{x-a}{L}\right)F \qquad L^2 = (x-a)^2 + y^2 + z^2$$

$$\begin{pmatrix} x \\ y \end{pmatrix} = \text{Find}(x, y) \qquad \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 8.67 \\ 1.89 \end{pmatrix} \text{ft}$$

Problem 2-103

Each of the four forces acting at E has magnitude F. Express each force as a Cartesian vector and determine the resultant force.

Units used:

$\text{kN} = 10^3$ N

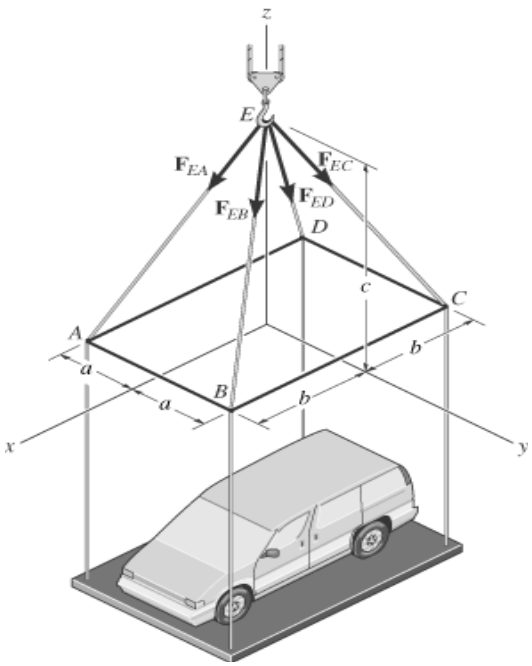
Given:

$F = 28$ kN

$a = 4$ m

$b = 6$ m

$c = 12$ m



Solution:

Find the position vectors and then the forces

$$\mathbf{r_{EA}} = \begin{pmatrix} b \\ -a \\ -c \end{pmatrix} \qquad \mathbf{F_{EA}} = F\frac{\mathbf{r_{EA}}}{|\mathbf{r_{EA}}|} \qquad \mathbf{F_{EA}} = \begin{pmatrix} 12 \\ -8 \\ -24 \end{pmatrix} \text{kN}$$

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$$\mathbf{r_{EB}} = \begin{pmatrix} b \\ a \\ -c \end{pmatrix} \qquad \mathbf{F_{EB}} = F\frac{\mathbf{r_{EB}}}{|\mathbf{r_{EB}}|} \qquad \mathbf{F_{EB}} = \begin{pmatrix} 12 \\ 8 \\ -24 \end{pmatrix} \text{kN}$$

$$\mathbf{r_{EC}} = \begin{pmatrix} -b \\ a \\ -c \end{pmatrix} \qquad \mathbf{F_{EC}} = F\frac{\mathbf{r_{EC}}}{|\mathbf{r_{EC}}|} \qquad \mathbf{F_{EC}} = \begin{pmatrix} -12 \\ 8 \\ -24 \end{pmatrix} \text{kN}$$

$$\mathbf{r_{ED}} = \begin{pmatrix} -b \\ -a \\ -c \end{pmatrix} \qquad \mathbf{F_{ED}} = F\frac{\mathbf{r_{ED}}}{|\mathbf{r_{ED}}|} \qquad \mathbf{F_{ED}} = \begin{pmatrix} -12 \\ -8 \\ -24 \end{pmatrix} \text{kN}$$

Find the resultant sum

$$\mathbf{F_R} = \mathbf{F_{EA}} + \mathbf{F_{EB}} + \mathbf{F_{EC}} + \mathbf{F_{ED}} \qquad \mathbf{F_R} = \begin{pmatrix} 0 \\ 0 \\ -96 \end{pmatrix} \text{kN}$$

Problem 2-104

The tower is held in place by three cables. If the force of each cable acting on the tower is shown, determine the magnitude and coordinate direction angles α, β, γ of the resultant force.

Units Used:

$\text{kN} = 10^3 \text{ N}$

Given:

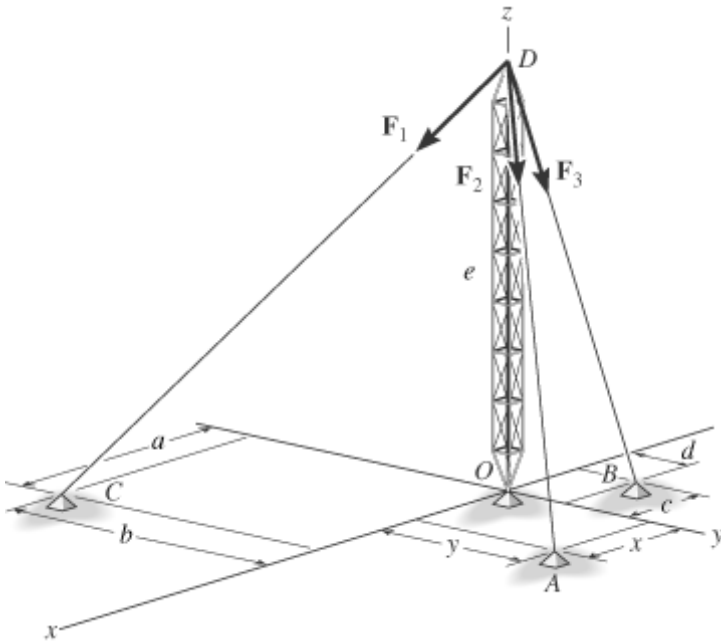
$x = 20$ m $\quad a = 16$ m

$y = 15$ m $\quad b = 18$ m

$F_1 = 600$ N $\quad c = 6$ m

$F_2 = 400$ N $\quad d = 4$ m

$F_3 = 800$ N $\quad e = 24$ m



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Solution:

Find the position vectors, then the force vectors

$$\mathbf{r_{DC}} = \begin{pmatrix} a \\ -b \\ -e \end{pmatrix} \qquad \mathbf{F_{1v}} = F_1 \frac{\mathbf{r_{DC}}}{|\mathbf{r_{DC}}|} \qquad \mathbf{F_{1v}} = \begin{pmatrix} 282.4 \\ -317.6 \\ -423.5 \end{pmatrix} \text{N}$$

$$\mathbf{r_{DA}} = \begin{pmatrix} x \\ y \\ -e \end{pmatrix} \qquad \mathbf{F_{2v}} = F_2 \frac{\mathbf{r_{DA}}}{|\mathbf{r_{DA}}|} \qquad \mathbf{F_{2v}} = \begin{pmatrix} 230.8 \\ 173.1 \\ -277 \end{pmatrix} \text{N}$$

$$\mathbf{r_{DB}} = \begin{pmatrix} -c \\ d \\ -e \end{pmatrix} \qquad \mathbf{F_{3v}} = F_3 \frac{\mathbf{r_{DB}}}{|\mathbf{r_{DB}}|} \qquad \mathbf{F_{3v}} = \begin{pmatrix} -191.5 \\ 127.7 \\ -766.2 \end{pmatrix} \text{N}$$

Find the resultant, magnitude, and direction angles

$$\mathbf{F_R} = \mathbf{F_{1v}} + \mathbf{F_{2v}} + \mathbf{F_{3v}} \qquad \mathbf{F_R} = \begin{pmatrix} 0.322 \\ -0.017 \\ -1.467 \end{pmatrix} \text{kN} \qquad |\mathbf{F_R}| = 1.502\,\text{kN}$$

$$\begin{pmatrix} \alpha \\ \beta \\ \gamma \end{pmatrix} = \text{acos}\left(\frac{\mathbf{F_R}}{|\mathbf{F_R}|}\right) \qquad \begin{pmatrix} \alpha \\ \beta \\ \gamma \end{pmatrix} = \begin{pmatrix} 77.6 \\ 90.6 \\ 167.6 \end{pmatrix} \text{deg}$$

Problem 2-105

The chandelier is supported by three chains which are concurrent at point *O*. If the force in each chain has magnitude *F*, express each force as a Cartesian vector and determine the magnitude and coordinate direction angles of the resultant force.

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Given:

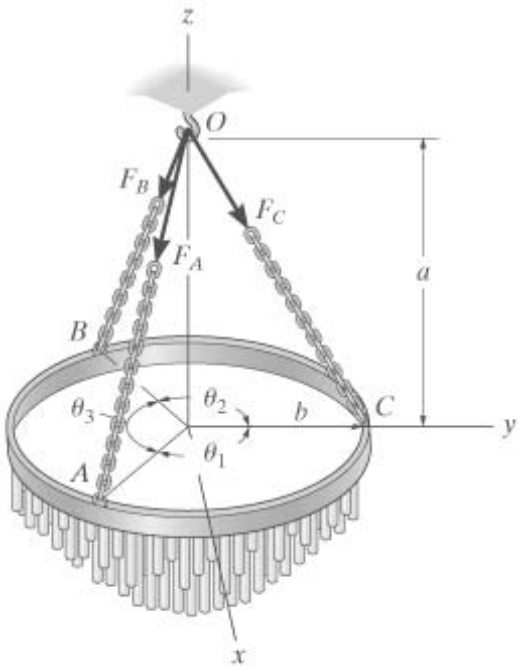
$F = 60$ lb

$a = 6$ ft

$b = 4$ ft

$\theta_1 = 120$ deg

$\theta_2 = 120$ deg



Solution:

$\theta_3 = 360 \text{ deg} - \theta_1 - \theta_2$

$$\mathbf{r_{OA}} = \begin{pmatrix} b\sin(\theta_1) \\ b\cos(\theta_1) \\ -a \end{pmatrix} \qquad \mathbf{F_A} = F\frac{\mathbf{r_{OA}}}{|\mathbf{r_{OA}}|} \qquad \mathbf{F_A} = \begin{pmatrix} 28.8 \\ -16.6 \\ -49.9 \end{pmatrix} \text{lb}$$

$$\mathbf{r_{OB}} = \begin{pmatrix} b\sin(\theta_1 + \theta_2) \\ b\cos(\theta_1 + \theta_2) \\ -a \end{pmatrix} \qquad \mathbf{F_B} = F\frac{\mathbf{r_{OB}}}{|\mathbf{r_{OB}}|} \qquad \mathbf{F_B} = \begin{pmatrix} -28.8 \\ -16.6 \\ -49.9 \end{pmatrix} \text{lb}$$

$$\mathbf{r_{OC}} = \begin{pmatrix} 0 \\ b \\ -a \end{pmatrix} \qquad \mathbf{F_C} = F\frac{\mathbf{r_{OC}}}{|\mathbf{r_{OC}}|} \qquad \mathbf{F_C} = \begin{pmatrix} 0 \\ 33.3 \\ -49.9 \end{pmatrix} \text{lb}$$

$$\mathbf{F_R} = \mathbf{F_A} + \mathbf{F_B} + \mathbf{F_C} \qquad |\mathbf{F_R}| = 149.8 \text{ lb}$$

$$\begin{pmatrix} \alpha \\ \beta \\ \gamma \end{pmatrix} = \text{acos}\left(\frac{\mathbf{F_R}}{|\mathbf{F_R}|}\right) \qquad \begin{pmatrix} \alpha \\ \beta \\ \gamma \end{pmatrix} = \begin{pmatrix} 90 \\ 90 \\ 180 \end{pmatrix} \text{deg}$$

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Problem 2-106

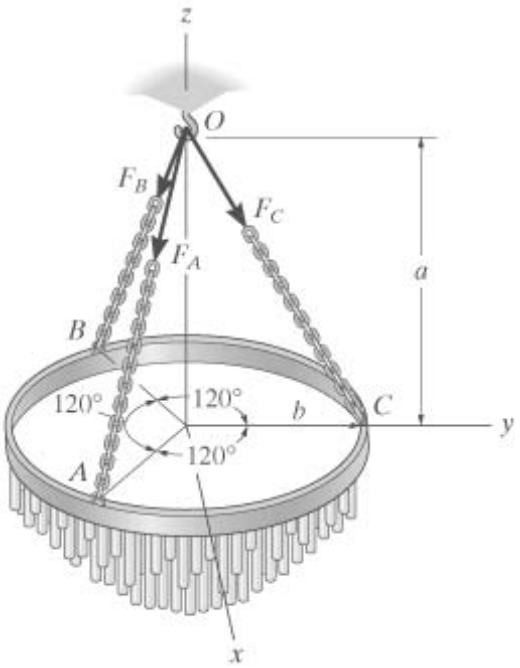
The chandelier is supported by three chains which are concurrent at point O. If the resultant force at O has magnitude F_R and is directed along the negative z axis, determine the force in each chain assuming $F_A = F_B = F_C = F$.

Given:

$a = 6$ ft

$b = 4$ ft

$F_R = 130$ lb



Solution:

$$F = \frac{\sqrt{a^2 + b^2}}{3a} F_R$$

$F = 52.1$ lb

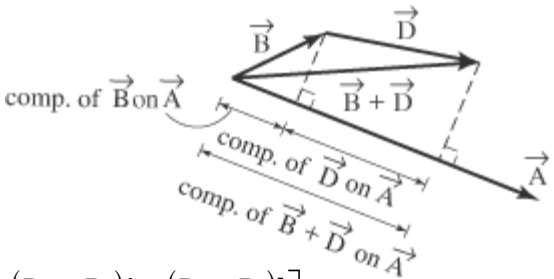
Problem 2-107

Given the three vectors **A**, **B**, and **D**, show that $\mathbf{A}\cdot(\mathbf{B} + \mathbf{D}) = (\mathbf{A}\cdot\mathbf{B}) + (\mathbf{A}\cdot\mathbf{D})$.

Solution:

Since the component of **(B + D)** is equal to the sum of the components of **B** and **D**, then

$\mathbf{A}\cdot(\mathbf{B} + \mathbf{D}) = \mathbf{A}\cdot\mathbf{B} + \mathbf{A}\cdot\mathbf{D}$ (QED)



Also,

$$\mathbf{A}\cdot(\mathbf{B} + \mathbf{D}) = (A_x\mathbf{i} + A_y\mathbf{j} + A_z\mathbf{k})[(B_x + D_x)\mathbf{i} + (B_y + D_y)\mathbf{j} + (B_z + D_z)\mathbf{k}]$$

$$= A_x(B_x + D_x) + A_y(B_y + D_y) + A_z(B_z + D_z)$$

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$$= \left(A_x B_x + A_y B_y + A_z B_z\right) + \left(A_x D_x + A_y D_y + A_z D_z\right)$$

$$= \left(\mathbf{A}\cdot\mathbf{B}\right) + \left(\mathbf{A}\cdot\mathbf{D}\right) \qquad \text{(QED)}$$

Problem 2-108

Cable *BC* exerts force **F** on the top of the flagpole. Determine the projection of this force along the *z* axis of the pole.

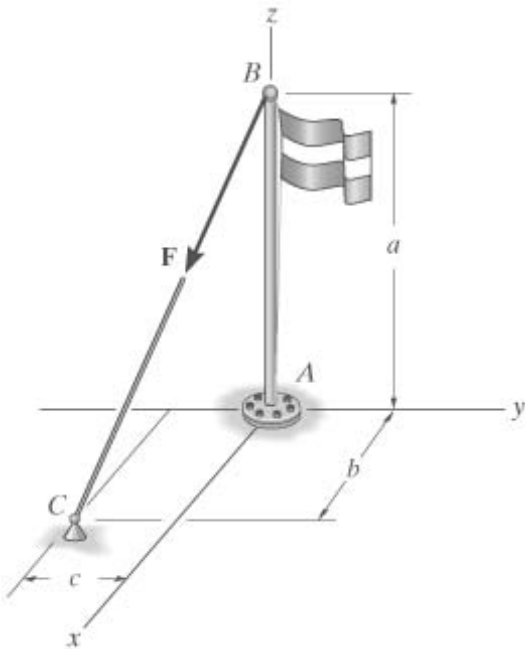
Given:

$F = 28$ N

$a = 12$ m

$b = 6$ m

$c = 4$ m



Solution:

$$\mathbf{r_{BC}} = \begin{pmatrix} b \\ -c \\ -a \end{pmatrix} \qquad \mathbf{k} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

$$\mathbf{F_V} = F\frac{\mathbf{r_{BC}}}{\left|\mathbf{r_{BC}}\right|}$$

$$F_z = -\mathbf{F_V}\,\mathbf{k} \qquad F_z = 24 \text{ N}$$

Problem 2-109

Determine the angle θ between the tails of the two vectors.

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Given:

$r_1 = 9$ m

$r_2 = 6$ m

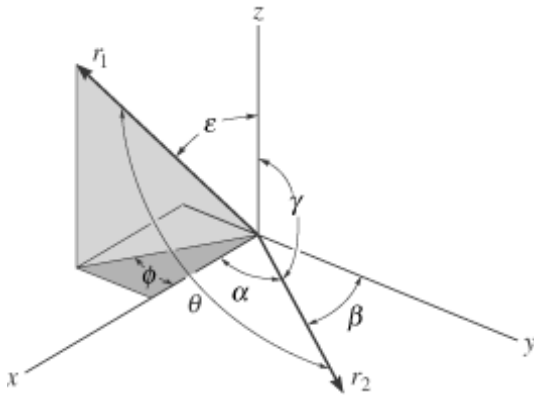
$\alpha = 60$ deg

$\beta = 45$ deg

$\gamma = 120$ deg

$\phi = 30$ deg

$\varepsilon = 40$ deg



Solution:

Determine the two position vectors and use the dot product to find the angle

$$\mathbf{r_{1v}} = r_1 \begin{pmatrix} \sin(\varepsilon)\cos(\phi) \\ -\sin(\varepsilon)\sin(\phi) \\ \cos(\varepsilon) \end{pmatrix} \qquad \mathbf{r_{2v}} = r_2 \begin{pmatrix} \cos(\alpha) \\ \cos(\beta) \\ \cos(\gamma) \end{pmatrix}$$

$$\theta = \operatorname{acos}\left(\frac{\mathbf{r_{1v}} \cdot \mathbf{r_{2v}}}{|\mathbf{r_{1v}}|\ |\mathbf{r_{2v}}|} \right) \qquad \theta = 109.4\,\text{deg}$$

Problem 2-110

Determine the magnitude of the projected component of $\mathbf{r_1}$ along $\mathbf{r_2}$, and the projection of $\mathbf{r_2}$ along $\mathbf{r_1}$.

Given:

$r_1 = 9$ m

$r_2 = 6$ m

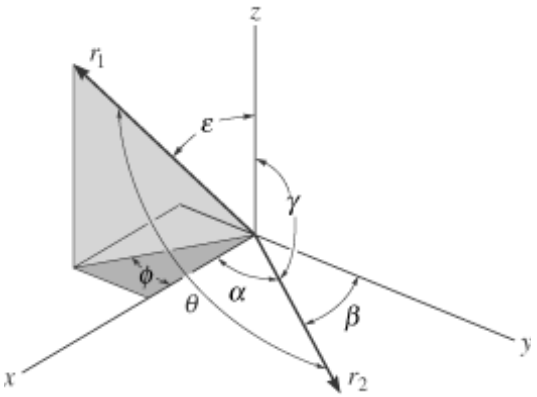
$\alpha = 60$ deg

$\beta = 45$ deg

$\gamma = 120$ deg

$\phi = 30$ deg

$\varepsilon = 40$ deg



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Solution:

Write the vectors and unit vectors

$$\mathbf{r_{1v}} = r_1 \begin{pmatrix} \sin(\varepsilon)\cos(\phi) \\ -\sin(\varepsilon)\sin(\phi) \\ \cos(\varepsilon) \end{pmatrix} \qquad \mathbf{r_{1v}} = \begin{pmatrix} 5.01 \\ -2.89 \\ 6.89 \end{pmatrix} \text{ m}$$

$$\mathbf{r_{2v}} = r_2 \begin{pmatrix} \cos(\alpha) \\ \cos(\beta) \\ \cos(\gamma) \end{pmatrix} \qquad \mathbf{r_{2v}} = \begin{pmatrix} 3 \\ 4.24 \\ -3 \end{pmatrix} \text{ m}$$

$$\mathbf{u_1} = \frac{\mathbf{r_{1v}}}{|\mathbf{r_{1v}}|} \qquad \mathbf{u_2} = \frac{\mathbf{r_{2v}}}{|\mathbf{r_{2v}}|} \qquad \mathbf{u_1} = \begin{pmatrix} 0.557 \\ -0.321 \\ 0.766 \end{pmatrix} \qquad \mathbf{u_2} = \begin{pmatrix} 0.5 \\ 0.707 \\ -0.5 \end{pmatrix}$$

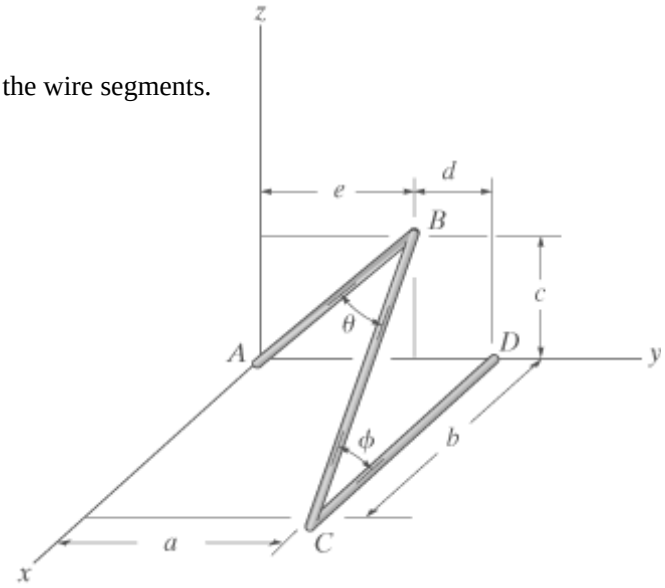
The magnitude of the projection of $\mathbf{r_1}$ along $\mathbf{r_2}$. $|\mathbf{r_{1v}} \cdot \mathbf{u_2}| = 2.99$ m

The magnitude of the projection of $\mathbf{r_2}$ along $\mathbf{r_1}$. $|\mathbf{r_{2v}} \cdot \mathbf{u_1}| = 1.99$ m

---

Problem 2-111

Determine the angles θ and ϕ between the wire segments.



Given:

$a = 0.6$

$b = 0.8$

$c = 0.5$

$d = 0.2$

Solution:

$e = a - d$

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$$\mathbf{r_{BA}} = \begin{pmatrix} 0 \\ -e \\ -c \end{pmatrix} \text{ m} \qquad \mathbf{r_{BC}} = \begin{pmatrix} b \\ d \\ -c \end{pmatrix} \text{ ft} \qquad \theta = \operatorname{acos}\left(\frac{\mathbf{r_{BA}} \cdot \mathbf{r_{BC}}}{|\mathbf{r_{BA}}|\ |\mathbf{r_{BC}}|}\right) \qquad \theta = 74.0\,\text{deg}$$

$$\mathbf{r_{CB}} = \begin{pmatrix} -b \\ -d \\ c \end{pmatrix} \text{ m} \qquad \mathbf{r_{CD}} = \begin{pmatrix} -b \\ 0 \\ 0 \end{pmatrix} \text{ ft} \qquad \phi = \operatorname{acos}\left(\frac{\mathbf{r_{CB}} \cdot \mathbf{r_{CD}}}{|\mathbf{r_{CB}}|\ |\mathbf{r_{CD}}|}\right) \qquad \phi = 33.9\,\text{deg}$$

Problem 2-112

Determine the angle θ between the two cords.

Given:

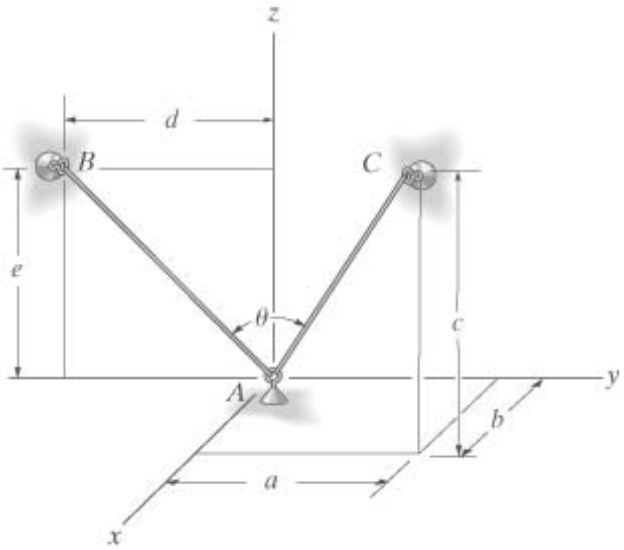
$a = 3$ m

$b = 2$ m

$c = 6$ m

$d = 3$ m

$e = 4$ m



Solution:

$$\mathbf{r_{AC}} = \begin{pmatrix} b \\ a \\ c \end{pmatrix} \text{ ft} \qquad \mathbf{r_{AB}} = \begin{pmatrix} 0 \\ -d \\ e \end{pmatrix} \text{ ft} \qquad \theta = \operatorname{acos}\left(\frac{\mathbf{r_{AC}} \cdot \mathbf{r_{AB}}}{|\mathbf{r_{AC}}|\ |\mathbf{r_{AB}}|}\right) \qquad \theta = 64.6\,\text{deg}$$

Problem 2-113

Determine the angle θ between the two cables.

Given:

$a = 8$ ft

$b = 10$ ft

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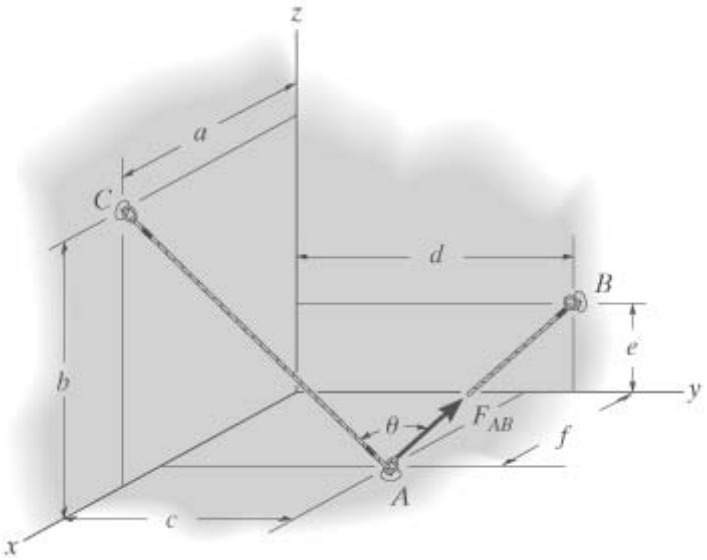
$c = 8$ ft

$d = 10$ ft

$e = 4$ ft

$f = 6$ ft

$F_{AB} = 12$ lb



Solution:

$$\mathbf{r_{AC}} = \begin{pmatrix} a - f \\ -c \\ b \end{pmatrix} \text{ ft} \qquad \mathbf{r_{AB}} = \begin{pmatrix} -f \\ d - c \\ e \end{pmatrix} \text{ ft} \qquad \theta = \operatorname{acos}\left(\frac{\mathbf{r_{AC}} \cdot \mathbf{r_{AB}}}{|\mathbf{r_{AC}}|\ |\mathbf{r_{AB}}|} \right) \qquad \theta = 82.9\,\text{deg}$$

Problem 2-114

Determine the projected component of the force **F** acting in the direction of cable *AC*. Express the result as a Cartesian vector.

Given:

$F = 12$ lb

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$a = 8$ ft

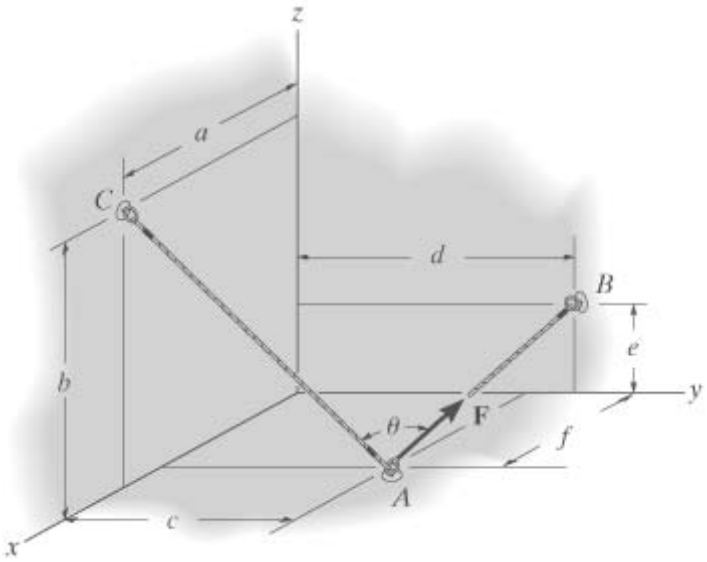
$b = 10$ ft

$c = 8$ ft

$d = 10$ ft

$e = 4$ ft

$f = 6$ ft



Solution:

$$\mathbf{r_{AC}} = \begin{pmatrix} a - f \\ -c \\ b \end{pmatrix} \text{ m} \qquad \mathbf{u_{AC}} = \frac{\mathbf{r_{AC}}}{|\mathbf{r_{AC}}|} \qquad \mathbf{u_{AC}} = \begin{pmatrix} 0.2 \\ -0.6 \\ 0.8 \end{pmatrix}$$

$$\mathbf{r_{AB}} = \begin{pmatrix} -f \\ d - c \\ e \end{pmatrix} \qquad \mathbf{F_{AB}} = F \frac{\mathbf{r_{AB}}}{|\mathbf{r_{AB}}|} \qquad \mathbf{F_{AB}} = \begin{pmatrix} -9.6 \\ 3.2 \\ 6.4 \end{pmatrix} \text{ lb}$$

$$\mathbf{F_{AC}} = \left(\mathbf{F_{AB}} \cdot \mathbf{u_{AC}}\right) \mathbf{u_{AC}} \qquad \mathbf{F_{AC}} = \begin{pmatrix} 0.229 \\ -0.916 \\ 1.145 \end{pmatrix} \text{ lb}$$

Problem 2-115

Determine the components of $\mathbf{F}$ that act along rod AC and perpendicular to it. Point B is located at the midpoint of the rod.

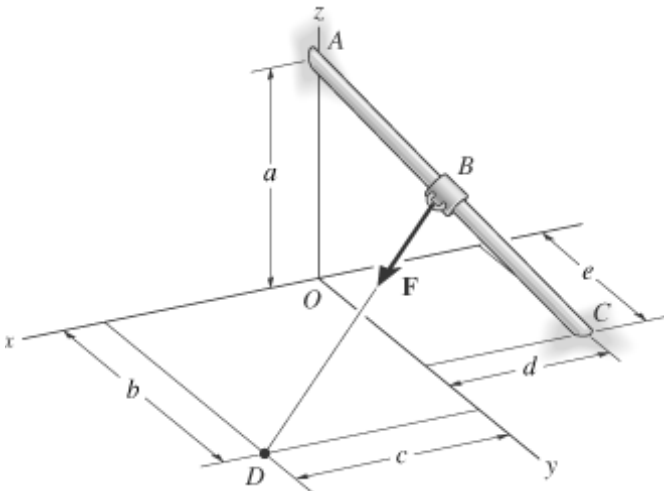
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Given:

$F = 600$ N $\quad c = 4$ m

$a = 4$ m $\quad d = 3$ m

$b = 6$ m $\quad e = 4$ m



Solution:

Find the force vector and the unit vector $\mathbf{u}_{\mathbf{AC}}$.

$$\mathbf{r_{BD}} = \begin{pmatrix} c + \frac{d}{2} \\ b - \frac{e}{2} \\ \frac{-a}{2} \end{pmatrix} \qquad \mathbf{r_{BD}} = \begin{pmatrix} 5.5 \\ 4 \\ -2 \end{pmatrix} \text{ m} \qquad \mathbf{F_V} = F \frac{\mathbf{r_{BD}}}{|\mathbf{r_{BD}}|} \qquad \mathbf{F_V} = \begin{pmatrix} 465.5 \\ 338.6 \\ -169.3 \end{pmatrix} \text{ N}$$

$$\mathbf{r_{AC}} = \begin{pmatrix} -d \\ e \\ -a \end{pmatrix} \qquad \mathbf{r_{AC}} = \begin{pmatrix} -3 \\ 4 \\ -4 \end{pmatrix} \text{ m} \qquad \mathbf{u_{AC}} = \frac{\mathbf{r_{AC}}}{|\mathbf{r_{AC}}|} \qquad \mathbf{u_{AC}} = \begin{pmatrix} -0.5 \\ 0.6 \\ -0.6 \end{pmatrix}$$

Now find the component parallel to *AC*.

$$F_{parallel} = \mathbf{F_V} \cdot \mathbf{u_{AC}} \qquad F_{parallel} = 99.1 \text{ N}$$

The perpendicular component is now found

$$F_{perpendicular} = \sqrt{\mathbf{F_V} \cdot \mathbf{F_V} - F_{parallel}^2} \qquad F_{perpendicular} = 591.8 \text{ N}$$

Problem 2-116

Determine the components of $\mathbf{F}$ that act along rod *AC* and perpendicular to it. Point *B* is located a distance *f* along the rod from end *C*.

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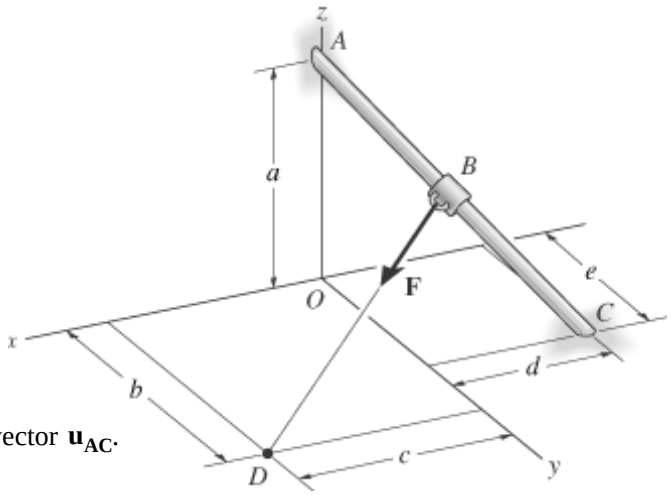
Given:

$F = 600$ N $\quad c = 4$ m

$a = 4$ m $\quad d = 3$ m

$b = 6$ m $\quad e = 4$ m

$f = 3$ m



Solution:

$$r = \frac{f}{\sqrt{d^2 + e^2 + a^2}}$$

Find the force vector and the unit vector $\mathbf{u_{AC}}$.

$$\mathbf{r_{BD}} = \begin{bmatrix} c + d(1 - r) \\ b - e(1 - r) \\ -a\,r \end{bmatrix} \qquad \mathbf{r_{BD}} = \begin{pmatrix} 5.5944 \\ 3.8741 \\ -1.8741 \end{pmatrix} \text{ m}$$

$$\mathbf{F_V} = F\frac{\mathbf{r_{BD}}}{|\mathbf{r_{BD}}|} \qquad \mathbf{F_V} = \begin{pmatrix} 475.6 \\ 329.3 \\ -159.3 \end{pmatrix} \text{ N}$$

$$\mathbf{r_{AC}} = \begin{pmatrix} -d \\ e \\ -a \end{pmatrix} \qquad \mathbf{r_{AC}} = \begin{pmatrix} -3 \\ 4 \\ -4 \end{pmatrix} \text{ m} \qquad \mathbf{u_{AC}} = \frac{\mathbf{r_{AC}}}{|\mathbf{r_{AC}}|} \qquad \mathbf{u_{AC}} = \begin{pmatrix} -0.5 \\ 0.6 \\ -0.6 \end{pmatrix}$$

Now find the component parallel to *AC*.

$$F_{parallel} = \mathbf{F_V} \cdot \mathbf{u_{AC}} \qquad F_{parallel} = 82.4 \text{ N}$$

The perpendicular component is now found

$$F_{perpendicular} = \sqrt{\mathbf{F_V} \cdot \mathbf{F_V} - F_{parallel}^2} \qquad F_{perpendicular} = 594.3 \text{ N}$$

Problem 2-117

Determine the magnitude of the projected component of the length of cord *OA* along the *Oa* axis.

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Given:

$a = 10$ ft

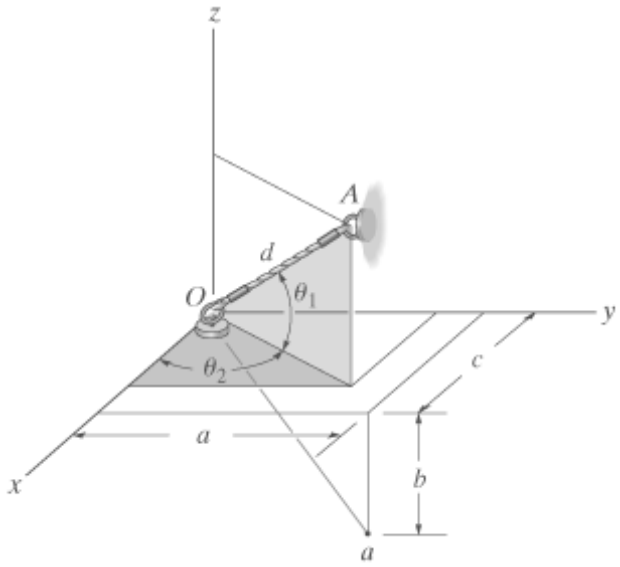
$b = 5$ ft

$c = 15$ ft

$d = 5$ ft

$\theta_1 = 45$ deg

$\theta_2 = 60$ deg



Solution:

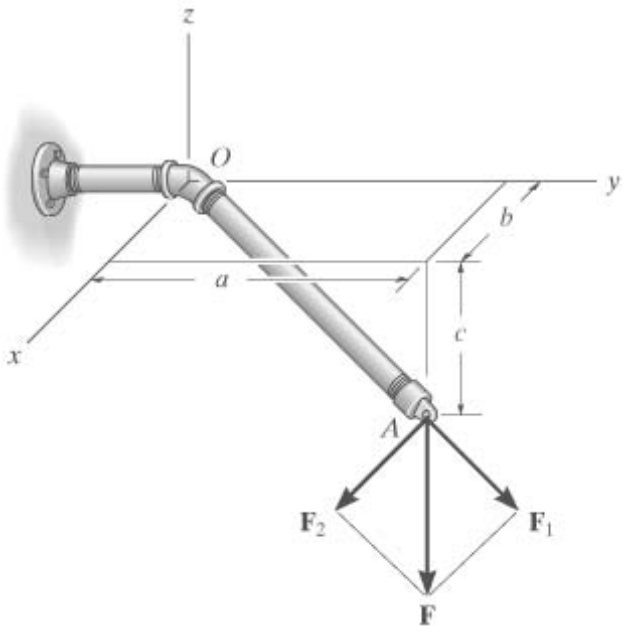
$$\mathbf{r_{OA}} = d\begin{pmatrix}\cos(\theta_1)\cos(\theta_2)\\ \cos(\theta_1)\sin(\theta_2)\\ \sin(\theta_1)\end{pmatrix} \qquad \mathbf{r_{Oa}} = \begin{pmatrix}c\\ a\\ -b\end{pmatrix} \qquad \mathbf{u_{Oa}} = \frac{\mathbf{r_{Oa}}}{|\mathbf{r_{Oa}}|}$$

$r_{Oa} = \mathbf{r_{OA}} \cdot \mathbf{u_{Oa}}$ $\qquad r_{Oa} = 2.1$ ft

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Problem 2-118

Force **F** acts at the end of the pipe. Determine the magnitudes of the components $\mathbf{F_1}$ and $\mathbf{F_2}$ which are directed along the pipe's axis and perpendicular to it.



Given:

$$\mathbf{F} = \begin{pmatrix} 0 \\ 0 \\ -40 \end{pmatrix} \text{ lb} \qquad a = 5 \text{ ft} \qquad b = 3 \text{ ft} \qquad c = 3 \text{ ft}$$

Solution:

$$\mathbf{r} = \begin{pmatrix} b \\ a \\ -c \end{pmatrix} \qquad \mathbf{u} = \frac{\mathbf{r}}{|\mathbf{r}|}$$

$$F_1 = \mathbf{F}\cdot\mathbf{u} \qquad F_1 = 18.3 \text{ lb}$$

$$F_2 = \sqrt{\mathbf{F}\cdot\mathbf{F} - F_1^2} \qquad F_2 = 35.6 \text{ lb}$$

Problem 2-119

Determine the projected component of the force **F** acting along the axis *AB* of the pipe.

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Given:

$F = 80$ N

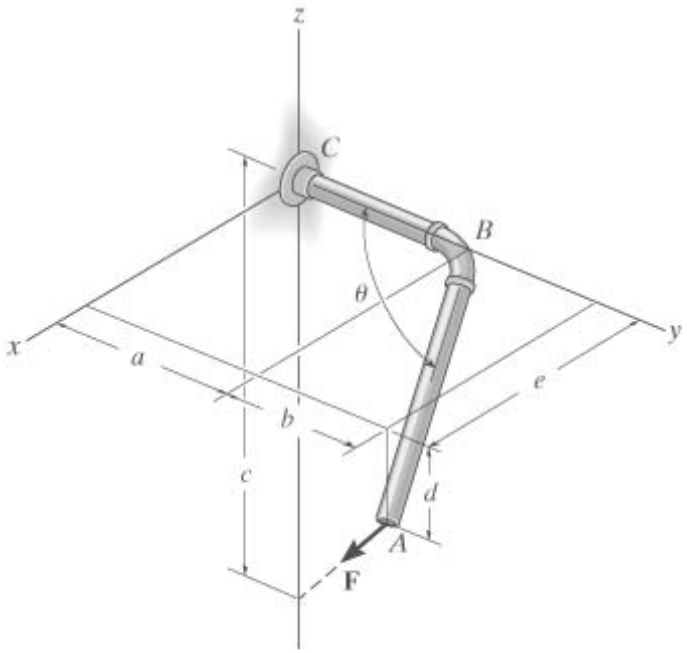
$a = 4$ m

$b = 3$ m

$c = 12$ m

$d = 2$ m

$e = 6$ m



Solution:

Find the force and the unit vector

$$\mathbf{r_A} = \begin{pmatrix} -e \\ -a-b \\ d-c \end{pmatrix} \qquad \mathbf{r_A} = \begin{pmatrix} -6 \\ -7 \\ -10 \end{pmatrix} \text{ m} \qquad \mathbf{F_V} = F\frac{\mathbf{r_A}}{|\mathbf{r_A}|} \qquad \mathbf{F_V} = \begin{pmatrix} -35.3 \\ -41.2 \\ -58.8 \end{pmatrix} \text{ N}$$

$$\mathbf{r_{AB}} = \begin{pmatrix} -e \\ -b \\ d \end{pmatrix} \qquad \mathbf{r_{AB}} = \begin{pmatrix} -6 \\ -3 \\ 2 \end{pmatrix} \text{ m} \qquad \mathbf{u_{AB}} = \frac{\mathbf{r_{AB}}}{|\mathbf{r_{AB}}|} \qquad \mathbf{u_{AB}} = \begin{pmatrix} -0.9 \\ -0.4 \\ 0.3 \end{pmatrix}$$

Now find the projection using the Dot product.

$F_{AB} = \mathbf{F_V} \cdot \mathbf{u_{AB}}$ $F_{AB} = 31.1$ N

Problem 2-120

Determine the angles θ and ϕ between the axis OA of the pole and each cable, AB and AC.

Given:

$F_1 = 50$ N

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$F_2 = 35$ N

$a = 1$ m

$b = 3$ m

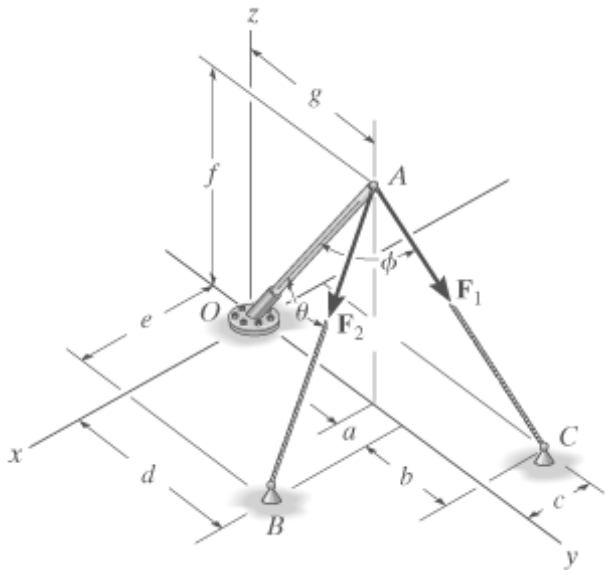
$c = 2$ m

$d = 5$ m

$e = 4$ m

$f = 6$ m

$g = 4$ m



Solution:

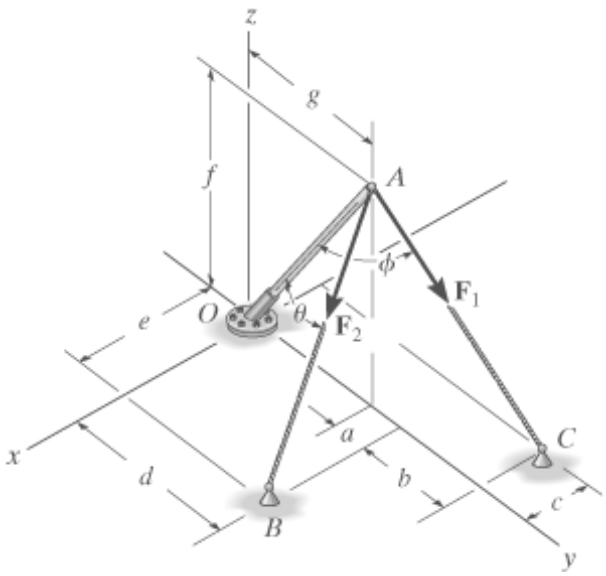
$$\mathbf{r_{AO}} = \begin{pmatrix} 0 \\ -g \\ -f \end{pmatrix} \qquad \mathbf{r_{AB}} = \begin{pmatrix} e \\ a \\ -f \end{pmatrix} \qquad \mathbf{r_{AC}} = \begin{pmatrix} -c \\ a+b \\ -f \end{pmatrix}$$

$$\theta = \operatorname{acos}\left(\frac{\mathbf{r_{AO}} \cdot \mathbf{r_{AB}}}{\left|\mathbf{r_{AO}}\right| \left|\mathbf{r_{AB}}\right|}\right) \qquad \theta = 52.4\,\text{deg}$$

$$\phi = \operatorname{acos}\left(\frac{\mathbf{r_{AO}} \cdot \mathbf{r_{AC}}}{\left|\mathbf{r_{AO}}\right| \left|\mathbf{r_{AC}}\right|}\right) \qquad \phi = 68.2\,\text{deg}$$

Problem 2-121

The two cables exert the forces shown on the pole. Determine the magnitude of the projected component of each force acting along the axis *OA* of the pole.



Given:

$F_1 = 50$ N

$F_2 = 35$ N

$a = 1$ m

$b = 3$ m

$c = 2$ m

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$d = 5$ m

$e = 4$ m

$f = 6$ m

$g = 4$ m

Solution:

$$\mathbf{r_{AB}} = \begin{pmatrix} e \\ a \\ -f \end{pmatrix} \qquad \mathbf{r_{AC}} = \begin{pmatrix} -c \\ a+b \\ -f \end{pmatrix} \qquad \mathbf{r_{AO}} = \begin{pmatrix} 0 \\ -g \\ -f \end{pmatrix} \qquad \mathbf{u_{AO}} = \frac{\mathbf{r_{AO}}}{|\mathbf{r_{AO}}|}$$

$$\mathbf{F_{1v}} = F_1 \frac{\mathbf{r_{AC}}}{|\mathbf{r_{AC}}|} \qquad F_{1AO} = \mathbf{F_{1v}} \cdot \mathbf{u_{AO}} \qquad F_{1AO} = 18.5 \text{ N}$$

$$\mathbf{F_{2v}} = F_2 \frac{\mathbf{r_{AB}}}{|\mathbf{r_{AB}}|} \qquad F_{2AO} = \mathbf{F_{2v}} \cdot \mathbf{u_{AO}} \qquad F_{2AO} = 21.3 \text{ N}$$

Problem 2-122

Force **F** is applied to the handle of the wrench. Determine the angle θ between the tail of the force and the handle *AB*.

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Given:

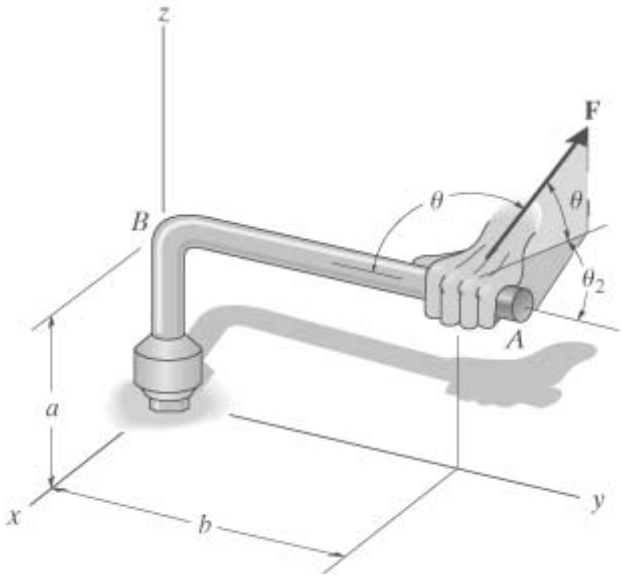
$a = 300$ mm

$b = 500$ mm

$F = 80$ N

$\theta_1 = 30$ deg

$\theta_2 = 45$ deg



Solution:

$$\mathbf{F_v} = F\begin{pmatrix} -\cos(\theta_1)\sin(\theta_2) \\ \cos(\theta_1)\cos(\theta_2) \\ \sin(\theta_1) \end{pmatrix} \qquad \mathbf{u_{ab}} = \begin{pmatrix} 0 \\ -1 \\ 0 \end{pmatrix}$$

$$\theta = \operatorname{acos}\left(\frac{\mathbf{F_v}\cdot\mathbf{u_{ab}}}{F}\right) \qquad \theta = 127.8\,\text{deg}$$

Problem 2-123

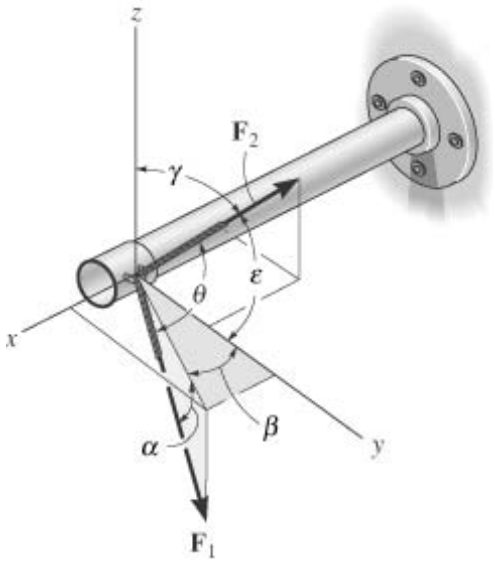
Two cables exert forces on the pipe. Determine the magnitude of the projected component of $\mathbf{F_1}$ along the line of action of $\mathbf{F_2}$.

Given:

$F_1 = 30$ lb $\quad \beta = 30$ deg

$F_2 = 25$ lb $\quad \gamma = 60$ deg

$\alpha = 30$ deg $\quad \varepsilon = 60$ deg



Solution:

We first need to find the third angle (> 90 deg) that locates force $\mathbf{F_2}$.

Initial Guess: $\phi = 120$ deg

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Given

$$\cos(\varepsilon)^2 + \cos(\gamma)^2 + \cos(\phi)^2 = 1$$

$$\phi = \text{Find}(\phi) \qquad \phi = 135\,\text{deg}$$

Find the force $\mathbf{F_{1v}}$ and the unit vector $\mathbf{u_2}$.

$$\mathbf{F_{1v}} = F_1 \begin{pmatrix} \cos(\alpha)\sin(\beta) \\ \cos(\alpha)\cos(\beta) \\ -\sin(\alpha) \end{pmatrix} \qquad \mathbf{F_{1v}} = \begin{pmatrix} 13 \\ 22.5 \\ -15 \end{pmatrix} \text{lb}$$

$$\mathbf{u_2} = \begin{pmatrix} \cos(\phi) \\ \cos(\varepsilon) \\ \cos(\gamma) \end{pmatrix} \qquad \mathbf{u_2} = \begin{pmatrix} -0.7 \\ 0.5 \\ 0.5 \end{pmatrix}$$

Now find the projection $F_{12} = \left|\mathbf{F_{1v}} \cdot \mathbf{u_2}\right|$ $F_{12} = 5.4\,\text{lb}$

Problem 2-124

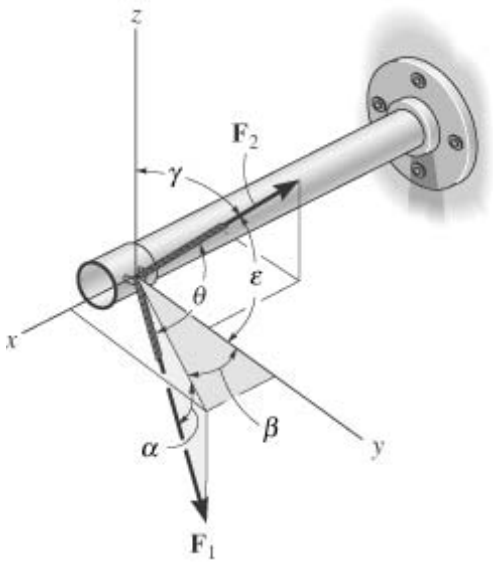
Determine the angle θ between the two cables attached to the pipe.

Given:

$F_1 = 30$ lb $\beta = 30$ deg

$F_2 = 25$ lb $\gamma = 60$ deg

$\alpha = 30$ deg $\varepsilon = 60$ deg



Solution:

We first need to find the third angle (> 90 deg) that locates force F_2.

Initial Guesses: $\phi = 120$ deg

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Given

$$\cos(\varepsilon)^2 + \cos(\gamma)^2 + \cos(\phi)^2 = 1$$

$\phi = \text{Find}(\phi)$ $\phi = 135\,\text{deg}$

Find the unit vectors $\mathbf{u_1}$ and $\mathbf{u_2}$.

$$\mathbf{u_1} = \begin{pmatrix} \cos(\alpha)\sin(\beta) \\ \cos(\alpha)\cos(\beta) \\ -\sin(\alpha) \end{pmatrix} \quad \mathbf{u_1} = \begin{pmatrix} 0.4 \\ 0.8 \\ -0.5 \end{pmatrix}$$

$$\mathbf{u_2} = \begin{pmatrix} \cos(\phi) \\ \cos(\varepsilon) \\ \cos(\gamma) \end{pmatrix} \quad \mathbf{u_2} = \begin{pmatrix} -0.7 \\ 0.5 \\ 0.5 \end{pmatrix}$$

Find the angle using the dot product $\theta = \text{acos}(\mathbf{u_1} \cdot \mathbf{u_2})$ $\theta = 100.4\,\text{deg}$

Problem 2-125

Determine the angle θ between the two cables.

Given:

$a = 7.5$ ft

$b = 2$ ft

$c = 3$ ft

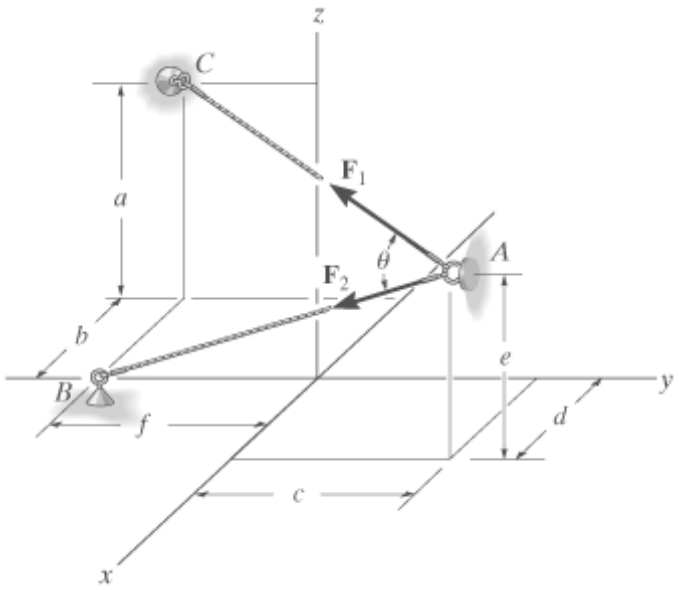
$d = 2$ ft

$e = 3$ ft

$f = 3$ ft

$F_1 = 60$ lb

$F_2 = 30$ lb



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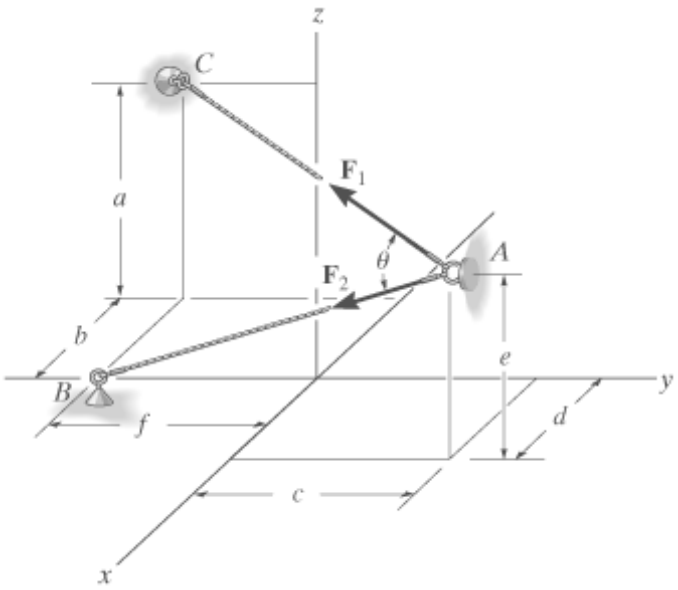
Solution:

$$\mathbf{r_{AC}} = \begin{pmatrix} -d-b \\ -c-f \\ a-e \end{pmatrix} \qquad \mathbf{r_{AB}} = \begin{pmatrix} -d \\ -c-f \\ -e \end{pmatrix}$$

$$\theta = \operatorname{acos}\left(\frac{\mathbf{r_{AC}}\cdot\mathbf{r_{AB}}}{|\mathbf{r_{AC}}|\,|\mathbf{r_{AB}}|}\right) \qquad \theta = 59.2\,\text{deg}$$

Problem 2-126

Determine the projection of the force $\mathbf{F}_1$ along cable *AB*. Determine the projection of the force $\mathbf{F}_2$ along cable *AC*.



Given:

$a = 7.5$ ft

$b = 2$ ft

$c = 3$ ft

$d = 2$ ft

$e = 3$ ft

$f = 3$ ft

$F_1 = 60$ lb

$F_2 = 30$ lb

Solution:

$$\mathbf{r_{AC}} = \begin{pmatrix} -d-b \\ -c-f \\ a-e \end{pmatrix} \qquad \mathbf{r_{AB}} = \begin{pmatrix} -d \\ -c-f \\ -e \end{pmatrix} \qquad \mathbf{u_{AC}} = \frac{\mathbf{r_{AC}}}{|\mathbf{r_{AC}}|} \qquad \mathbf{u_{AB}} = \frac{\mathbf{r_{AB}}}{|\mathbf{r_{AB}}|}$$

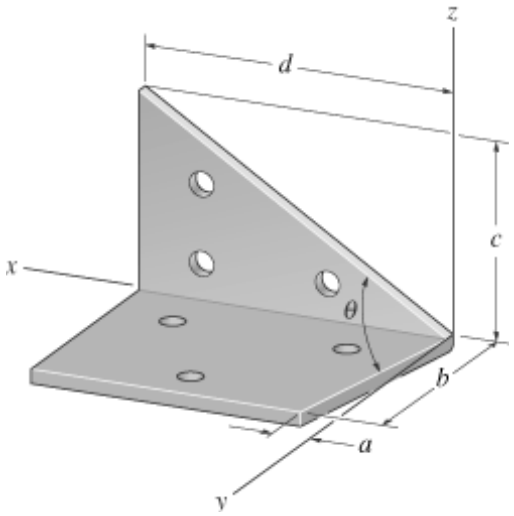
$$\mathbf{F_{1v}} = F_1\mathbf{u_{AC}} \qquad F_{1AB} = \mathbf{F_{1v}}\cdot\mathbf{u_{AB}} \qquad F_{1AB} = 30.8\,\text{lb}$$

$$\mathbf{F_{2v}} = F_2\mathbf{u_{AB}} \qquad F_{2AC} = \mathbf{F_{2v}}\cdot\mathbf{u_{AC}} \qquad F_{2AC} = 15.4\,\text{lb}$$

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Problem 2-127

Determine the angle θ between the edges of the sheet-metal bracket.



Given:

$a = 50$ mm

$b = 300$ mm

$c = 250$ mm

$d = 400$ mm

Solution:

Find the unit vectors and use the dot product

$$\mathbf{r_1} = \begin{pmatrix} d \\ 0 \\ c \end{pmatrix} \qquad \mathbf{u_1} = \frac{\mathbf{r_1}}{|\mathbf{r_1}|}$$

$$\mathbf{r_2} = \begin{pmatrix} a \\ b \\ 0 \end{pmatrix} \qquad \mathbf{u_2} = \frac{\mathbf{r_2}}{|\mathbf{r_2}|}$$

$$\mathbf{u_1} = \begin{pmatrix} 0.848 \\ 0.000 \\ 0.530 \end{pmatrix} \quad \mathbf{u_2} = \begin{pmatrix} 0.164 \\ 0.986 \\ 0.000 \end{pmatrix} \qquad \theta = \operatorname{acos}(\mathbf{u_1} \cdot \mathbf{u_2}) \qquad \theta = 82\,\text{deg}$$

Problem 2-128

Determine the magnitude of the projected component of the force **F** acting along the axis *BC* of the pipe.

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Given:

$F = 100$ lb

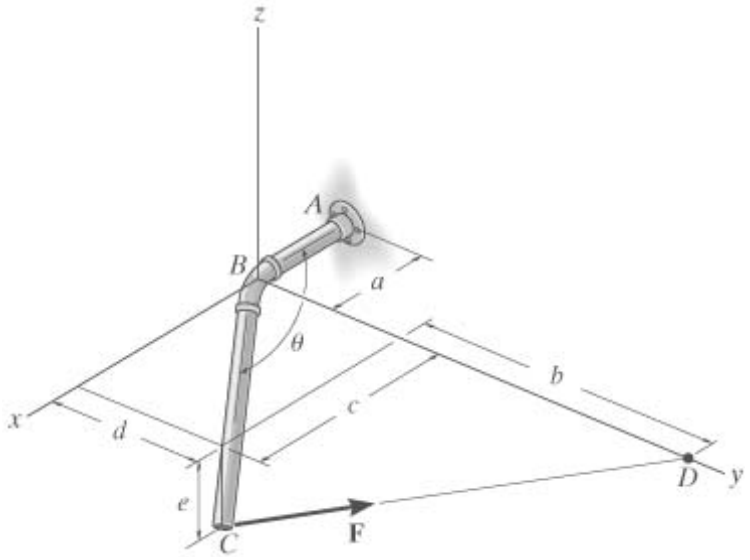
$a = 2$ ft

$b = 8$ ft

$c = 6$ ft

$d = 4$ ft

$e = 2$ ft



Solution:

$$\mathbf{r_{CD}} = \begin{pmatrix} -c \\ b \\ e \end{pmatrix} \qquad \mathbf{u_{CD}} = \frac{\mathbf{r_{CD}}}{|\mathbf{r_{CD}}|} \qquad \mathbf{r_{CB}} = \begin{pmatrix} -c \\ -d \\ e \end{pmatrix} \qquad \mathbf{u_{CB}} = \frac{\mathbf{r_{CB}}}{|\mathbf{r_{CB}}|}$$

$$F_{BC} = \left(F\mathbf{u_{CD}}\right)\cdot\mathbf{u_{CB}} \qquad F_{BC} = 10.5\,\text{lb}$$

Problem 2-129

Determine the angle θ between pipe segments *BA* and *BC*.

Given:

$F = 100$ lb

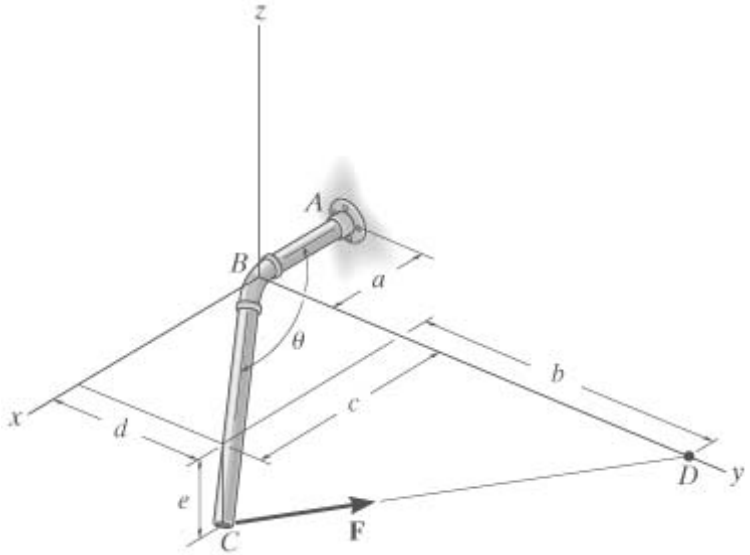
$a = 3$ ft

$b = 8$ ft

$c = 6$ ft

$d = 4$ ft

$e = 2$ ft



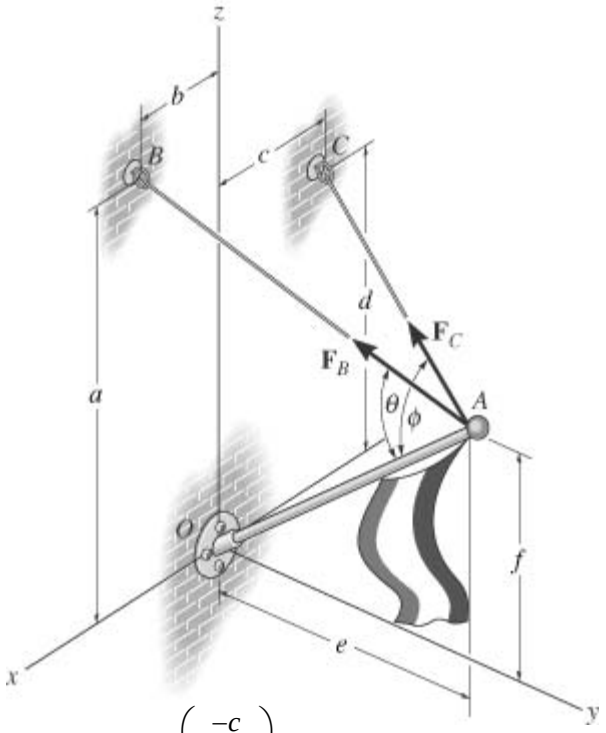
Solution:

$$\mathbf{r_{BC}} = \begin{pmatrix} c \\ d \\ -e \end{pmatrix} \qquad \mathbf{r_{BA}} = \begin{pmatrix} -a \\ 0 \\ 0 \end{pmatrix} \qquad \theta = \operatorname{acos}\left(\frac{\mathbf{r_{BC}}\cdot\mathbf{r_{BA}}}{|\mathbf{r_{BC}}|\,|\mathbf{r_{BA}}|}\right) \qquad \theta = 143.3\,\text{deg}$$

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Problem 2-130

Determine the angles θ and ϕ made between the axes *OA* of the flag pole and *AB* and *AC*, respectively, of each cable.



Given:

$F_B = 55$ N $\quad c = 2$ m

$F_C = 40$ N $\quad d = 4$ m

$a = 6$ m $\quad e = 4$ m

$b = 1.5$ m $\quad f = 3$ m

Solution:

$$\mathbf{r_{AO}} = \begin{pmatrix} 0 \\ -e \\ -f \end{pmatrix} \qquad \mathbf{r_{AB}} = \begin{pmatrix} b \\ -e \\ a - f \end{pmatrix} \qquad \mathbf{r_{AC}} = \begin{pmatrix} -c \\ -e \\ d - f \end{pmatrix}$$

$$\theta = \operatorname{acos}\left(\frac{\mathbf{r_{AB}} \cdot \mathbf{r_{AO}}}{|\mathbf{r_{AB}}|\,|\mathbf{r_{AO}}|}\right) \qquad \theta = 74.4\,\text{deg}$$

$$\phi = \operatorname{acos}\left(\frac{\mathbf{r_{AC}} \cdot \mathbf{r_{AO}}}{|\mathbf{r_{AC}}|\,|\mathbf{r_{AO}}|}\right) \qquad \phi = 55.4\,\text{deg}$$

Problem 2-131

Determine the magnitude and coordinate direction angles of $\mathbf{F_3}$ so that resultant of the three forces acts along the positive *y* axis and has magnitude F_R.

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Given:

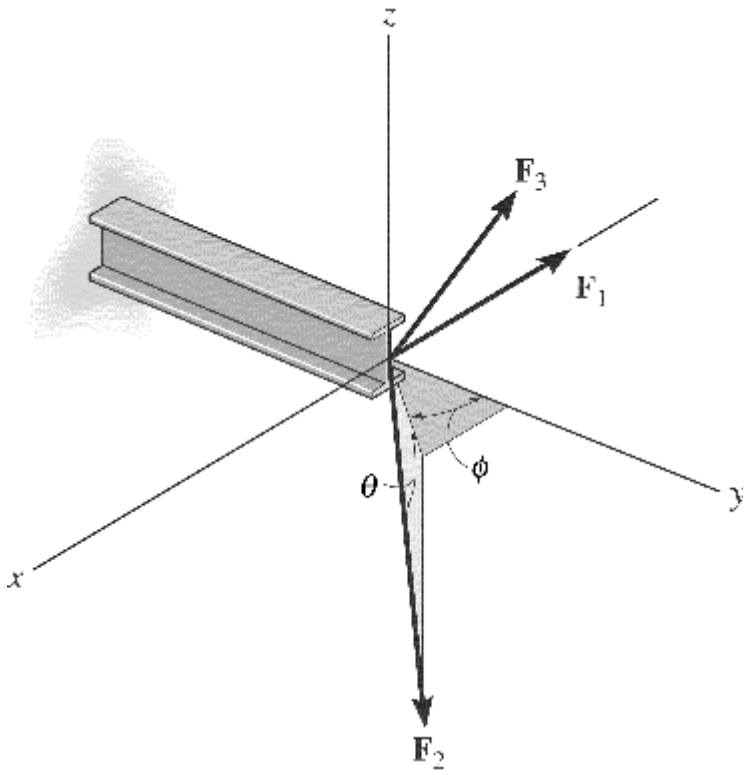
$F_R = 600$ lb

$F_1 = 180$ lb

$F_2 = 300$ lb

$\phi = 40$ deg

$\theta = 30$ deg



Solution:

The initial guesses**:**

$F_3 = 100$ lb $\quad \beta = 30$ deg

$\alpha = 10$ deg $\quad \gamma = 60$ deg

Given

$F_{Rx} = \Sigma F_x; \quad -F_1 + F_2\cos(\theta)\sin(\phi) + F_3\cos(\alpha) = 0$

$F_{Ry} = \Sigma F_y; \quad F_2\cos(\theta)\cos(\phi) + F_3\cos(\beta) = F_R$

$F_{Rz} = \Sigma F_z; \quad -F_2\sin(\theta) + F_3\cos(\gamma) = 0$

$\cos(\alpha)^2 + \cos(\beta)^2 + \cos(\gamma)^2 = 1$

Solving:

$$\begin{pmatrix} F_3 \\ \alpha \\ \beta \\ \gamma \end{pmatrix} = \text{Find}(F_3, \alpha, \beta, \gamma) \qquad \begin{pmatrix} \alpha \\ \beta \\ \gamma \end{pmatrix} = \begin{pmatrix} 88.3 \\ 20.6 \\ 69.5 \end{pmatrix} \text{deg} \qquad F_3 = 428.3\,\text{lb}$$

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Problem 2-132

Determine the magnitude and coordinate direction angles of $\mathbf{F_3}$ so that resultant of the three forces is zero.

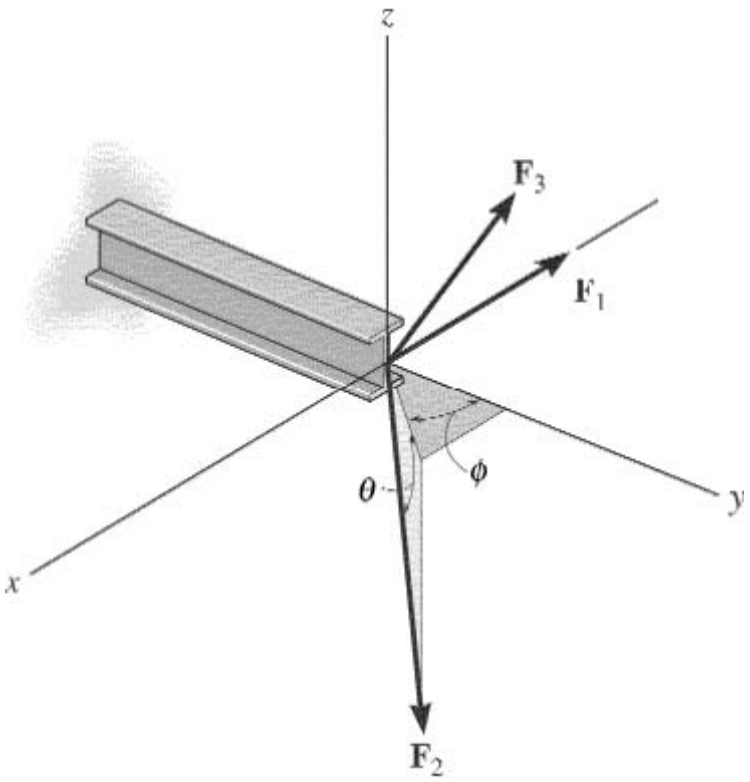
Given:

$F_1 = 180$ lb

$F_2 = 300$ lb

$\phi = 40$ deg

$\theta = 30$ deg



Solution:

The initial guesses**:**

$\alpha = 10$ deg

$\beta = 30$ deg

$\gamma = 60$ deg

$F_3 = 100$ lb

Given

$F_{Rx} = \Sigma F_x; \quad -F_1 + F_2\cos(\theta)\sin(\phi) + F_3\cos(\alpha) = 0$

$F_{Ry} = \Sigma F_y; \quad F_2\cos(\theta)\cos(\phi) + F_3\cos(\beta) = 0$

$F_{Rz} = \Sigma F_z; \quad -F_2\sin(\theta) + F_3\cos(\gamma) = 0$

$\cos(\alpha)^2 + \cos(\beta)^2 + \cos(\gamma)^2 = 1$

Solving:

$$\begin{pmatrix} F_3 \\ \alpha \\ \beta \\ \gamma \end{pmatrix} = \text{Find}(F_3, \alpha, \beta, \gamma) \qquad \begin{pmatrix} \alpha \\ \beta \\ \gamma \end{pmatrix} = \begin{pmatrix} 87 \\ 142.9 \\ 53.1 \end{pmatrix} \text{deg} \qquad F_3 = 249.6\,\text{lb}$$

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Problem 2-133

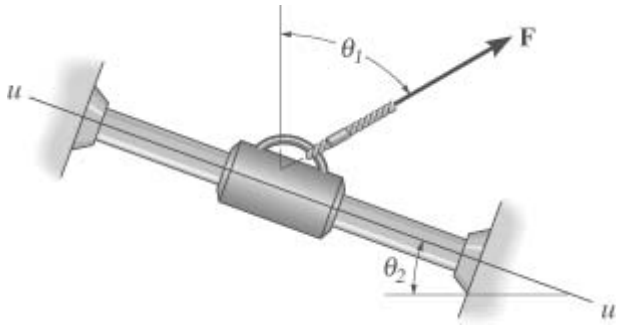
Resolve the force **F** into two components, one acting parallel and the other acting perpendicular to the *u* axis.

Given:

$F = 600$ lb

$\theta_1 = 60$ deg

$\theta_2 = 20$ deg



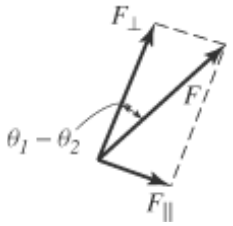
Solution:

$F_{perpendicular} = F\cos(\theta_1 - \theta_2)$

$F_{perpendicular} = 460\,\text{lb}$

$F_{parallel} = F\sin(\theta_1 - \theta_2)$

$F_{parallel} = 386\,\text{lb}$



Problem 2-134

The force **F** has a magnitude *F* and acts at the midpoint *C* of the thin rod. Express the force as a Cartesian vector.

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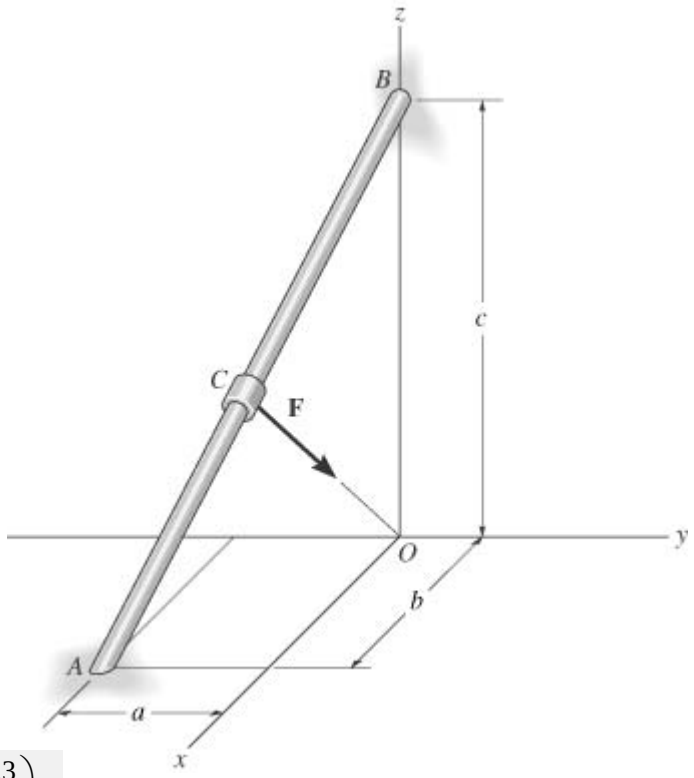
Given:

$F = 80$ lb

$a = 2$ ft

$b = 3$ ft

$c = 6$ ft



Solution:

$$\mathbf{r_{CO}} = \begin{pmatrix} \frac{-b}{2} \\ \frac{a}{2} \\ \frac{-c}{2} \end{pmatrix}$$

$$\mathbf{F_V} = F\frac{\mathbf{r_{CO}}}{|\mathbf{r_{CO}}|} \qquad \mathbf{F_V} = \begin{pmatrix} -34.3 \\ 22.9 \\ -68.6 \end{pmatrix} \text{lb}$$

Problem 2-135

Determine the magnitude and direction of the resultant $\mathbf{F_R} = \mathbf{F_1} + \mathbf{F_2} + \mathbf{F_3}$ of the three forces by first finding the resultant $\mathbf{F'} = \mathbf{F_1} + \mathbf{F_3}$ and then forming $\mathbf{F_R} = \mathbf{F'} + \mathbf{F_2}$. Specify its direction measured counterclockwise from the positive x axis.

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Given:

$F_1 = 80$ N

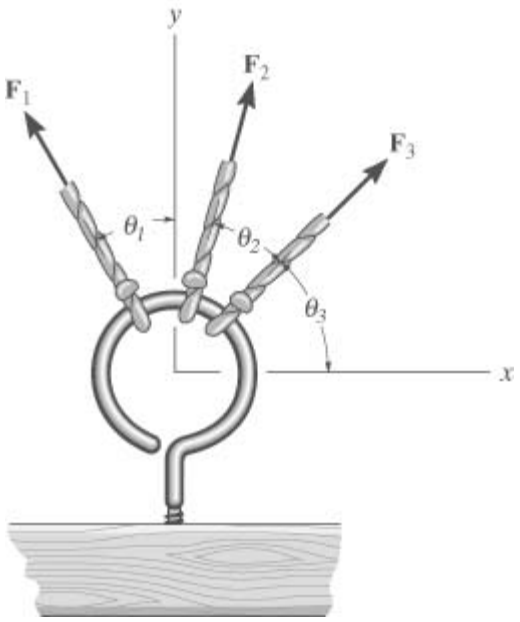
$F_2 = 75$ N

$F_3 = 50$ N

$\theta_1 = 30$ deg

$\theta_2 = 30$ deg

$\theta_3 = 45$ deg



Solution:

$$\mathbf{F_{1v}} = F_1\begin{pmatrix}-\sin(\theta_1)\\ \cos(\theta_1)\end{pmatrix} \qquad \mathbf{F_{2v}} = F_2\begin{pmatrix}\cos(\theta_2+\theta_3)\\ \sin(\theta_2+\theta_3)\end{pmatrix} \qquad \mathbf{F_{3v}} = F_3\begin{pmatrix}\cos(\theta_3)\\ \sin(\theta_3)\end{pmatrix}$$

$$\mathbf{F'} = \mathbf{F_{1v}} + \mathbf{F_{3v}} \qquad \mathbf{F'} = \begin{pmatrix}-4.6\\ 104.6\end{pmatrix}\text{ N} \qquad \mathbf{i} = \begin{pmatrix}1\\ 0\end{pmatrix} \qquad \mathbf{j} = \begin{pmatrix}0\\ 1\end{pmatrix}$$

$$\mathbf{F_R} = \mathbf{F'} + \mathbf{F_{2v}} \qquad \mathbf{F_R} = \begin{pmatrix}14.8\\ 177.1\end{pmatrix}\text{ N} \qquad |\mathbf{F_R}| = 177.7\text{ N}$$

$$\theta = \operatorname{atan}\left(\frac{\mathbf{F_R}\mathbf{j}}{\mathbf{F_R}\mathbf{i}}\right) \qquad \theta = 85.2\text{ deg}$$

Problem 2-136

The leg is held in position by the quadriceps *AB*, which is attached to the pelvis at *A*. If the force exerted on this muscle by the pelvis is **F**, in the direction shown, determine the stabilizing force component acting along the positive *y* axis and the supporting force component acting along the negative *x* axis.

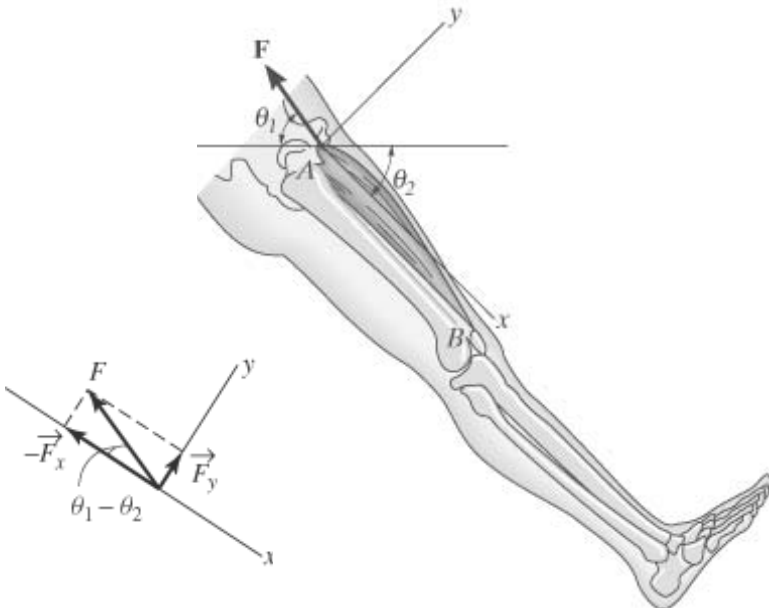
Given:

$F = 85$ N

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$\theta_1 = 55 \text{ deg}$

$\theta_2 = 45 \text{ deg}$



Solution:

$F_x = F\cos(\theta_1 - \theta_2)$

$F_x = 83.7 \text{ N}$

$F_y = F\sin(\theta_1 - \theta_2)$

$F_y = 14.8 \text{ N}$

Problem 2-137

Determine the magnitudes of the projected components of the force **F** in the direction of the cables *AB* and *AC* .

Given:

$$\mathbf{F} = \begin{pmatrix} 60 \\ 12 \\ -40 \end{pmatrix} \text{ N}$$

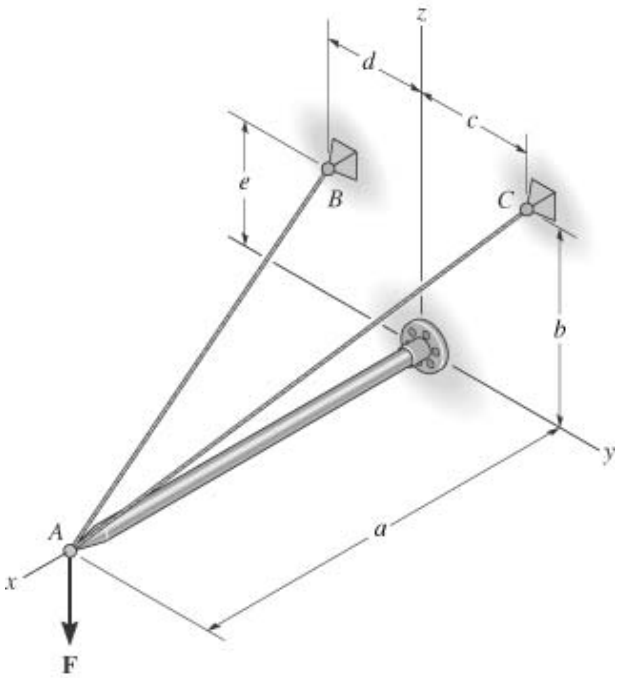
$a = 3 \text{ m}$

$b = 1.5 \text{ m}$

$c = 1 \text{ m}$

$d = 0.75 \text{ m}$

$e = 1 \text{ m}$



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Solution:

Find the unit vectors, then use the dot product

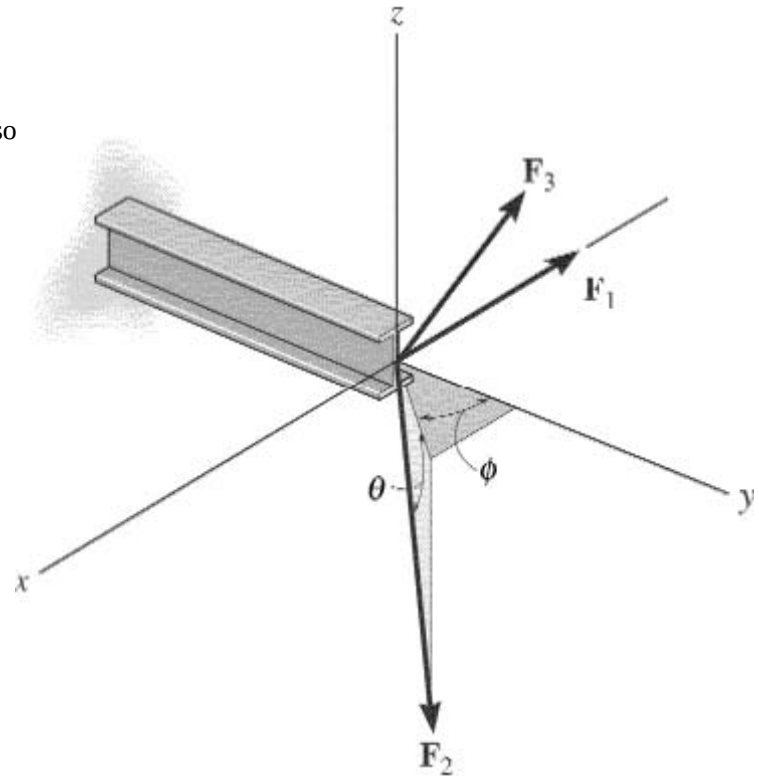
$$\mathbf{r_{AB}} = \begin{pmatrix} -a \\ -d \\ e \end{pmatrix} \quad \mathbf{r_{AB}} = \begin{pmatrix} -3 \\ -0.8 \\ 1 \end{pmatrix} \text{ m} \quad \mathbf{u_{AB}} = \frac{\mathbf{r_{AB}}}{|\mathbf{r_{AB}}|} \quad \mathbf{u_{AB}} = \begin{pmatrix} -0.9 \\ -0.2 \\ 0.3 \end{pmatrix}$$

$$\mathbf{r_{AC}} = \begin{pmatrix} -a \\ c \\ b \end{pmatrix} \quad \mathbf{r_{AC}} = \begin{pmatrix} -3 \\ 1 \\ 1.5 \end{pmatrix} \text{ m} \quad \mathbf{u_{AC}} = \frac{\mathbf{r_{AC}}}{|\mathbf{r_{AC}}|} \quad \mathbf{u_{AC}} = \begin{pmatrix} -0.9 \\ 0.3 \\ 0.4 \end{pmatrix}$$

$$\mathbf{F_{AB}} = F\mathbf{u_{AB}} \quad \mathbf{F_{AB}} = \begin{pmatrix} -78.5 \\ -19.6 \\ 26.2 \end{pmatrix} \text{ N} \quad \mathbf{F_{AC}} = F\mathbf{u_{AC}} \quad \mathbf{F_{AC}} = \begin{pmatrix} -72.9 \\ 24.3 \\ 36.4 \end{pmatrix} \text{ N}$$

Problem 2-138

Determine the magnitude and coordinate direction angles of $\mathbf{F_3}$ so that resultant of the three forces is zero.



Given:

$F_1 = 180$ lb $\quad \phi = 40$ deg

$F_2 = 300$ lb $\quad \theta = 30$ deg

Solution:

The initial guesses**:**

$\alpha = 10$ deg

$\beta = 30$ deg

$\gamma = 60$ deg

$F_3 = 100$ lb

Given

$$F_{Rx} = \Sigma F_x; \qquad -F_1 + F_2\cos(\theta)\sin(\phi) + F_3\cos(\alpha) = 0$$

$$F_{Ry} = \Sigma F_y; \qquad F_2\cos(\theta)\cos(\phi) + F_3\cos(\beta) = 0$$

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$F_{Rz} = \Sigma F_z; \qquad -F_2 \sin(\theta) + F_3 \cos(\gamma) = 0$

$\cos(\alpha)^2 + \cos(\beta)^2 + \cos(\gamma)^2 = 1$

Solving:

$$\begin{pmatrix} F_3 \\ \alpha \\ \beta \\ \gamma \end{pmatrix} = \text{Find}(F_3, \alpha, \beta, \gamma) \qquad \begin{pmatrix} \alpha \\ \beta \\ \gamma \end{pmatrix} = \begin{pmatrix} 87 \\ 142.9 \\ 53.1 \end{pmatrix} \text{deg} \qquad F_3 = 249.6\,\text{lb}$$

Problem 2-139

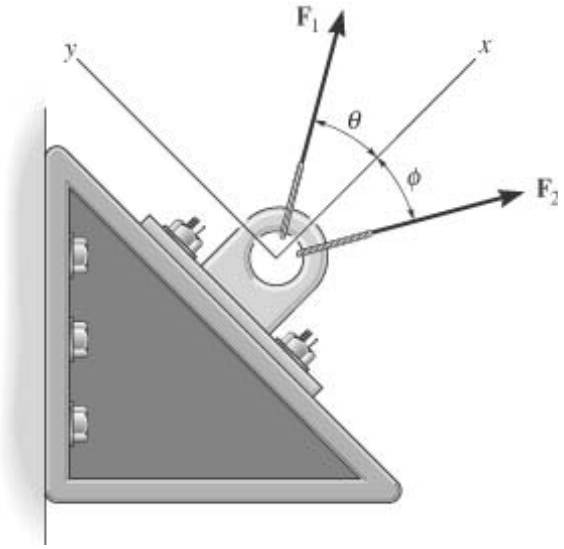
Determine the angles θ and ϕ so that the resultant force is directed along the positive x axis and has magnitude F_R.

Given:

$F_1 = 30$ lb

$F_2 = 30$ lb

$F_R = 20$ lb



Solution:

Initial Guesses:

$\theta = 20$ deg $\qquad \phi = 20$ deg

Given

$$\frac{F_1}{\sin(\phi)} = \frac{F_2}{\sin(\theta)}$$

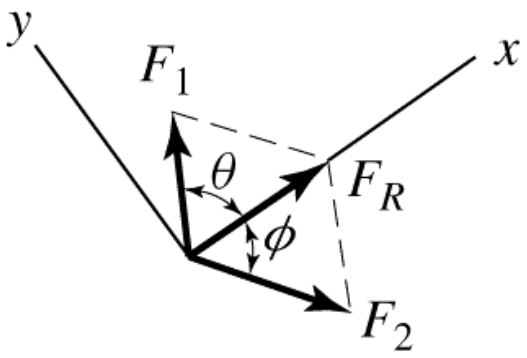
$$F_R{}^2 = F_1{}^2 + F_2{}^2 - 2\,F_1\,F_2 \cos(180\ \text{deg} - \theta - \phi)$$

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$$\begin{pmatrix} \theta \\ \phi \end{pmatrix} = \text{Find}(\theta, \phi)$$

$\theta = 70.5\,\text{deg}$

$\phi = 70.5\,\text{deg}$



Problem 2-140

Determine the magnitude of the resultant force and its direction measured counterclockwise from the *x* axis.

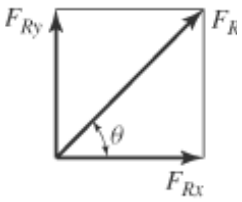
Given:

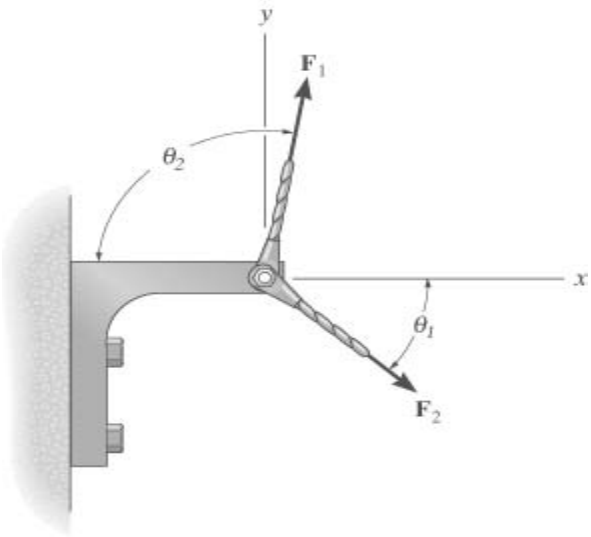
$F_1 = 300\ \text{lb}$

$F_2 = 200\ \text{lb}$

$\theta_1 = 40\ \text{deg}$

$\theta_2 = 100\ \text{deg}$





Solution:

$$F_{Rx} = F_1 \cos(180\ \text{deg} - \theta_2) + F_2 \cos(\theta_1) \qquad F_{Rx} = 205.3\,\text{lb}$$

$$F_{Ry} = F_1 \sin(180\ \text{deg} - \theta_2) - F_2 \sin(\theta_1) \qquad F_{Ry} = 166.9\,\text{lb}$$

$$F_R = \sqrt{F_{Rx}^2 + F_{Ry}^2} \qquad F_R = 265\,\text{lb}$$

$$\theta = \text{atan}\left(\frac{F_{Ry}}{F_{Rx}}\right) \qquad \theta = 39.1\,\text{deg}$$

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